11

DLEARN—A Linear Static and Dynamic Finite Element Analysis Program

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11.1. INTRODUCTION

This chapter describes DLEARN, a finite element code for linear static and dynamic analysis. For instructional purposes it may be considered a "first example" in that the use of auxiliary disk storage to reduce memory requirements is omitted, and blocking (segmenting large jobs via auxiliary storage) and overlay structures are avoided. Also, the class of capabilties—linear static and dynamic analysis—is relatively simple. Nevertheless, many sophisticated procedures are employed (e.g., compacted column solution and data structures, dynamic storage allocation, a memory manager, element routines, data generation routines, etc.) that are used in more advanced situations.

An important feature of DLEARN is that it has been written to be an integral constituent of this book. Methods and data-processing techniques developed throughout the first 10 chapters are employed in DLEARN. The names of variables and arrays are kept as close as possible to those used in the development of the theory. It is believed that this will expedite the learning process. Finite element methods find their ultimate manifestations in computer programs. Unless the student comprehends the intricacies of finite element programs, he or she will never fully appreciate the finite element method.

The major features and capabilities of DLEARN are listed below along with relevant subsections of the Input Instructions (these are denoted by "Sec. In." followed by a decimal number):

- Modular program structure to enhance readability and extendability.
- Memory pointer dictionary printed with each execution to provide storage information and illustrate the concept of dynamic storage allocation.
- Simple, but labor-saving, nodal generation capabilities in one, two, and three

dimensions (Sec. In.5.0) as well as element generation capabilities (Sec. In.10.0).

- Complex loadings can be synthesized by combination of multiple load vectors and load-time functions (Secs. In.7.0 and In.8.0).
- Static analysis with full stress recovery (Sec. In.2.0).
- Dynamic analysis with an α-method predictor/corrector algorithm allowing calculation of consistent initial accelerations (Sec. In.2.0). Includes Rayleigh viscous damping with mixed implicit-explicit element groups (Sec. In.10.0).
- Four-node quadrilaterial, elastic continuum element, with plane stress, plane strain, and torsionless axisymmetric capabilities. Includes a mean-dilatational $\overline{B}$ capability for applications to nearly incompressible media. (Sec. In.10.1).
- Three-dimensional. elastic truss element (Sec. In.10.2).

DLEARN is not intended to be a "full-service" analysis package with a broad range of capabilities. For instance, the number of element types included has been purposely limited. However, major emphasis has been placed on providing a flexible and comprehensible program structure to which the student or researcher can profitably add new features of his or her own choosing. Moreover, the code has been structured around techniques having natural extensions to nonlinear analysis.

DLEARN may be used for problem solution in finite element courses and serves the following additional purposes:

1. It is felt that the first step in learning to implement finite element methods is to add a new element to an existing program. In this regard, DLEARN is written so that it is very easy to add new elements. (Instructions are given in Sec.11.4 entitled "Adding an Element to DLEARN.") Students may elect to perform projects of this kind. For this reason, only two sets of element routines are included with the listing so that many standard element types (e.g., three-dimensional elasticity; beam, plate, and shell elements; heat conduction elements; etc.) are available for project topics.

2. DLEARN makes available to students a good, first, finite element "software package." The subroutines are efficiently and simply coded and, as the student becomes more advanced, the subroutines may be used as a set of building blocks for more-sophisticated applications. With this idea in mind it was decided to include a very efficient equation solver in DLEARN, which employs a compacted column storage mode.

The reader is forwarned that the level of documentation is not considered self-sufficient (nor is it intended to be) for fully understanding all aspects of the program but rather is supplementary to the remainder of the text.

1.2. DESCRIPTION OF CODING TECHNIQUES USED IN DLEARN

In keeping with contemporary programming practice, DLEARN has been written with the objectives of readability, maintainability and extendability. The 1000-line subroutines of an earlier generation of finite element codes have been eschewed in

favor of compact subroutines with a limited number of tasks to perform. While large subroutines have a place in codes dedicated to efficient solution of large-scale problems, we believe smaller program units will be more easily comprehended by students first learning how to implement finite element methodologies. To this end, utility subroutines are widely used to remove "in-line" code, which, though faster, might obscure the algorithmic outline of a procedure. To understand DLEARN and ultimately extend its capabilities, the user must become familiar with the data structure.

Definition. The *data structure* is the mode of storage, location, and history of existence of the data employed.

In DLEARN, all the data are stored in central ("core") memory in the unnamed common block COMMON A(MTOT), where the size parameter MTOT is set in the main program. Many operating systems, such as those for Control Data Corporation (CDC) and Cray Research computers, allow greater run-time flexibility in memory allocation if the storage vector A is kept in an unnamed COMMON block. This fact is responsible for the commonly made remark that finite element programs store data in "blank common." On virtual memory machines such as the Digital Equipment Corporation (DEC) VAX-11/7xx series, however, no penalty is incurred by using a named common block. The basic features of the data structure are:

Compacted column architecture This feature is engendered by the method of equation solving employed. The solver is called variously a "profile," "skyline," or "active column" solver. Since, in large problems, the dominant portion of storage is usually that devoted to the effective mass matrix M^*, the storage scheme for M^* is the most important feature of the storage.

Dynamic storage allocation Because of the variable sizes of the arrays needed in different problems, it would be very inefficient to "fix" the dimensioning. In the dynamic storage allocation concept the dimensions of arrays are set in the program at the time of execution.

Element groups Many features of finite element computer programs are the same even though the intended applications may be quite different. To some extent this is also true of the individual finite element subroutines. However, the data structure and options available may vary considerably from one element to another. To accommodate these differences, the elements are read in, stored and operated upon in groups. The data structure for the group is set up via dynamic storage allocation in individual element subroutines.

11.2.1 Compacted Column Storage Scheme

Assume the symmetric effective mass matrix M^* has dimension 8×8 and has zero and nonzero elements indicated by 0 and X, respectively, in the figure on page 634. The dashed line enveloping the nonzero terms is sometimes called the "profile" or "skyline."

$$M^* = [M^*_{ij}] = \begin{bmatrix} X & X & 0 & 0 & 0 & 0 & 0 & X \\ & X & X & 0 & X & 0 & 0 & 0 \\ & & X & X & X & 0 & 0 & 0 \\ & & & X & 0 & X & 0 & X \\ & & & & X & X & 0 & 0 \\ & & & & & X & X & 0 \\ & & & & & & X & X \\ & & & & & & & X \end{bmatrix}$$

M^* is stored in a one-dimensional array, ALHS, column-wise beginning with the first nonzero element in each column and ending with the diagonal term. The locations of the diagonal terms in ALHS are stored in a one-dimensional integer array IDIAG. The dimension of IDIAG is NEQ, the number of equations (e.g., for M^*, NEQ = 8). The dimension of ALHS is NALHS, the sum of the "column heights." The column height is the number of terms in a column beginning with the first nonzero term and ending with the diagonal term (e.g., for the above matrix M^*, column 1 has height 1, column 4 has height 2, column 6 has height 3, etc.). NALHS for M^* is 24. The arrays ALHS and IDIAG corresponding to M^* are given explicitly as follows:

ALHS(1) = M^*_{11}	column 1
ALHS(2) = M^*_{12}	column 2
ALHS(3) = M^*_{22}	
ALHS(4) = M^*_{23}	column 3
ALHS(5) = M^*_{33}	
ALHS(6) = M^*_{34}	column 4
ALHS(7) = M^*_{44}	
ALHS(8) = M^*_{25}	column 5
ALHS(9) = M^*_{35}	
ALHS(10) = M^*_{45}	
ALHS(11) = M^*_{55}	
ALHS(12) = M^*_{46}	column 6
ALHS(13) = M^*_{56}	
ALHS(14) = M^*_{66}	
ALHS(15) = M^*_{67}	column 7
ALHS(16) = M^*_{77}	

$$
\left.\begin{aligned}
\text{ALHS}(17) &= M^*_{18}\\
\text{ALHS}(18) &= M^*_{28}\\
\text{ALHS}(19) &= M^*_{38}\\
\text{ALHS}(20) &= M^*_{48}\\
\text{ALHS}(21) &= M^*_{58}\\
\text{ALHS}(22) &= M^*_{68}\\
\text{ALHS}(23) &= M^*_{78}\\
\text{ALHS}(24) &= M^*_{88}
\end{aligned}\right\} \text{column 8}
$$

$$
\begin{aligned}
\text{IDIAG}(1) &= 1\\
\text{IDIAG}(2) &= 3\\
\text{IDIAG}(3) &= 5\\
\text{IDIAG}(4) &= 7\\
\text{IDIAG}(5) &= 11\\
\text{IDIAG}(6) &= 14\\
\text{IDIAG}(7) &= 16\\
\text{IDIAG}(8) &= 24
\end{aligned}
$$

The zero terms beneath the skyline must be stored because they become nonzero during the factorization process. All information in the original M^* is now contained in the arrays ALHS and IDIAG in compact fashion. For large finite element effective mass matrices, the saving of storage is considerable.

The heights of the columns in M^* are determined by subroutine COLHT from the element group LM arrays. The column heights are initially stored in IDIAG. When the calculation of column heights is completed, subroutine DIAG determines the diagonal address and "overwrites" them into IDIAG.

The right-hand-side vector (residual, or out-of-balance, force) in

$$M^* \Delta a = R$$

is stored in an array BRHS of dimension NEQ in the obvious way:

$$
\begin{aligned}
\text{BRHS}(1) &= R_1\\
\text{BRHS}(2) &= R_2\\
&\vdots\\
\text{BRHS}(8) &= R_8
\end{aligned}
$$

The elimination scheme employed is based upon the following theorem.

Theorem. Let M^* be symmetric and positive-definite. Then there exists a

nonsingular upper triangular matrix U, with unit diagonal entries, and a diagonal matrix D such that

$$M^* = U^T D U$$

Remarks

1. In the case when M^* is positive-definite, the diagonal entries of D are all strictly positive. Thus examining the diagonal entries of D enables us to determine the rank of M^*. This is the purpose of the ***rank check*** option in the program.

2. The matrix U has the same profile as M^*. The fact that U has diagonal entries equal to one allows us to save storage by placing the diagonal entries of D in the diagonal entries of the array ALHS. Thus after factorization the nonzero entries of both U and D are stored in array ALHS. During the factorization procedure *no* auxiliary storage is needed.

3. Upper triangular matrices with unit diagonal entries are sometimes referred to as ***unit upper triangular matrices***.

11.2.2 Crout Elimination

The procedure used for performing the factorization of the left-hand-side matrix is called ***Crout elimination***, a convenient variant of Gauss elimination. In the Crout algorithm, one column at a time is completely factorized, beginning with the first column and not involving subsequent columns. This feature proves convenient in nonlinear analysis in which only a small zone of the entire matrix is nonlinear. The linear portion can be located in the first columns, factorized once and retained for the duration of the calculation.

After factorization, the solution is carried out in three steps:

$$U^T z = R \quad \text{forward reduction}$$

$$Dy = z \quad \text{diagonal scaling}$$

$$U \Delta a = y \quad \text{back substitution}$$

In each step the solution is an explicit process, the coefficient matrix being triangular or diagonal. The intermediate vectors y and z are in turn stored in the array BRHS. Thus again no additional storage besides that for ALHS and BRHS (and IDIAG) is needed to carry out the factorization and solution processes.

Derivation of Computational Formulas for Crout Elimination

To simplify the notation, let us write

$$Ax = b$$

and assume x is n-dimensional. We wish to derive algorithms for

$$A = U^T D U$$

$$U^T z = b$$

$$Dy = z$$

and

$$Ux = y$$

Factorization

The index version of the Crout factorization is expressed as

$$A_{ij} = \sum_{k=1}^{j} U_{ki} D_{kk} U_{kj} \qquad (1 \le i \le j)$$

where

$$U_{ii} = 1,$$
$$U_{ij} = 0, \qquad \text{for } i > j$$

Useful formulas for programming may be derived by expanding the above for $j = 1, 2, \ldots,$ as follows:

$$A_{11} = U_{11} D_{11} U_{11} = D_{11}$$

$$\begin{aligned} A_{12} &= U_{11} D_{11} U_{12} + U_{21} D_{22} U_{22} \\ &= D_{11} U_{12} \\ A_{22} &= U_{12} D_{11} U_{12} + U_{22} D_{22} U_{22} \\ &= U_{12} D_{11} U_{12} + D_{22} \end{aligned}$$

$$\begin{aligned} A_{13} &= U_{11} D_{11} U_{13} + U_{21} D_{22} U_{23} + U_{31} D_{33} U_{33} \\ &= D_{11} U_{13} \\ A_{23} &= U_{12} D_{11} U_{13} + U_{22} D_{22} U_{23} + U_{32} D_{33} U_{33} \\ &= U_{12} D_{11} U_{13} + D_{22} U_{23} \\ A_{33} &= U_{13} D_{11} U_{13} + U_{23} D_{22} U_{23} + U_{33} D_{33} U_{33} \\ &= U_{13} D_{11} U_{13} + U_{23} D_{22} U_{23} + D_{33} \end{aligned}$$

$$\begin{aligned} A_{14} &= U_{11} D_{11} U_{14} + U_{21} D_{22} U_{24} + U_{31} D_{33} U_{34} + U_{41} D_{44} U_{44} \\ &= D_{11} U_{14} \\ A_{24} &= U_{12} D_{11} U_{14} + U_{22} D_{22} U_{24} + U_{32} D_{33} U_{34} + U_{42} D_{44} U_{44} \\ &= U_{12} D_{11} U_{14} + D_{22} U_{24} \\ A_{34} &= U_{13} D_{11} U_{14} + U_{23} D_{22} U_{24} + U_{33} D_{33} U_{34} + U_{43} D_{44} U_{44} \\ &= U_{13} D_{11} U_{14} + U_{23} D_{22} U_{24} + D_{33} U_{34} \\ A_{44} &= U_{14} D_{11} U_{14} + U_{24} D_{22} U_{24} + U_{34} D_{33} U_{34} + U_{44} D_{44} U_{44} \\ &= U_{14} D_{11} U_{14} + U_{24} D_{22} U_{24} + U_{34} D_{33} U_{34} + D_{44} \end{aligned}$$

And so on. These formulas may be solved for the U_{ij}'s and D_{jj}'s:

$$D_{11} = A_{11}$$

$$U_{12} = \frac{A_{12}}{D_{11}}$$

$$D_{22} = A_{22} - D_{11}U_{12}^2$$

$$U_{13} = \frac{A_{13}}{D_{11}}$$

$$U_{23} = \frac{A_{23} - U_{12}D_{11}U_{13}}{D_{22}}$$

$$D_{33} = A_{33} - D_{11}U_{13}^2 - D_{22}U_{23}^2$$

$$U_{14} = \frac{A_{14}}{D_{11}}$$

$$U_{24} = \frac{A_{24} - U_{12}D_{11}U_{14}}{D_{22}}$$

$$U_{34} = \frac{A_{34} - U_{13}D_{11}U_{14} - U_{23}D_{22}U_{24}}{D_{33}}$$

$$D_{44} = A_{44} - D_{11}U_{14}^2 - D_{22}U_{24}^2 - D_{33}U_{34}^2$$

And so on. For purposes of efficient programming, it is worthwhile to introduce the auxiliary variable

$$L_{ji} \overset{\text{def}}{=} D_{ii}U_{ij}$$

Then, equivalent to the above, we have

$$D_{11} = A_{11}$$

$$L_{21} = A_{12}$$

$$U_{12} = L_{21}/D_{11}$$

$$D_{22} = A_{22} - L_{21}U_{12}$$

$$L_{31} = A_{13}$$

$$L_{32} = A_{23} - U_{12}L_{31}$$

$$U_{13} = L_{31}/D_{11}$$

$$U_{23} = L_{32}/D_{22}$$

$$D_{33} = A_{33} - L_{31}U_{13} - L_{32}U_{23}$$

$$L_{41} = A_{14}$$

$$L_{42} = A_{24} - U_{12}L_{41}$$

$$L_{43} = A_{34} - U_{13}L_{41} - U_{23}L_{42}$$
$$U_{14} = L_{41}/D_{11}$$
$$U_{24} = L_{42}/D_{22}$$
$$U_{34} = L_{43}/D_{33}$$
$$D_{44} = A_{44} - L_{41}U_{14} - L_{42}U_{24} - L_{43}U_{34}$$

And so on. Summarizing, for $j = 1, 2, \ldots, n$

$$L_{ji} = A_{ij} - \sum_{k=1}^{i-1} U_{ki}L_{jk}, \qquad 1 \le i \le j - 1$$
$$U_{ij} = L_{ji}/D_{ii}$$
$$D_{jj} = A_{jj} - \sum_{i=1}^{j-1} L_{ji}U_{ij}$$

Observe that no additional storage besides that needed for A is necessary in the factorization process if A is overwritten by U and D, viz.,

For $i = 2, 3, \ldots, j - 1$

$$A_{ij} \leftarrow A_{ij} - \sum_{k=1}^{i-1} A_{ki}A_{kj}$$

For $i = 1, 2, \ldots, j - 1$

$$T \leftarrow A_{ij}$$
$$A_{ij} \leftarrow T/A_{ii}$$
$$A_{jj} \leftarrow A_{jj} - TA_{ij}$$

This is the procedure coded in subroutine FACTOR in DLEARN. Additionally, FACTOR takes account of the profile storage of the coefficient matrix. The indexing necessary to account for the profile somewhat complicates the procedure but is necessary to achieve optimal efficiency. Figures 11.2.1(a) and 11.2.1(b) are presented as an aid for understanding the indexing and one-dimensional storage of the coefficient matrix in FACTOR. Note that if $A_{ii} = 0$, subroutine FACTOR skips the operations in the second do loop.

Forward Reduction

The indicial form of the forward reduction is

$$\sum_{i=1}^{n} U_{ij}z_i = b_j$$

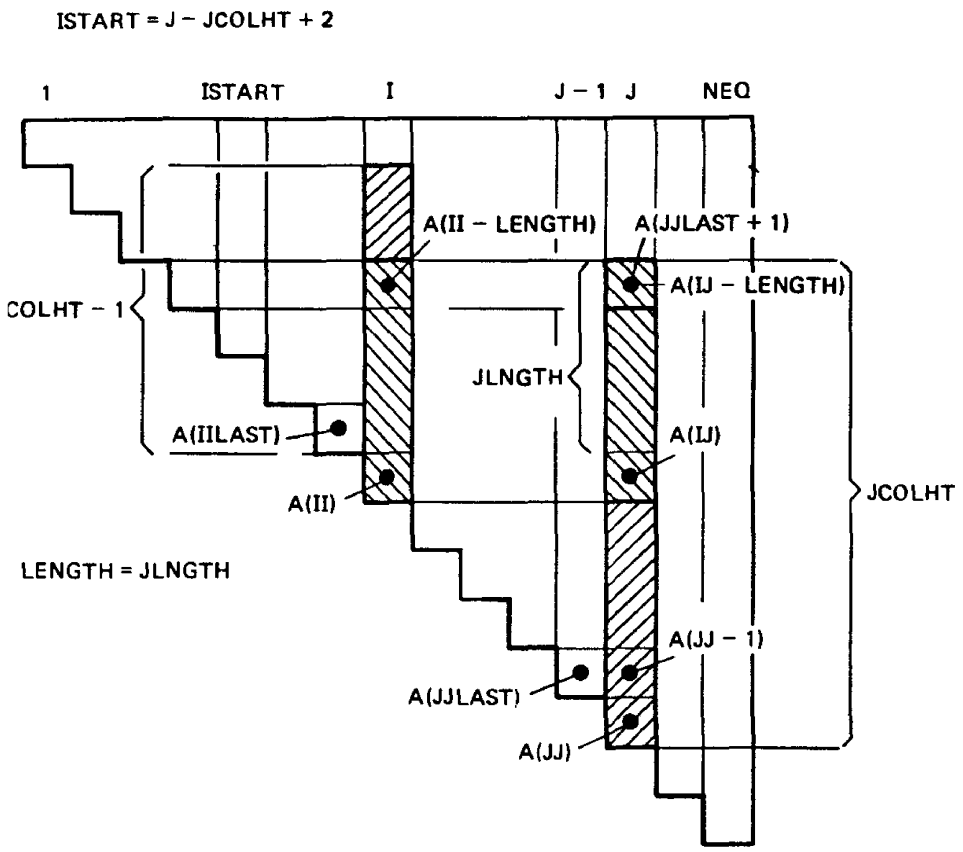


Figure 11.2.1(a) Indexing and storage for subroutine FACTOR; the case ICOLHT − 1 > JLNGTH.

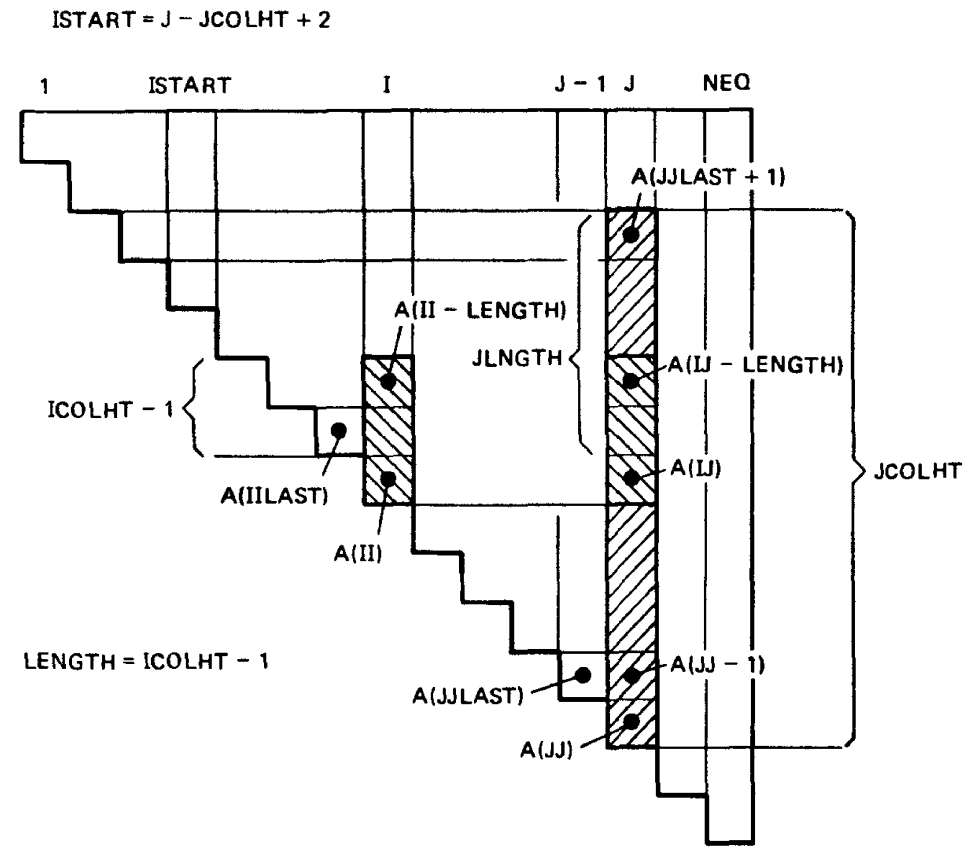


Figure 11.2.1(b) Indexing and storage for subroutine FACTOR; the case ICOLHT − 1 < JLNGTH.

Using the properties of the U_{ij}'s, we can immediately derive

$$z_1 = b_1$$
$$z_2 = b_2 - U_{12}z_1$$
$$z_3 = b_3 - U_{13}z_1 - U_{23}z_2$$
$$\vdots$$

from which we conclude that

$$z_j = b_j - \sum_{i=1}^{j-1} U_{ij}z_i, \qquad (2 \le j \le n)$$

It is clear from this formula that b_j may be overwritten by z_j, viz.,

$$b_j \leftarrow b_j - \sum_{i=1}^{j-1} A_{ij}b_i \qquad (2 \le j \le n)$$

where we have used the fact that A_{ij} has been overwritten by U_{ij}. This procedure is coded in subroutine BACK, which takes account of the profile storage of the coefficient matrix. The indexing is illustrated in Fig. 11.2.2(a).

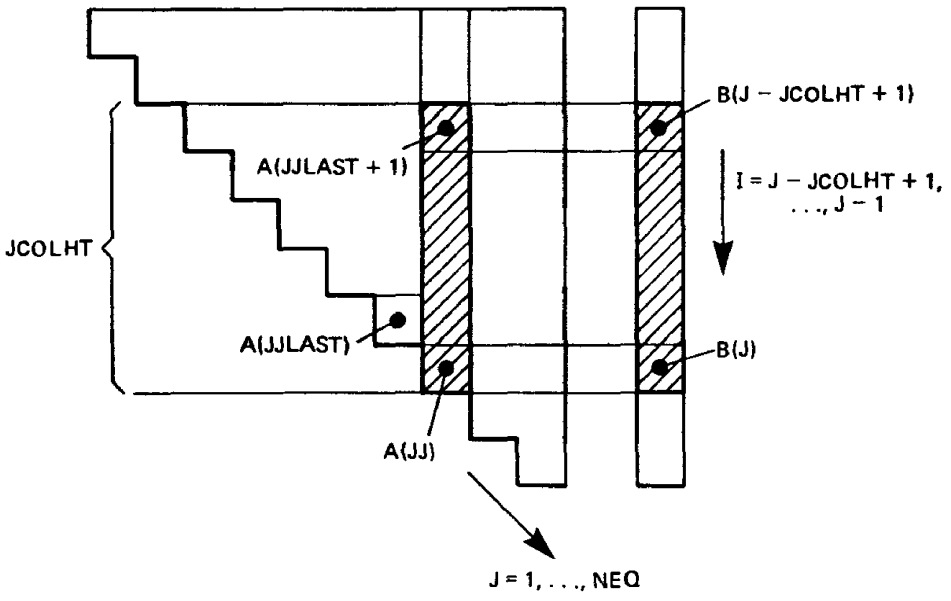


Figure 11.2.2(a) Indexing for forward reduction phase of subroutine BACK.

Diagonal Scaling

Diagonal scaling is also performed in subroutine BACK. Again b_j is overwritten:

$$y_j = \frac{z_j}{D_{jj}} \quad \Leftrightarrow \quad \boxed{b_j \leftarrow b_j / A_{jj} \qquad (1 \le j \le n)}$$

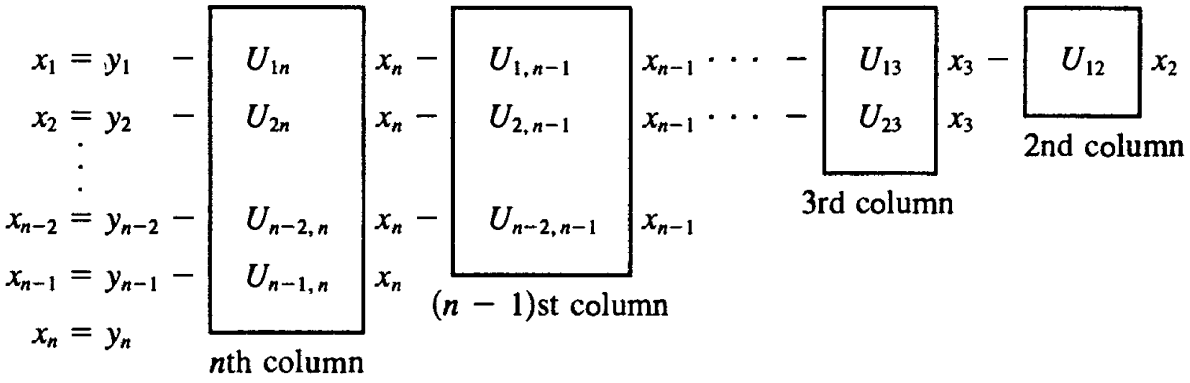
If $A_{jj} = 0$, diagonal scaling is skipped in subroutine BACK.

Back Substitution

The indicial form of the back substitution is

$$\sum_{j=1}^{n} U_{ij} x_j = y_i$$

Expanding this expression, we obtain



There are several ways of organizing the calculations. The most computationally efficient proceeds from left to right, subtracting multiples of the columns of U. Proceeding in this manner allows b to be overwritten with x:

```
For j = n, n − 1, . . . , 2
        For i = 1, 2, . . . , j − 1
        b_i ← b_i − A_ij b_j.
```

The indexing used in subroutine BACK is illustrated in Fig. 11.2.2(b). The number of "flops" (floating-point operations [1]) involved in Crout factorization is approximately $m^2 n/2$, where m is the mean half-bandwidth (defined to be the sum of the column heights divided by the number of equations). In obtaining this result we assume $1 \ll m \ll n$, and thus lower-order terms may be ignored. Given the factorized matrix, a "solution" (which consists of a forward reduction, diagonal scaling, and back

[1] Roughly speaking, one flop consists of a floating-point multiply, an addition, and some subscripting. For example, X = X + Y(I)*Z(J) would amount to one flop.

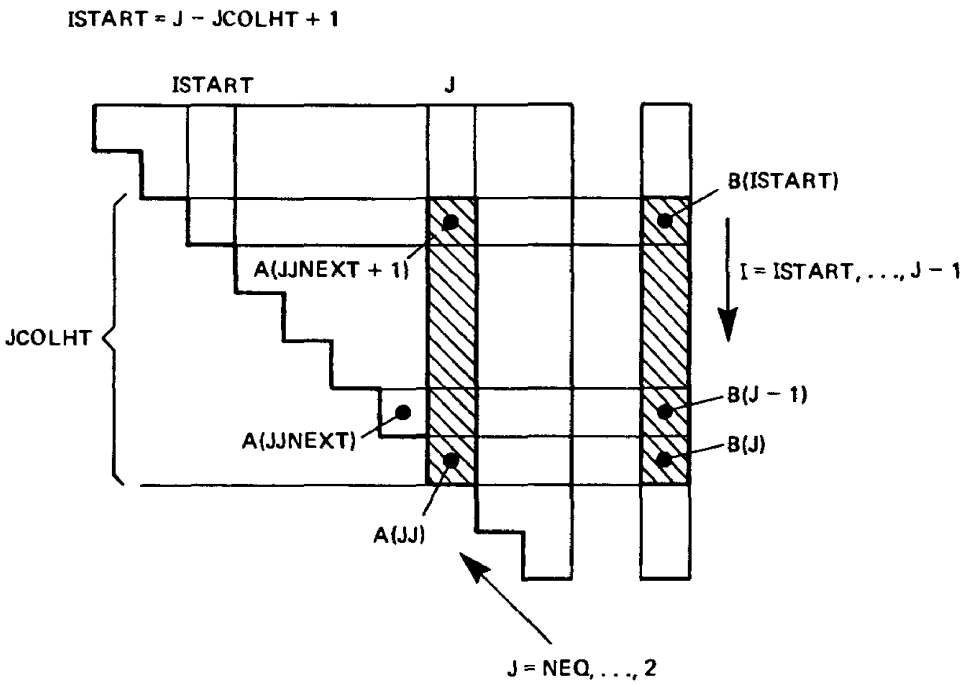


Figure 11.2.2(b) Indexing for back substitution phase of subroutine BACK. (Note: ISTART has a different significance in this routine than in subroutine FACTOR.)

substitution) involves approximately $2mn$ flops. Clearly, for large systems, factorization of the coefficient matrix is much more expensive than solution. Factorization costs vary quadratically with m and thus there is considerable interest in schemes for ordering the equations to minimize m. For additional information on this subject, see [1–4]. An excellent reference for equation solving in general is [5].

Exercise 1. Consider the system

$$\boldsymbol{K}\boldsymbol{d} = \boldsymbol{F}$$

where

$$\boldsymbol{K} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$$\boldsymbol{d} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{Bmatrix} \qquad \boldsymbol{F} = \begin{Bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

i. Obtain the Crout factorization of $\boldsymbol{K}$ (i.e., define the diagonal matrix $\boldsymbol{D}$ and the unit upper triangular matrix $\boldsymbol{U}$).

ii. Obtain the solution $\boldsymbol{d}$ by performing the following steps:

$$\boldsymbol{U}^T\boldsymbol{z} = \boldsymbol{F} \qquad \text{forward reduction}$$

$$\boldsymbol{D}\boldsymbol{y} = \boldsymbol{z} \qquad \text{diagonal scaling}$$

$$\boldsymbol{U}\boldsymbol{d} = \boldsymbol{y} \qquad \text{back substitution}$$

iii. Check parts (i) and (ii) by verifying $\boldsymbol{U}^T\boldsymbol{D}\boldsymbol{U} = \boldsymbol{K}$ and $\boldsymbol{K}\boldsymbol{d} = \boldsymbol{F}$.

Exercise 2. The ***Cholesky decomposition*** of a symmetric positive-definite matrix A takes the form

$$A = G^T G$$

where G is an upper triangular matrix with positive diagonal entries. The Cholesky decomposition may be obtained from the Crout factorization by way of

$$G = D^{1/2} U$$

Develop an algorithm for directly determining G along the lines of the Crout factorization presented above.

11.2.3 Dynamic Storage Allocation

Examples

In the initialization subroutine STATIN, memory storage is allocated for the global arrays required for static analysis. On lines 24 and 25 the "pointers" are determined for the X and ID arrays:

```
MPX = MPOINT('X          ',NSD ,NUMNP,0,IPREC)
MPID = MPOINT('ID        ',NDOF,NUMNP,0,1)
```

FUNCTION MPOINT manages memory allocation within the unnamed or "blank" COMMON A(MTOT). The first argument is a 2A4 string, which serves as the array's label when the storage dictionary is printed. The next three arguments are array dimensions; in this case we see that both ID and X are two-dimensional arrays. The last argument determines the precision of the array (IPREC = 1 or 2 corresponding to single or double precision, respectively). Notice that as an integer array, ID is always defined to be single precision.

The calling statements:

```
CALL COORD(A(MPX),NSD,NUMNP,IPRTIN)
CALL BC(A(MPID),NDOF,NUMNP,NEQ,IPRTIN)
```

and the corresponding subroutine statements:

```
SUBROUTINE COORD(X,NSD,NUMNP, IPRTIN)
SUBROUTINE BC(ID,NDOF,NUMNP,NEQ,IPRTIN)
```

mean *array* X *begins at address* MPX *of blank common and array* ID *begins at address* MPID *of blank common,* respectively.

In COORD and BC, the arrays X and ID are treated as *two-dimensonal,* as evidenced by the DIMENSION statements:

```
DIMENSION X(NSD,1)
```

and

```
DIMENSION ID(NDOF,1)
```

respectively. It does not matter that X and ID correspond to portions of a one-dimensional array in the calling program STATIN. The significance of the dimensioning is as follows: *The arrays* X *and* ID *are stored in blank common in consecutively addressed cells by columns*. Consequently the addresses of X(M,N) and ID(M,N) in A are given by

$$\text{IPREC}*(\text{NSD}*(\text{N} - 1)+\text{M} - 1) + \text{MPX}$$

and

$$\text{NDOF}*(\text{N} - 1)+\text{M} - 1 + \text{MPID}$$

Thus the only information necessary to determine the appropriate addresses in A are the starting addresses (i.e., MPX and MPID, respectively), which are provided by the CALL statements, and the number of rows (i.e., NSD and NDOF, respectively), which are provided by the DIMENSION statements in the subroutines. This explains why it is not necessary to define the number of columns of two-dimensional arrays "passed" to subroutines. The 1's in the column slots could be replaced by any positive integer; they only serve notice that these arrays are to be regarded as two-dimensional in the subroutine—the integer is never used. For three-dimensional arrays, the first two dimensions must be precisely defined, whereas the third may be set to 1, etc.

The array names, pointers, and other dictionary entries are stored at the end of blank common. Thus for the first two pointers declared during execution:

$$\text{A(MTOT} - 4) = \text{MPSTEP} = 1$$

$$\text{A(MTOT} - 11) = \text{MPDPRT}, \ldots .$$

Note that seven words of storage are required for each entry in the dictionary. The memory FUNCTION MPOINT monitors the blank common allocation to ensure that data arrays starting from the beginning of blank common do not overwrite the array pointer information. If the user receives a memory error, he or she must reduce the size of the problem or increase MTOT. Given the availability of sufficient machine memory size, *the latter action requires only recompiling the main program (which amounts to only a few statements), followed by linking to create a new executable file.*

The following tables present the storage arrays and associated pointers for all global data used during static or dynamic analysis. Also, an example of local element data storage requirements is presented for the four-node quadrilateral, elastic continuum element QUADC.

TABLE 11.2.1 Storage in Blank Common—Time Sequence and Time History Data*

Array Pointer	Array
MPSTEP	NSTEP(NUMSEQ)
MPDPRT	NDPRT(NUMSEQ)
MPSPRT	NSPRT(NUMSEQ)
MPHPLT	NHPLT(NUMSEQ)
MPITER	NITER(NUMSEQ)
MPALPH	ALPHA(NUMSEQ)
MPBETA	BETA(NUMSEQ)

* Required for static and dynamic analysis, storage allocated in SUBROUTINE TSEQ.

TABLE 11.2.1 (cont.) Storage in Blank Common—Time Sequence and Time History Data

Array Pointer	Array
MPGAMM	GAMMA(NUMSEQ)
MPDT	DT(NUMSEQ)
MPIDHS	IDHIST(3,NDOUT)†
MPDOUT	DOUT(NDOUT+1,NPLPTS)†

where, for each time sequence I=1,NUMSEQ:

NSTEP(I)	=	number of time steps
NDPRT(I)	=	number of time steps between printing of kinematic output
NSPRT(I)	=	number of time steps between printing of element stress/strain output
NHPLT(I)	=	number of time steps between plotting of output time-history data
NITER(I)	=	number of iterations in corrector loop
ALPHA(I)	=	α coefficient for time-integration algorithm
BETA(I)	=	β coefficient for time-integration algorithm
GAMMA(I)	=	γ coefficient for time-integration algorithm
DT(I)	=	time step Δt

End time sequence data.

IDHIST(1,N)	=	node number for output history N
IDHIST(2,N)	=	degree of freedom number for output history N
IDHIST(3,N)	=	kinematic-variable-type number for output history N
DOUT(1,J)	=	value of time at plot step J (stored as single precision for IPREC = 1 or 2)
DOUT(N+1,J)	=	output history N at plot step J (stored as single precision for IPREC = 1 or 2)
NDOUT	=	number of nodal output time-histories
NPLPTS	=	number of points in output time-history plots
IPREC	=	precision flag: 1 ⇒ single, 2 ⇒ double

†If NDOUT=0, no storage is allocated for IDHIST and DOUT.

TABLE 11.2.2 Storage in Blank Common—Dynamic Analysis Data*

Array Pointer	Array
MPVPRD	VPRED(NDOF,NUMNP)
MPDPRD	DPRED(NDOF,NUMNP)
MPA	A(NDOF,NUMNP)
MPV	V(NDOF,NUMNP)

where:

VPRED(I,J)	=	Ith predictor velocity component of node J
DPRED(I,J)	=	Ith predictor displacement component of node J
A(I,J)	=	Ith acceleration component of node J
V(I,J)	=	Ith velocity component of node J
NDOF	=	number of degrees of freedom per node
NUMNP	=	number of nodal points

*Required only for dynamic analysis; storage allocated in SUBROUTINE DYNPTS.

TABLE 11.2.3 Storage in Blank Common—Static Analysis Data*

Array Pointer	Array
MPD	D(NDOF,NUMNP)
MPX	X(NSD,NUMNP)
MPID	ID(NDOF,NUMNP)
MPF	F(NDOF,NUMNP,NLVECT)†
MPG	G(NPTSLF,2,NLTFTN)‡
MPG1	G1(NLTFTN)‡
MPDIAG	IDIAG(NEQ)
MPNGRP	NGRP(NUMEG)

where:

D(I,J)	=	Ith displacement component of node J
X(I,J)	=	Ith coordinate of node J
ID(I,J)	=	equation number for degree of freedom I, node J
F(I,J,L)	=	Ith prescribed force/kinematic-variable§ component for node J, load vector L
G(K,1,L)	=	time t_k for load-time function L
G(K,2,L)	=	function value at time t_k, load-time function L
G1(L)	=	value of load-time function L at current time t
IDIAG(N)	=	location of Nth diagonal element in coefficient array ALHS
NGRP(N)	=	pointer to beginning address of element group N
NDOF	=	number of degrees of freedom per node
NUMNP	=	number of nodal points
NSD	=	number of space dimensions
NLVECT	=	number of load vectors
NPTSLF	=	number of points in a load-time function
NLTFTN	=	number of load-time functions
NEQ	=	number of equations
NUMEG	=	number of element groups

*Required for static and dynamic analysis; storage allocated in SUBROUTINE STATIN.

†If NLVECT = 0, no storage is allocated for F.

‡If NLTFTN = 0, no storage is allocated for G and G1.

§Displacements for static analysis; displacements, velocities or accelerations for dynamic analysis.

TABLE 11.2.4 Storage Requirements for Four-Node Quadrilateral, Elastic Continuum Element (NTYPE = 1)

Each element group has an associated array NPAR with pointer MPNPAR; group control data can be stored in positions 1 through 16. This is followed immediately in blank common by the MP array, which stores the memory pointers for the element group data. This allows each element type to have a unique data structure. For the present element, MP contains 35 words and stores the following pointers:

Array Pointer	Array
MP(MW)	W(NINT)
MP(MDET)	DET(NINT)
MP(MR)	R(NINT)
MP(MSHL)	SHL(NROWSH,NEN,NINT)

TABLE 11.2.4 (cont.) Storage Requirements for Four-Node Quadrilateral, Elastic Continuum Element (NTYPE = 1)

Array Pointer	Array
MP(MSHG)	SHG(NROWSH,NEN,NINT)
MP(MSHGBR)	SHGBAR(NROWSH,NEN,NRINT)
MP(MRHO)	RHO(NUMAT)
MP(MRDPM)	RDAMPM(NUMAT)
MP(MRDPK)	RDAMPK(NUMAT)
MP(MTH)	TH(NUMAT)
MP(MC)	C(NROWB,NROWB,NUMAT)
MP(MGRAV)	GRAV(NESD)
MP(MIEN)	IEN(NEN,NUMEL)
MP(MMAT)	MAT(NUMEL)
MP(MLM)	LM(NED,NEN,NUMEL)
MP(MIELNO)	IELNO(NSURF)
MP(MISIDE)	ISIDE(NSURF)
MP(MPRESS)	PRESS(2,NSURF)
MP(MSHEAR)	SHEAR(2,NSURF)
MP(MISHST)	ISHIST(3,NSOUT)*
MP(MSOUT)	SOUT(NSOUT+1,NPLPTS)*
MP(MELEFM)	ELEFFM(NEE,NEE)
MP(MXL)	XL(NESD,NEN)
MP(MWORK)	WORK(16)
MP(MB)	B(NROWB,NEE)
MP(MDMAT)	DMAT(NROWB,NROWB)
MP(MDB)	DB(NROWB,NEE)
MP(MVL)	VL(NED,NEN)
MP(MAL)	AL(NED,NEN)
MP(MELRES)	ELRESF(NEE)
MP(MDL)	DL(NED,NEN)
MP(MSTRN)	STRAIN(NROWB)
MP(MSTRS)	STRESS(NROWB)
MP(MPSTRN)	PSTRN(NROWB)
MP(MPSTRS)	PSTRS(NROWB)

where, for each quadrature point L=1,NINT:

W(L)	=	integration-rule weight
DET(L)	=	Jacobian determinant
R(L)	=	radius at quadrature point (axisymmetric option)
SHL(1,J,L)	=	local derivative $N_{j,\xi}$
SHL(2,J,L)	=	local derivative $N_{j,\eta}$
SHL(3,J,L)	=	N_j
SHG(1,J,L)	=	global derivative N_{j,x_1}
SHG(2,J,L)	=	global derivative N_{j,x_2}
SHG(3,J,L)	=	N_j

End quadrature-point data.

*If NSOUT = 0, no storage is allocated for ISHIST and SOUT.

TABLE 11.2.4 (cont.) Storage Requirements for Four-Node Quadrilateral, Elastic Continuum Element (NTYPE = 1)

For mean-dilatational $\bar{B}$ formulation:

SHGBAR(1,J) = $\int_{\Omega^e} N_{j,x_1}\, d\Omega \Big/ \int_{\Omega^e} d\Omega$

SHGBAR(2,J) = $\int_{\Omega^e} N_{j,x_2}\, d\Omega \Big/ \int_{\Omega^e} d\Omega$

SHGBAR(3,J) = $\int_{\Omega^e} N_j\, d\Omega \Big/ \int_{\Omega^e} d\Omega$

For each material set M=1,NUMAT:

RHO(M)	=	mass density
RDAMPM(M)	=	Rayleigh mass-proportional damping coefficient
RDAMPK(M)	=	Rayleigh stiffness-proportional damping coefficient
TH(M)	=	thickness
C(I,J,M)	=	I,Jth component of material moduli matrix

End material data.

GRAV(I)	=	gravity-load multiplier in x_i direction
IEN(J,NEL)	=	global node number for local node J of element NEL
MAT(NEL)	=	material set number for element NEL
LM(I,J,NEL)	=	global equation number for Ith degree of freedom, Jth local node of element NEL
IELNO(N)	=	element number of Nth applied surface force
ISIDE(N)	=	element-side number of Nth applied surface force
PRESS(1,N)	=	pressure at local node I for Nth applied surface force†
PRESS(2,N)	=	pressure at local node J for Nth applied surface force†
SHEAR(1,N)	=	shear at local node I for Nth applied surface force†
SHEAR(2,N)	=	shear at local node J for Nth applied surface force†
ISHIST(1,N)	=	element number for output history N
ISHIST(2,N)	=	quadrature-point number for output history N
ISHIST(3,N)	=	stress/strain-component number for output history N
SOUT(1,J)	=	time at plot step J (stored as single precision for IPREC = 1 or 2)
SOUT(N+1,J)	=	output history N at plot step J (stored as single precision for IPREC = 1 or 2)
ELEFFM(I,J)	=	element effective mass matrix
XL(I,J)	=	Ith coordinate component of Jth element node
WORK(I)	=	work space used for temporary calculations
B(I,J)	=	strain-displacement matrix
DMAT(I,J)	=	scalar multiple of element material moduli matrix
DB(I,J)	=	matrix product of DMAT and B
VL(I,J)	=	Ith velocity component of Jth element node
AL(I,J)	=	Ith acceleration component of Jth element node
ELRESF(I)	=	Ith entry of element residual force vector
DL(I,J)	=	Ith displacement component of Jth element node
STRAIN(I)	=	strains
STRESS(I)	=	stresses
PSTRN(I)	=	principal strains

†For the four-node quadrilateral, $1 \le I \le 4$; if $I \le 3$, $J = I + 1$; if $I = 4$, $J = 1$.

TABLE 11.2.4 (cont.) Storage Requirements for Four-Node Quadrilateral, Elastic Continuum Element (NTYPE = 1)

PSTRS(I)	=	principal stresses
NINT	=	number of integration points
NUMAT	=	number of materials in this group
NROWSH	=	number of rows in shape-function arrays (=3)
NEN	=	number of element nodes (=4)
NRINT	=	number of reduced-integration points; for mean-dilatational $\bar{B}$ formulation = 1
NROWB‡	=	number of rows in strain-displacement matrix (=4)
NUMEL	=	number of elements in this group
NSURF	=	number of element surface loads in this group
NSOUT	=	number of stress/strain time-histories in this group
NPLPTS	=	number of points in output time-histories.
NEE	=	number of element equations (=8)
NED	=	number of element degrees of freedom per node (=2)

‡ Note that while the strain-displacement matrix B has a constant row dimension of 4, the parameter NSTR specified in SUBROUTINE QUADC denotes the number of nonzero strain-energy components in the element. For plane stress (IOPT = 0), and plane strain (IOPT = 1) without mean-dilatational $\bar{B}$ formulation (IBBAR = 0), NSTR equals 3. However, for plane strain with $\bar{B}$ (IBBAR = 1), and torsionless axisymmetry (IOPT = 2) with or without $\bar{B}$, NSTR equals 4.

TABLE 11.2.5 Storage in Blank Common—Equation System Data*

Array Pointer	Array
MPALHS	ALHS(NALHS)
MPBRHS	BRHS(NEQ)

where:

ALHS(N)	=	Nth term of coefficient matrix stored in compacted column form
BRHS(N)	=	Nth active residual (dynamic analysis) or force (static analysis) component†
NALHS	=	number of terms in ALHS
NEQ	=	number of equations

* Required for static and dynamic analysis; storage allocated in SUBROUTINE EQSET.

† Upon back substitution BRHS contains acceleration increments (dynamic analysis) or displacements (static analysis) for active degrees of freedom.

1.3 PROGRAM STRUCTURE

In this section we present an organizational description of DLEARN, including algorithmic equations when they are deemed illustrative. All FORTRAN subroutines appear in boldface and variable names are capitalized. In the example element loops, the term "generic subroutine" identifies a fundamental program unit which could be accessed by any element type, e.g., **COLHT**, or by a class of elements, e.g., **PROP2D** for two-dimensional, elastic continuum elements.

11.3.1 Global Control

G1. MAIN PROGRAM

1. Set blank common capacity MTOT, and initialize addresses of first and last available words.
2. Specify numerical precision IPREC.
3. Specify input/output unit numbers and open files.
4. Call global driver **DLEARN.**
5. Close files and exit program.

G2. DLEARN: Global driver

1. **ECHO:** IF (IECHO.EQ.1) write copy of input data (uninterpreted) to output file.
2. Read TITLE: IF (TITLE(1).EQ.'*END') RETURN to G1.5.
3. Read execution control card. If static analysis LDYN = .FALSE., otherwise .TRUE. .
4. Call initialization drivers, GO TO I1.
5. IF (IEXEC.EQ.1) **DRIVER:** solution driver, GO TO S1.
6. IF (IWRITR.EQ.1) **RSOUT:** write restart file.
7. **PRTDC:** print memory pointer dictionary.
8. GO TO 2 (begin new problem).

11.3.2 Initialization Phase

I1. TSEQ: driver to input time sequence control data.

1. Set memory pointers for time sequence data.
2. **TSEQIN:** read time sequence data and compute storage for time histories. Set defaults for static analysis.
3. Set memory pointers for nodal time history data arrays.

I2. IF (LDYN) **DYNPTS:** set pointers for dynamic analysis data arrays.

I3. IF (NDOUT.GT.0) **DHIST:** read nodal output time-history data, store in IDHIST(3,NDOUT).

I4. STATIN: driver to set pointers for static analysis data arrays and read geometric and boundary condition input data. IF (.NOT.LDYN) MPDPRD = MPD, then element task 'FORM_RHS' accesses proper displacement array during static analysis. (cf. S10.1.a).

1. **COORD:** read/generate nodal coordinates, store in X(NSD,NUMNP).

2. **BC:** read/generate nodal boundary condition codes into ID(NDOF,NUMNP), then overwrite array with equation numbers.
3. IF (NLVECT.GT.0) **INPUT:** read/generate prescribed nodal force/ kinematic boundary condition data and store in F(NDOF, NUMNP, NLVECT).
4. IF (NLTFTN.GT.0) **LTIMEF:** read load-time functions, store in G(NPTSLF,2,NLTFTN).

I5. IF (LDYN): input kinematic initial conditions.

1. IF (IREADR .EQ. 1) THEN
   RSIN: read initial conditions from restart file.
   ELSE
   INPUT: read initial conditions from input file.
   ENDIF

I6. IF (LDYN .AND. NDOUT.GT.0) **STORED:** store kinematic initial conditions for time histories in DOUT(NDOUT+1,NPLPTS).

I7. **ELEMNT('INPUT__'):** loop over element groups NEG=1,NUMEG and read/generate element definition data; element task 1.

Example

QUADC: four-node quadrilateral, elastic continuum element.

1. Set memory pointers.
2. **QDCT1:** element task 1.
   a. **QDCSHL:** compute bilinear shape functions N_a and their local derivatives $N_{a,\xi}$, $N_{a,\eta}$ at integration points, store in SHL(NROWSH,NEN,NINT); generic subroutine.
   b. **PROP2D:** read material property data for two-dimensional elastic continuum. Compute material moduli matrix, store in C(NROWB,NROWB, NUMAT); generic subroutine.
   c. Read body-force vector GRAV(NESD).
   d. **GENEL:** read/generate element data; store element node numbers in IEN(NEN,NUMEL) and material set numbers in MAT(NUMEL).
   e. **FORMLM:** form assembly mapping LM(NED,NEN,NUMEL); generic subroutine.
   f. If element group contributes to global left-hand-side matrix, then **COLHT:** determine column heights of each element; generic subroutine.
   g. IF (NSURF.GT.0) **QDCRSF:** read surface force data.
   h. IF (NSOUT.GT.0) **SHIST:** read stress/strain output time-history data, store in ISHIST(3,NSOUT); generic subroutine.

I8. EQSET: complete specification of data structure for global equation system.

1. **DIAG:** determine addresses of diagonal entries of global left-hand-side matrix ALHS, store in IDIAG(NEQ). Set NALHS equal to number of terms in ALHS.
2. Allocate memory for ALHS(NALHS) and global right-hand-side vector BRHS(NEQ).

I9. Resume global driver **DLEARN;** GO TO G2.5.

11.3.3 Solution Phase

Time sequence loop (DO S19 NSQ = 1,NUMSEQ)

S1. TIMCON: set current time sequence parameters: NSTEP1, NDPRT1, NSPRT1, NHPLT1, NITER1, ALPHA1, BETA1, GAMMA1, DT1 and compute coefficients for α-method transient algorithm:

$$\text{COEFF1} = (1 + \alpha),$$

$$\text{COEFF2} = \gamma\Delta t,$$

$$\text{COEFF3} = \beta\Delta t^2$$

$$\text{COEFF4} = (1 + \alpha)\gamma\Delta t,$$

$$\text{COEFF5} = (1 + \alpha)\beta\Delta t^2,$$

$$\text{COEFF6} = (1 + \alpha)\Delta t,$$

$$\text{COEFF7} = \frac{1}{2}(1 + \alpha)(1 - 2\beta)\Delta t^2,$$

$$\text{COEFF8} = (1 + \alpha)(1 - \gamma)\Delta t.$$

S2. ELEMNT('FORM_LHS'): loop over element groups NEG=1,NUMEG and form effective mass matrix $\boldsymbol{M}^*$; element task 2.

$$\boldsymbol{M}^* = \boldsymbol{M}^E + \boldsymbol{M}^I + (1 + \alpha)\gamma\Delta t\boldsymbol{C}^I + (1 + \alpha)\beta\Delta t^2\boldsymbol{K}^I \qquad \text{(dynamic analysis)}$$

$$\boldsymbol{M}^* = \boldsymbol{K} \qquad \text{(static analysis)}$$

where superscripts E and I denote explicit and implicit element group contributions, respectively.

Example

QUADC: four-node quadrilateral, elastic continuum element.

1. **QDCT2:** element task 2.

DO h NEL=1,NUMEL

a. **CLEAR:** zero element effective mass array ELEFFM(NEE,NEE).

b. **LOCAL:** localize nodal coordinates X, store in XL(NESD,NEN); generic subroutine.

c. **QDCSHG:** compute shape function global derivatives $N_{a,x}$, $N_{a,y}$ at integration points, store in SHG(NROWSH,NEN,NINT); generic subroutine.

d. IF (IOPT.EQ.2), compute radial coordinates of integration points for torsionless axisymmetry option, store in R(NINT). Include radial weighting with Jacobian determinants DET(NINT).

e. IF (LDYN . AND. IMASS.NE.2) THEN

CONTM: form consistent or lumped mass matrix for continuum element and store in ELEFFM(NEE,NEE); generic subroutine.

$$\boldsymbol{m}^* = \boldsymbol{m}^* + [1 + \text{RDAMPM(MAT(NEL))}*(1 + \alpha)\gamma\Delta t]\boldsymbol{m}^e$$

ENDIF

f. IF ((NOT.LDYN) .OR. IMPEXP.EQ.0) THEN

QDCK: form stiffness matrix $\boldsymbol{k}^e = \int \boldsymbol{B}^T \boldsymbol{D}\boldsymbol{B}\, d\Omega$ and sum to ELEFFM(NEE,NEE).

$$\boldsymbol{m}^* = \boldsymbol{m}^* + [\text{RDAMPK(MAT(NEL))}*(1 + \alpha)\gamma\Delta t + (1 + \alpha)\beta\Delta t^2]\boldsymbol{k}^e$$

ENDIF

g. **ADDLHS:** assemble ELEFFM into ALHS; generic subroutine.

h. CONTINUE to next element.

S3. FACTOR: perform Crout factorization of ALHS.

$$\boldsymbol{M}^* = \boldsymbol{U}^T\boldsymbol{D}\boldsymbol{U}$$

S4. Perform rank check if requested.

1. IF (IRANK.EQ.1) **PIVOTS:** print numbers of negative and zero pivots; RETURN to S19.
2. IF (IRANK.EQ.2) **PRINTP:** print all pivots; RETURN to S19.

Time-step loop (DO S18 N=1,NSTEP1)

S5. IF (LDYN) THEN

PREDCT: predictor update of all degrees of freedom.

$$\tilde{\boldsymbol{d}}^{(1)} = \boldsymbol{d}_n + (1 + \alpha)\Delta t\boldsymbol{v}_n + \frac{1}{2}(1 + \alpha)(1 - 2\beta)\Delta t^2\boldsymbol{a}_n \qquad \text{DPRED} = \text{D} + \text{COEFF6} * \text{V} + \text{COEFF7} * \text{A}$$

$\tilde{v}^{(1)} = v_n + (1 + \alpha)(1 - \gamma)\Delta t a_n$ VPRED = V + COEFF8 * A

$\tilde{a}^{(1)} = \mathbf{0}$ A = ZERO

ELSE

CLEAR: zero displacement array D.

ENDIF

S6. IF (NLVECT.GT.0) **LFAC:** evaluate load-time functions at time t_{n+1}, store values in G1(NLTFTN).

S7. IF (DT1.NE.ZERO) **COMPBC:** overwrite predictors to account for kinematic boundary conditions.

1. If d_{n+1} given:

IF (LDYN) THEN

$d^* = (1 + \alpha)d_{n+1} - \alpha d_n$ TEMP = COEFF1 * VAL − ALPHA1 * D

$\tilde{a}^{(1)} = \dfrac{d^* - \tilde{d}^{(1)}}{(1 + \alpha)\beta\Delta t^2}$ A = (TEMP − DPRED)/COEFF5

$\tilde{d}^{(1)} = d^*$ DPRED = TEMP

$\tilde{v}^{(1)} = \tilde{v}^{(1)} + (1 + \alpha)\gamma\Delta t\tilde{a}^{(1)}$ VPRED = VPRED + COEFF4 * A

ELSE

$\tilde{d} = d_{n+1}$ DPRED = VAL

ENDIF

2. If v_{n+1} given:

$v^* = (1 + \alpha)v_{n+1} - \alpha v_n$ TEMP = COEFF1 * VAL − ALPHA1 * V

$\tilde{a}^{(1)} = \dfrac{v^* - \tilde{v}^{(1)}}{(1 + \alpha)\gamma\Delta t}$ A = (TEMP − VPRED)/COEFF4

$\tilde{v}^{(1)} = v^*$ VPRED = TEMP

$\tilde{d}^{(1)} = \tilde{d}^{(1)} + (1 + \alpha)\beta\Delta t^2\tilde{a}^{(1)}$ DPRED = DPRED + COEFF5 * A

3. If a_{n+1} given:

$\tilde{d}^{(1)} = \tilde{d}^{(1)} + (1 + \alpha)\beta\Delta t^2 a_{n+1}$ DPRED = DPRED + COEFF5 * VAL

$\tilde{v}^{(1)} = \tilde{v}^{(1)} + (1 + \alpha)\gamma\Delta t a_{n+1}$ VPRED = VPRED + COEFF4 * VAL

$\tilde{a}^{(1)} = a_{n+1}$ A = VAL

Multicorrector iteration loop (DO S13 I=1,NITER1)

S8. IF (NLTFTN.GT.0) **LFAC:** evaluate load-time functions at time $t_{n+1+\alpha}$, store values in G1(NLTFTN).

S9. IF (NLVECT.GT.0) **LOAD:** form nodal contribution to residual force, $F_{n+1+\alpha}$, store in BRHS(NEQ).

S10. **ELEMNT('FORM_RHS'):** loop over element groups NEG=1,NUMEG and form element contributions to residual force R; element task 3.

$$R = \underbrace{F_{n+1+\alpha}}_{\text{Nodal contribution}} \quad \underbrace{-M\tilde{a}^{(i)} - C\tilde{v}^{(i)} - K\tilde{d}^{(i)} + \mathop{\mathsf{A}}_{e=1}^{n_{el}} f^e_{n+1+\alpha}}_{\text{Element contribution}} \qquad \text{(dynamic analysis)}$$

$$R = \underbrace{F}_{\text{Nodal contribution}} + \underbrace{\mathop{\mathsf{A}}_{e=1}^{n_{el}} f^e}_{\text{Element contribution}} \qquad \text{(static analysis)}$$

Example

QUADC: four-node quadrilateral, elastic continuum element.

1. **QDCT3:** element task 3.

DO h NEL=1,NUMEL

a. **LOCAL:** localize displacements DPRED, store in DL(NED,NEN); generic subroutine. Note: *For static analysis* MPDPRD = MPD, *so reference to* DPRED *accesses the array* D.

b. IF (LDYN) THEN

   i. **LOCAL:** localize VPRED and A, store in VL(NED,NEN) and AL(NED,NEN), respectively; generic subroutine.

   ii. Compute effective displacement and acceleration accounting for Rayleigh damping:

$$\tilde{d}^e = \tilde{d}^e + \text{RDAMPK} * \tilde{v}^e,$$

$$\tilde{a}^e = \tilde{a}^e + \text{RDAMPM} * \tilde{v}^e.$$

   ENDIF

c. Determine if element makes inertial (FORMMA = .TRUE.) and/or stiffness (FORMKD = .TRUE.) contribution to right-hand-side vector.

d. IF (FORMMA .OR.FORMKD) THEN

   i. **LOCAL:** localize nodal coordinates X, store in XL(NESD,NEN); generic subroutine.

   ii. **QDCSHG:** compute shape function global derivatives $N_{a,x}$, $N_{a,y}$ at integration points, store in SHG(NROWSH,NEN,NINT); generic subroutine.

iii. IF (IOPT.EQ.2) compute radial coordinates of integration points for torsionless axisymmetry option, store in R(NINT). Include radial weighting with Jacobian determinants DET(NINT).

ENDIF

e. IF (FORMMA) **CONTMA:** compute $-m^e(\tilde{a}^e - g^e)$ for continuum element and store in ELRESF(NEE); generic subroutine.

f. IF (FORMKD) **QDCKD:** compute internal force $-k^e\tilde{d}^e = -\int B^T\sigma\, d\Omega$, and add to ELRESF(NEE).

g. **ADDRHS:** assemble ELRESF into BLHS; generic subroutine.

h. CONTINUE to next element.

i. IF (NSURF.GT.0) **QDCSUF:** compute element surface forces and assemble into BRHS.

S11. BACK: perform forward reduction, diagonal scaling, and back substitution to find $\Delta a^{(i)}$ (dynamic analysis) or $d^{(1)}$ (static analysis).

S12. ITERUP: intermediate update of active degrees of freedom.

IF (LDYN) THEN

$\tilde{d}^{(i+1)} = \tilde{d}^{(i)} + (1+\alpha)\beta\Delta t^2\Delta a^{(i)}$	DPRED = DPRED + COEFF5 * BRHS
$\tilde{v}^{(i+1)} = \tilde{v}^{(i)} + (1+\alpha)\gamma\Delta t\Delta a^{(i)}$	VPRED = VPRED + COEFF4 * BRHS
$\tilde{a}^{(i+1)} = \tilde{a}^{(i)} + \Delta a^{(i)}$	A = A + BRHS

ELSE

$d = d^{(1)}$	D = BRHS

ENDIF

S13. CONTINUE to next iteration (GO TO S8).

S14. IF (LDYN) **CORRCT:** corrector update of all degrees of freedom.

$d_{n+1} = \dfrac{\tilde{d}^{(i+1)} - d_n}{1+\alpha} + d_n,$	D = (DPRED − D)/COEFF1 + D
$v_{n+1} = \dfrac{\tilde{v}^{(i+1)} - v_n}{1+\alpha} + v_n$	V = (VPRED − V)/COEFF1 + V
$a_{n+1} = \tilde{a}^{(i+1)}$	

S15. IF (LOUT(N,NDPRT1)) **PRINTD:** write kinematic output.

S16. IF (LOUT(N,NSPRT1)) **ELEMNT('STR_PRNT'):** loop over element groups NEG=1,NUMEG and perform stress recovery for printing; element task 4.

Example

QUADC: four-node quadrilateral, elastic continuum element.

1. **QDCT4:** element task 4.

DO h NEL=1,NUMEL

a. **LOCAL:** localize coordinates X and displacements D, store in XL(NESD,NEN) and DL(NED,NEN), respectively; generic subroutine.
b. **QDCSHG:** compute shape function global derivatives $N_{a,x}$, $N_{a,y}$ at integration points, store in SHG(NROWSH,NEN,NINT); generic subroutine.
c. Compute coordinates of integration points, store in XINT(NESD,NINT). IF (IOPT.EQ.2) set R(L) = XINT(1,L) and include radial weighting in Jacobian determinants DET(NINT).
d. IF (IBBAR.EQ.1) **MEANSH:** compute volume average of global shape function derivatives for mean-dilatational $\bar{B}$; IF (IOPT.EQ.2), compute volume average of shape functions; store in SHGBR(NROWSH,NEN).

DO g L=1,NINT

e. **QDCSTR:** compute stress and strain for two-dimensional continuum element, store in STRESS(NROWB) and STRAIN(NROWB), respectively.
f. **PRTS2D:** print stresses and strains for two-dimensional element; generic subroutine.
g. CONTINUE to next quadrature point.

h. CONTINUE to next element.

S17. IF (LDYN .AND. **LOUT**(N,NHPLT1)) THEN

1. **STORED:** store kinematic time history data in DOUT(NDOUT+1,NPLPTS).
2. **ELEMNT**('STR_STOR'): loop over element groups NEG=1,NUMEG and perform stress recovery for element time-histories; element task 5.

Example

QUADC: four-node quadrilateral, elastic continuum element.

1. **QDCT5:** element task 5.

a. Store current time value in SOUT(1,LOCPLT).

DO h I=1,NSOUT

b. For active element **LOCAL:** localize coordinates X and displacements D, store in XL(NESD,NEN) and DL(NED,NEN), respectively; generic subroutine.
c. **QDCSHG:** Compute shape function global derivatives $N_{a,x}$, $N_{a,y}$ at integration points, store in SHG(NROWSH,NEN,NINT); generic subroutine.
d. IF (IOPT.EQ.2) compute radial coordinates of integration points, store in R(NINT). Include radial weighting in Jacobian determinants DET(NINT).
e. IF (IBBAR.EQ.1) **MEANSH:** compute volume average of global shape function derivatives for mean-dilatational $\bar{B}$; IF (IOPT.EQ.2), compute volume average of shape functions; store in SHGBR(NROWSH,NEN).
f. For active integration point **QDCSTR:** compute stress and strain for two-dimensional continuum element, store in STRESS(NROWB) and STRAIN(NROWB), respectively.
g. Store required stress/strain component in SOUT(I+1,LOCPLT).
h. CONTINUE to next active element/quadrature point pair.

ENDIF

S18. CONTINUE to next time step (GO TO S5).

S19. CONTINUE to next time sequence (GO TO S1).

S20. IF (LDYN) THEN

1. **HPLOT:** plot nodal time-histories.
2. **ELEMNT**('STR_PLOT'): loop over element groups NEG=1,NUMEG and plot element time-histories with **HPLOT**; element task 6.

ENDIF

S21. RETURN to **DLEARN** (GO TO G2.6).

11.4 ADDING AN ELEMENT TO DLEARN

Example

Suppose we wish to add a shell element to the program. Let us call the subroutine that sets element group parameters and allocates blank common storage SUBROUTINE SHELL. The steps are as follows:

1. Study and understand thoroughly the subroutines QUADC and TRUSS before beginning. Notice the great similarity in their structure. Although a two-node truss

could be programmed more explicitly, i.e, in the spirit of matrix structural analysis, we feel the present implementation emphasizes more generalized finite element methodologies.

2. When studying the elements QUADC and TRUSS, note that the elements are broken into six calls to different subroutines; the variable ITASK defines which routine will be called. In the global routines the call to ELEMNT defines the value of ITASK. For example, in DRIVER, the statement CALL ELEMNT ('FORM_LHS') results in ITASK having the value 2 (see routine ELEMNT). In the routine QUADC, if ITASK has the value 2, the routine QDCT2 is executed. Therefore, to make any new element compatible with the "global element functions," the new element should be designed with these six functions in mind. However, if the user is extending the capabilities of DLEARN, he or she is totally free to define additional element functions. The six current functions are defined as follows:

   1. 'INPUT_____' : Read/generate element input data.
   2. 'FORM_LHS' : Compute element contribution to global left-hand-side matrix.
   3. 'FORM_RHS' : Compute element contribution to global right-hand-side vector.
   4. 'STR_PRNT' : Compute and print element stress/strain output.
   5. 'STR_STOR' : Compute element stress/strain for output time histories.
   6. 'STR_PLOT' : Plot element time histories.

   As an example, we present an outline of SUBROUTINE QUADC.

```
SUBROUTINE QUADC(ITASK,NPAR,MP,NEG)
PROGRAM TO SET STORAGE AND CALL TASKS FOR THE
   FOUR-NODE QUADRILATERAL, ELASTIC CONTINUUM ELEMENT
      *
      *
DEFINE STORAGE POINTER NAMES
      *
      *
EQUIVALENCE GROUP PARAMETERS FROM NPAR ARRAY
      *
      *
SET ELEMENT PARAMETERS
      *
      *
IF (ITASK.EQ.1) THEN
.. SET MEMORY POINTERS
      *
      *
ENDIF
TASK CALLS
IF (ITASK.GT.6) RETURN
GO TO (100,200,300,400,500,600),ITASK
```

```
C
  100 CONTINUE
C
C.... INPUT ELEMENT DATA ('INPUT___')
C
      CALL QDCT1(A(MP(MSHL   )),A(MP(MW     )),A(MP(MRHO   )),
                 *
                 *
C
      RETURN
C
  200 CONTINUE
C
C.... FORM ELEMENT EFFECTIVE MASS AND ASSEMBLE INTO GLOBAL
C        LEFT-HAND-SIDE MATRIX  ('FORM_LHS')
C
      CALL QDCT2(A(MP(MELEFM)),A(MP(MIEN   )),A(MPX        ),
                 *
                 *
C
      RETURN
C
  300 CONTINUE
C
C.... FORM ELEMENT RESIDUAL-FORCE VECTOR AND ASSEMBLE INTO GLOBAL
C        RIGHT-HAND-SIDE VECTOR ('FORM_RHS')
C
      CALL QDCT3(A(MP(MMAT   )),A(MP(MIEN   )),A(MPDPRD     ),
                 *
                 *
C
      RETURN
C
  400 CONTINUE
C
C.... CALCULATE AND PRINT ELEMENT STRESS/STRAIN OUTPUT ('STR_PRNT')
C
      IF (ISTPRT.EQ.0)
     &   CALL QDCT4(A(MP(MMAT   )),A(MP(MIEN   )),A(MPD        ),
                 *
                 *
C
      RETURN
C
  500 CONTINUE
C
C.... CALCULATE AND STORE ELEMENT TIME-HISTORIES ('STR_STOR')
C
      IF (NSOUT.GT.0)
     &   CALL QDCT5(A(MP(MISHST)),A(MP(MSOUT )),A(MP(MMAT   )),
                 *
                 *
C
      RETURN
C
  600 CONTINUE
C
C.... PLOT ELEMENT TIME-HISTORIES ('STR_PLOT')
C
      IF (NSOUT.GT.0)
     &   CALL HPLOT(A(MP(MISHST)),A(MP(MSOUT )),NSOUT ,3,NTYPE )
      RETURN
C
      END
```

3. When coding SHELL and its task routines, use existing utility routines, such as MULTAB, whenever possible.[2]
4. Code a new version of subroutine ELMLIB to include a call to SHELL. The necessary modifications are indicated:

[2]The experienced programmer will recognize that the use of utility routines such as MULTAB is highly inefficient. Nevertheless, this approach facilitates implementation and results in very readable coding. Significant gains in efficiency can be made by avoiding subroutine calls and explicitly writing out the matrix products line by line, taking account of zeros and repeated calculations. See Sec.3.10.

```
      GO TO (100,200,300),NTYPE
C
  100 CALL QUADC(ITASK,A(MPNPAR),A(MPNPAR+16),NEG)
      RETURN
C
  200 CALL TRUSS(ITASK,A(MPNPAR),A(MPNPAR+16),NEG)
      RETURN
```

```
C
  300 CALL SHELL(ITASK,A(MPNPAR),A(MPNPAR+16),NEG)
      RETURN
```

5. Define an array LABEL3 in BLOCK DATA and specify the desired labels for stress, strain, and any other time-histories. Include the appropriate WRITE statements in HPLOT and SHIST.
6. Write a description for the Input Instructions to be included in Sec.In.10, ELEMENT DATA. Mimic the style of Secs. In.10.1 and In.10.2, and use the same variable names as far as possible.

Instructions for Debugging[3] and Running the New Element

1. Compile all your new routines separately before including them in DLEARN.

2. Link all routines compiled and execute with simple, static, one-element problems (e.g., homogeneous deformation states), which are easy to check. Check all options (e.g., gravity loads, etc.) on the one-element problems.

3. Next, proceed to "patch tests," which also should be easy to check. Finally, after you are convinced your element is working properly, try some interesting static and dynamic test problems for which you have exact analytical, or reliable experimental data. Assess the accuracy of the element you have programmed.

11.5 DLEARN USER'S MANUAL

11.5.1 Remarks for the New User

This section presents the necessary input instructions for the user to execute DLEARN. The new user is advised to read *all* the instructions, paying particular

[3] Debugging is boring and frustrating, though the use of interactive debug utilities can reduce the tedium. When you are feeling particularly depressed by this experience, remember *everyone* who has ever programmed has experienced what you are feeling now. Misery loves company.

attention to the appended notes, to gain an initial overview of the data required to specify an analysis task to DLEARN. Readers without prior finite element computational experience will benefit from careful study of the example problems; they are intended to help clarify the use of the data generation features. The amount of data required to specify even a small problem can become burdensome if the user does not learn to exercise these capabilities.

Upon examining the input instructions, the reader should recognize that no system of units is assumed by the program. Rather, *it is the user's responsibility to specify data in the consistent units of his or her choice*. Overlooking this requirement is a common source of erroneous results, especially in dynamic analysis.

It is often the case that new users have difficulty in getting a program to successfully run. It is tempting to assume that there are "bugs" in the program and that the fault does not lie with oneself. DLEARN has undergone extensive testing to minimize the possibility of any "bugs" being present. Thus, the new user should normally avoid this alibi, for the vast majority of aborted runs are the result of input specification errors. Train yourself to look closely at generation sequences; a single slipped digit can lead to unbelievably chaotic input data and frequently the legendary "floating point overflow/divide by zero."

Unfortunately, input errors need not be this apparent, though nonetheless catastrophic. For example, an error in generating initial displacements for a dynamic analysis may lead INPUT to write beyond the storage allocated for the D array. The D array is followed in COMMON A(MTOT) by the nodal coordinates array X(NSD,NUMNP), thus INPUT would unknowingly overwrite the coordinate data. This may later lead to an error termination, e.g., calculation of a nonpositive Jacobian determinant in an element or the successful solution of the *wrong* initial/boundary-value problem. All this would occur after the coordinate data are written to the output file, making detection difficult. This can be a frustrating task for the new user. However, with a little experience, efficiency in recognizing errors will be gained. Take heart, even the most seasoned analyst occasionally wastes time (and money) searching for such anomalies.

11.5.2 Input Instructions

In.0.0 Input Data Echo Card[4] (I5)

Note	Columns	Variable	Description
(1)	1–5	IECHO	Input file data echo EQ.0, no data echo EQ.1, data echo

Notes

1. Data from input file are read and written, without interpretation, at beginning of output file before execution continues.

[4] Recognizing the preponderance of interactive computing today, the term "card" as used here refers to an 80-character line in the input file.

In.1.0 Title Card (20A4)

Note	Columns	Variable	Description
(1)	1–80	TITLE(20)	Job title for heading the output

Notes

1. Each data set begins with a title card. *To terminate execution, the last card in the input file, which will be read as a new title card, must contain the characters* *END *in the first four columns.*

In.2.0 Execution Control Card (15I5)

Note	Columns	Variable	Description
(1)	1–5	IEXEC	Execution code EQ.0, data check only EQ.1, execution
(2)	6–10	IACODE	Analysis code EQ.0, dynamic analysis EQ.1, static analysis
(3)	11–15	IREADR	Read restart file code EQ.0, do not read restart file EQ.1, read restart file
(4)	16–20	IWRITR	Write restart file code EQ.0, do not write restart file EQ.1, write restart file
(5)	21–25	IPRTIN	Input data print code EQ.0, print nodal and element input data EQ.1, do not print nodal and element input data
(6)	26–30	IRANK	Rank check code EQ.0, do not perform rank check EQ.1, print numbers of zero and negative pivots EQ.2, print all pivots
(7)	31–35	NUMSEQ	Number of time sequences, GE.1
(8)	36–40	NDOUT	Number of nodal output time histories
(9)	41–45	NSD	Number of space dimensions
	46–50	NUMNP	Number of nodal points
(10)	51–55	NDOF	Number of nodal degrees of freedom
(11)	56–60	NLVECT	Number of load vectors
(12)	61–65	NLTFTN	Number of load-time functions
(13)	66–70	NPTSLF	Number of points defining a load-time function
(14)	71–75	NUMEG	Number of element groups

Notes

1. In the data check mode, input data are printed and storage requirements are indicated. This mode should be employed prior to making expensive calculations.
2. A static analysis consists of solution of the equilibrium equations, resulting in displacement and stress output. A dynamic analysis consists of time-integration of the semi-discrete equations of motion, resulting in displacement, velocity, acceleration, and stress output.
3. If IREADR .EQ. 1, then initial displacement, velocity, and acceleration data will be read from the file associated with FORTRAN unit IRSIN at the beginning of a dynamic analysis (i.e. IACODE .EQ. 0).
4. If IWRITR .EQ. 1, then final displacement, velocity, and acceleration data will be written to the file associated with FORTRAN unit IRSOUT at the conclusion of a dynamic analysis (i.e., IACODE .EQ. 0).
5. If IPRTIN .EQ. 1, then only control input data, e.g., time-sequence data, is printed in the output file. This option is useful in reducing output for very large amounts of generated input data. However, it should only be employed after thorough checking of data has been performed prior to execution.
6. A rank check consists of factorization of the effective mass matrix, $\boldsymbol{M}^* = \boldsymbol{U}^T\boldsymbol{D}\boldsymbol{U}$, and either printing the numbers of zero and negative entries along the diagonal of $\boldsymbol{D}$ (IRANK .EQ. 1), or printing the entire diagonal of $\boldsymbol{D}$ (IRANK .EQ. 2), for each time sequence. The number of zero-energy modes can be deduced from the number of zeroes along the diagonal of $\boldsymbol{D}$. This option allows for the checking of elements and meshes suspected of containing spurious zero-energy modes, such as those arising in so-called "under-integrated" finite elements. If IRANK .GT. 0, then no other executable calculations are performed (i.e., dynamic or static analysis).
7. For static analysis, a single time sequence should be adequate, although the use of multiple sequences is not prohibited. A dynamic analysis can be subdivided into multiple contiguous time intervals in which different time-integration constants and output parameters may be specified. See Sec. In.3.0.
8. Output histories are ignored in static analysis as denoted by IACODE .EQ. 1. See Sec. In.4.0.
9. Consult individual element routines in Sec. In.10.0 for requirements.
10. In problems for which there are different numbers of degress of freedom at different nodes, NDOF should be set equal to the maximum number. Consult individual element routines in Sec. In.10.0 for requirements. Superfluous degrees of freedom can be eliminated by employing boundary condition codes. See Sec. In.6.0.
11. The prescribed nodal forces and kinematic boundary conditions are synthesized by NLVECT vectors and corresponding load-time functions. Details are presented in Secs. In.7.0 and In.8.0.
12. The user must define NLTFTN .GE. NLVECT. The first NLVECT load-time functions scale the corresponding load vectors. Additional load-time functions may be defined to scale element surface and body forces specified in the element group data.
13. The load-time functions are discretized by NPTSLF points. See Sec. In.8.0 for details.
14. The elements are read in groups.

In.3.0 Time Sequence Data Cards (6I5,4F10.0)

Note	Columns	Variable	Description
(1)	1–5	N	Sequence number
	6–10	NSTEP(N)	Number of time steps for sequence N
(2)	11–15	NDPRT(N)	Print kinematic output every NDPRT(N) time steps
(2)	16–20	NSPRT(N)	Print stress/strain output every NSPRT(N) time steps
(2)	21–25	NHPLT(N)	Plot time history data every NHPLT(N) time steps
	26–30	NITER(N)	Number of iterations in corrector loop; GE. 1
(3)	31–40	ALPHA(N)	α parameter for sequence N, LE. 0.0
(3)	41–50	BETA(N)	β parameter for sequence N, GT. 0.0
(3)	51–60	GAMMA(N)	γ parameter for sequence N, GE. 0.0
(3)	61–70	DT(N)	Time step Δt for sequence N

Notes

1. NUMSEQ time sequence data cards must be defined but need not be input in any particular order. NUMSEQ .GE. 1 is defined on the Execution Control Card; see Sec. In.2.0.
2. If set Eq. 0, no output. In each sequence, the final print/plot state is output at *sequence* time step NSTEP(N) − MOD(NSTEP(N), NDPRT(N)), etc., and the output logic will reinitialize at the beginning of the following sequence.
3. Many different transient algorithms can be realized by the proper choice of parameters α, β, and γ. The choice of $\alpha \in [-\frac{1}{3}, 0]$, $\beta = \frac{1}{4}(1 - \alpha)^2$, $\gamma = \frac{1}{2}(1 - 2\alpha)$ leads to a second-order accurate method, which is unconditionally stable *if all element groups are specified to be implicit*. The user should be cognizant of any stability restrictions upon the time step size Δt; *such limitations can be severe when using explicit element groups*.

   To compute the consistent initial acceleration for all *active* degrees of freedom, specify the first time sequence to consist of one time-step with $\alpha = 0$ and $\Delta t = 0$. The code will then solve $\boldsymbol{M}\Delta\boldsymbol{a} = \boldsymbol{F}_0 - \boldsymbol{M}\boldsymbol{a}_0 - \boldsymbol{C}\boldsymbol{v}_0 - \boldsymbol{K}\boldsymbol{d}_0$ and update: $\boldsymbol{a}_0 = \boldsymbol{a}_0 + \Delta\boldsymbol{a}$, $\boldsymbol{v}_0 = \boldsymbol{v}_0$, and $\boldsymbol{d}_0 = \boldsymbol{d}_0$. The values of β and γ are irrelevant, but the user should specify NDPRT = NSPRT = NHPLT = 1 if he or she wishes to examine the results of this calculation. Note that there is no rational way to *calculate* the initial acceleration for any node controlled by kinematic boundary conditions (see Sec. In.6.0): These data must be specified by the user (See Sec. In.9.0).

   For *static analysis*, the following defaults are set internally:

NSTEP(N) = MAX0(1, NSTEP(N))	
NDPRT(N) = 1	$\alpha = 0.0$
NSPRT(N) = 1	$\beta = 1.0$
NHPLT(N) = 0	$\gamma = 0.0$
NITER(N) = 1	$\Delta t = 1.0$

In.4.0 Nodal History Data Cards (3I5)

If there are no nodal output time-histories (i.e., NDOUT .EQ. 0 on the Execution Control Card; see Sec. In.2.0), then no nodal history data cards should be input; proceed directly to Sec. In.5.0.

If the analysis is static (i.e., IACODE .EQ. 1 on the Execution Control Card), then output history cards are read, but subsequently ignored.

Each component requested is plotted versus time in the output file. Plots of this type are useful in providing quick information concerning the time histories of important components. Data for line plots of nodal variables are stored in array DOUT. The user may wish to modify DLEARN to write this information to a disk file for subsequent off-line plotting.

Note	Columns	Variable	Description
(1)	1–5	IDHIST(1,N)	Node number; GE. 1 and LE. NUMNP
	6–10	IDHIST(2,N)	Degree of freedom number; LE. NDOF
(2)	11–15	IDHIST(3,N)	Kinematic quantity specifier

Notes

1. History output data must be input for each node/degree of freedom combination at which a time history output is desired. The index N in the array IDHIST corresponds to the order in which the cards are read: 1 .LE. N .LE. NDOUT. The cards need not be specified in any particular order. *The total number of degrees of freedom to be plotted for all nodes must equal* NDOUT, *defined on the Execution Control Card* (see Sec. In.2.0).
2. IDHIST(3,N) = 1, displacement
   IDHIST(3,N) = 2, velocity
   IDHIST(3,N) = 3, acceleration

In.5.0 Coordinate Data

In.5.1 Nodal Coordinate Cards (2I5,NSD×F10.0)

Note	Columns	Variable	Description
(1)	1–5	N	Node number; GE. 1 and LE. NUMNP
(2)	6–10	NUMGP	Number of generation points; EQ.0, no generation GT.0, generate nodal data
(3)	11–20	X(1,N)	x_1-coordinate of node N
	21–30	X(2,N)	x_2-coordinate of node N
	31–40	X(3,N)	x_3-coordinate of node N

Notes

1. The coordinates of each node must be defined, but need not be input in any particular

order. If NSD is specified as 2 on the Execution Control Card (See Sec. In.2.0), any nonzero x_3-coordinates will be ignored. *Terminate with a blank card.*

2. If NUMGP is greater than zero, this card initiates an isoparametric data generation sequence. Cards 2 to NUMGP of the sequence define the coordinates of the additional generation points (see Sec. In.5.2). The final card of the sequence defines the nodal increment information (see Sec. In.5.3). After the generation sequence is completed, additional nodal coordinate cards, or generation sequences, may follow.

   The generation may be performed along a line, over a surface, or within a volume. A description of each of these options follows.

Generation along a line

The line may be defined by two or three generation points (see Fig. 11.5.1), and the physical space may be one-, two-, or three-dimensional.

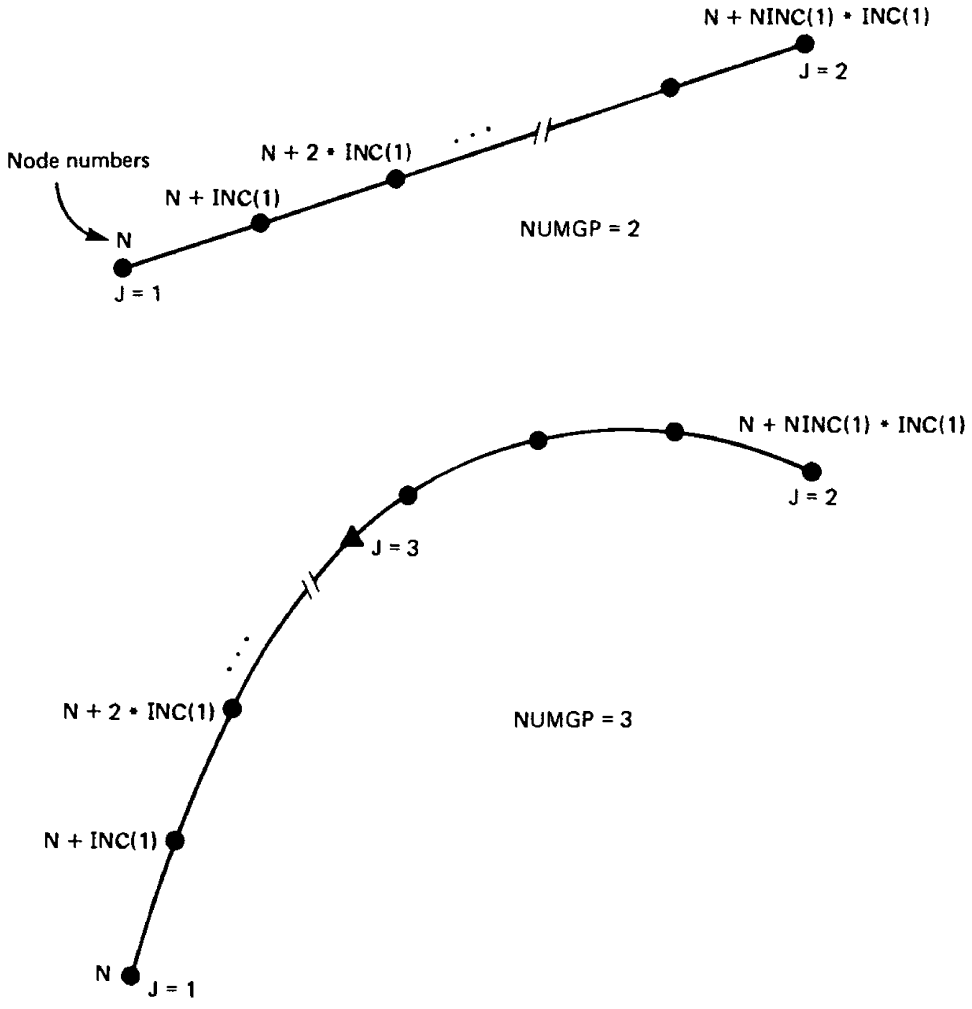


Figure 11.5.1 Nodal generation along a line.

In the case NUMGP = 2, linear interpolation takes place, resulting in equally spaced nodal points.

In the case NUMGP = 3, quadratic interpolation is employed and graded nodal spacing may be achieved by placing the third generation point (J = 3) off center. Note that the third generation point does not generally coincide with any nodal point. The spacing in this case may be determined from the following mapping:

$$x_A = x(\xi_A) = \frac{1}{2}\xi_A(\xi_A - 1)x_1^g + \frac{1}{2}\xi_A(\xi_A + 1)x_2^g + (1 - \xi_A^2)x_3^g$$

where ξ_A is the location of node number A in ξ-space. The nodes are placed at equal intervals in ξ-space; x_1^g, x_2^g and x_3^g are the coordinates of the three generation points in x-space; and x_A denotes the coordinates of the Ath node in x-space (see Fig. 11.5.2).

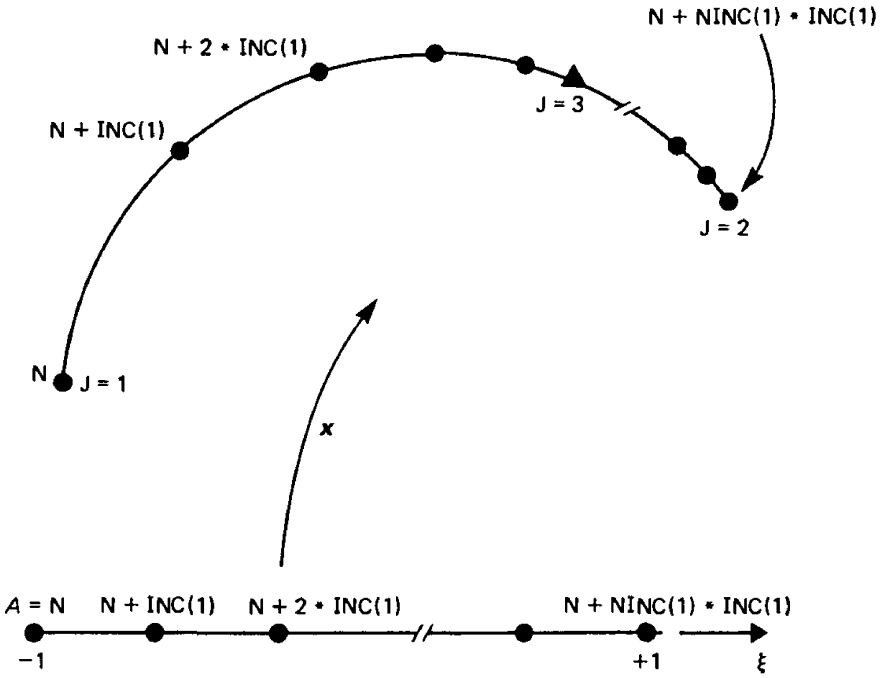


Figure 11.5.2 Nodal generation along a line: mapping from local interval to physical space.

Generation over a surface

The surface may be defined by four or eight generation points (see Fig. 11.5.3) and the physical space may be two- or three-dimensional. In the three-dimensional case, the surfaces may be curved.

In the case NUMGP = 4, bilinear interpolation is employed, resulting in equally spaced nodal points along generating lines.

In the case NUMGP = 8, biquadratic serendipity interpolation is employed, and graded nodal spacing may be achieved by placing generation points 5–8 off center. Note that generation points 5–8 do not generally coincide with any nodal points. The spacing of the nodal points may be determined from the serendipity mapping.

Generation within a volume

The volume is brick shaped and may be defined by 8 or 20 generation points (see Fig. 11.5.4). In this case the physical space must be three-dimensional.

If NUMGP = 8, trilinear interpolation is employed, resulting in equally spaced nodal points along generating lines.

If NUMGP = 20, triquadratic serendipity interpolation is employed and graded nodal spacing may be achieved by placing generation points 9–20 off center. Note that generation points 9–20 do not generally coincide with any nodal points. The spacing of the nodal points may be determined by the serendipity mapping.

3. If the coordinates of node N are input and/or generated more than one time, the last values take precedence.

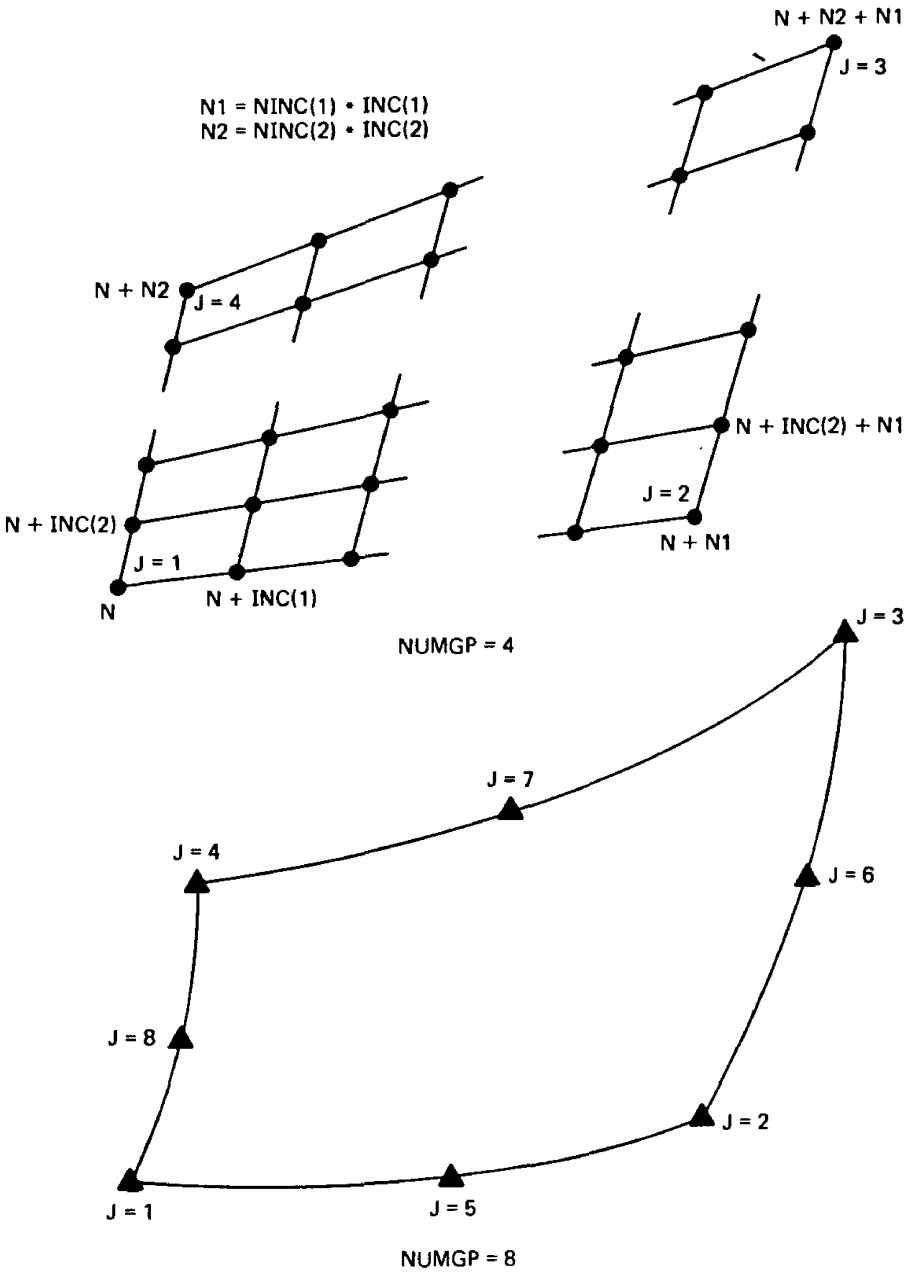
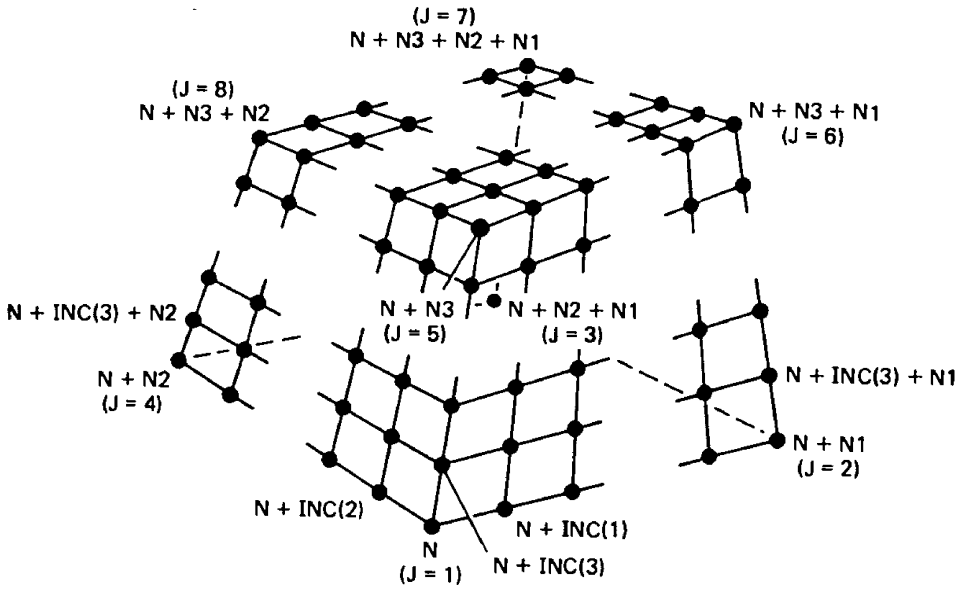


Figure 11.5.3 Nodal generation over a surface.

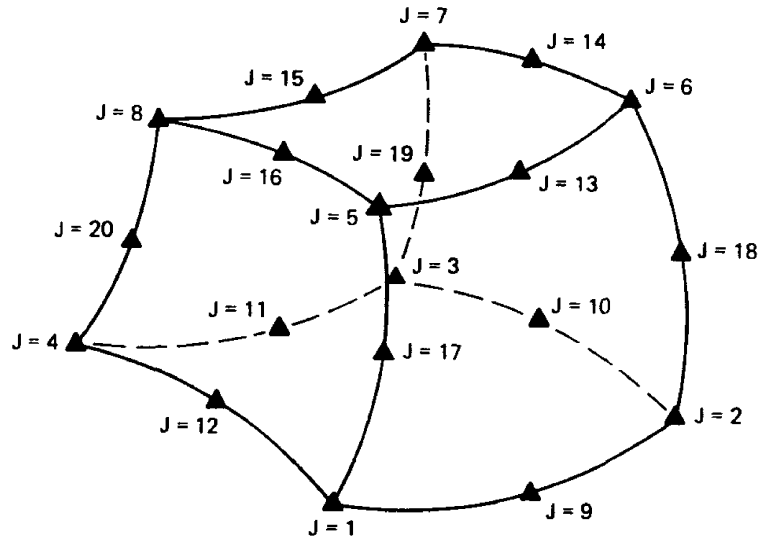
In.5.2 Generation Point Coordinate Cards (2I5,NSD×F10.0)

Note	Columns	Variable	Description
(1)	1–5	M	Node number
	6–10	MGEN	Generation parameter; EQ.0, coordinates of the Jth generation point are input on this card; M is ignored.

Note	Columns	Variable	Description
			EQ.1, coordinates of the Jth generation point are set equal to coordinates of the Mth node which was previously defined; coordinates on this card are ignored.
	11–20	TEMP(1,J)	x_1-coordinate of generation point J
	21–30	TEMP(2,J)	x_2-coordinate of generation point J
	31–40	TEMP(3,J)	x_3-coordinate of generation point J



NUMGP = 8



NUMGP = 20

Figure 11.5.4 Nodal generation within a volume.

Notes

1. The coordinates of each generation point are defined by a generation point coordinate card. The cards must be read in order (J = 2, 3, . . . , NUMGP) following the nodal coordinate card which initiated the generation sequence (J = 1). A nodal increments card (see Sec. In.5.3), which completes the sequence, follows the last generation point card (J = NUMGP).

In.5.3 Nodal Increments Card (6I5)

Note	Columns	Variable	Description
	1–5	NINC(1)	Number of node increments for direction 1; GE.0
	6–10	INC(1)	Node number increment for direction 1
(1)	11–15	NINC(2)	Number of node increments for direction 2; GE.0
	16–20	INC(2)	Node number increment for direction 2
(1)	21–25	NINC(3)	Number of node increments for direction 3; GE.0
	26–30	INC(3)	Node number increment for direction 3

Notes

1. Each generation option is assigned an option code (IOPT) as follows:

IOPT	Option
1	generation along a line
2	generation over a surface
3	generation within a volume

IOPT is determined by the following logic:

```
IOPT = 3
IF (NINC(3).EQ.0) IOPT = 2
IF (NINC(2).EQ.0) IOPT = 1
```

In.6.0 Boundary Conditions Data Cards((3+NDOF) × I5)

Note	Columns	Variable	Description
(1)	1–5	N	Number of first node in sequence GE.1 and LE. NUMNP
(2)	6–10	NE	Number of last node in generation sequence LE. NUMNP
	11–15	NG	Generation increment
(3)	16–20	ID(1,N)	Degree of freedom 1 boundary code
	21–25	ID(2,N)	Degree of freedom 2 boundary code
	⋮	⋮	⋮
		ID(NDOF,N)	Degree of freedom NDOF boundary code

Notes

1. Boundary condition data must be input for each node which has one or more specified displacements, velocities or accelerations. Cards need not be specified in any particular order. *Terminate with a blank card.*
2. Each data card will generate identical boundary condition codes for the following sequence of nodes:

   N, N+NG, N+2×NG, . . . , NE−MOD(NE−N,NG)

   If NG is blank or zero, no generation takes place and only boundary conditions for node N are defined. In this case NE is ignored.
3. Boundary condition codes may be assigned the following values:

   ID(I,N) = 0, unspecified displacement (active degree of freedom)

   ID(I,N) = 1, specified displacement

   ID(I,N) = 2, specified velocity

   ID(I,N) = 3, specified acceleration

   Upon exiting subroutine BC, the ID array holds the following values:

   ID(I,N) = M (> 0), global equation number for active degree of freedom I of node N

   ID(I,N) = 0, displacement specified for degree of freedom I of node N

   ID(I,N) = −1, velocity specified for degree of freedom I of node N

   ID(I,N) = −2, acceleration specified for degree of freedom I of node N

In.7.0 Prescribed Nodal Forces and Kinematic Boundary Conditions

If NLVECT .EQ. 0 on the Execution Control Card (see Sec. In.2.0), no prescribed nodal forces and boundary condition cards should be input; proceed directly to Sec. In.8.0.

Prescribed nodal forces and kinematic boundary conditions, i.e., displacements, velocities or accelerations, are defined by expansions of the form:

$$F(x_A, t) = \sum_{i=1}^{\text{NLVECT}} G_i(t) F_i(x_A)$$

where $F(x_A, t)$ is the resultant force or kinematic boundary condition at node A at time t; G_i is the ith load-time function; F_i is the ith load vector; and NLVECT is the total number of load vectors as specified on the Execution Control Card. The data preparation for the load-time functions is described in Sec. In.8.0. In this section, the data specification for each F_i is described.

The load vectors should be read in the order $F_1, F_2, \ldots, F_{\text{NLVECT}}$. Note that each entry of a single vector can be interpreted as a force or kinematic quantity depending on its corresponding boundary condition code defined in Sec. In.6.0.

Data cards for a typical load vector are described next.

In.7.1 Nodal Prescribed Forces/Kinematic Boundary Conditions Cards (2I5,NDOF×F10.0)

Note	Columns	Variable	Description
(1)	1–5	N	Node number; GE.1 and LE.NUMNP
(2)	6–10	NUMGP	Number of generation points EQ.0, no generation GT.1, generate data
(3)	11–20	F(1,N)	Degree of freedom 1 prescribed force/ kinematic boundary condition
	21–30	F(2,N)	Degree of freedom 2 prescribed force/ kinematic boundary condition
	⋮	⋮	⋮
		F(NDOF,N)	Degree of freedom NDOF prescribed force/ kinematic boundary condition

Notes

1. Data must be included for each node subjected to a nonzero prescribed force or kinematic boundary condition. Cards need not be specified in any particular order. *Terminate input for each load vector with a blank card.*
2. If NUMGP is greater than zero, this card initiates an isoparametric data generation sequence. The scheme used is the same as that employed for coordinate generation described previously in Sec. In.5. Cards 2 to NUMGP of the sequence define the prescribed forces/kinematic boundary condition data of the additional generation points (see Sec. In.7.2). The final card of the sequence defines the nodal increment information (see Sec. In.5.3). After the generation sequence is completed, additional nodal prescribed forces/kinematic boundary conditions cards, or generation sequences, may follow.

   The generation may be performed along a line, over a surface, or within a volume. A description of each of these options is given below.

   Generation along a line

   The line may be defined by two or three generation points (see Fig. 11.5.5), and the physical space (x-space) may be one-, two-, or three-dimensional.

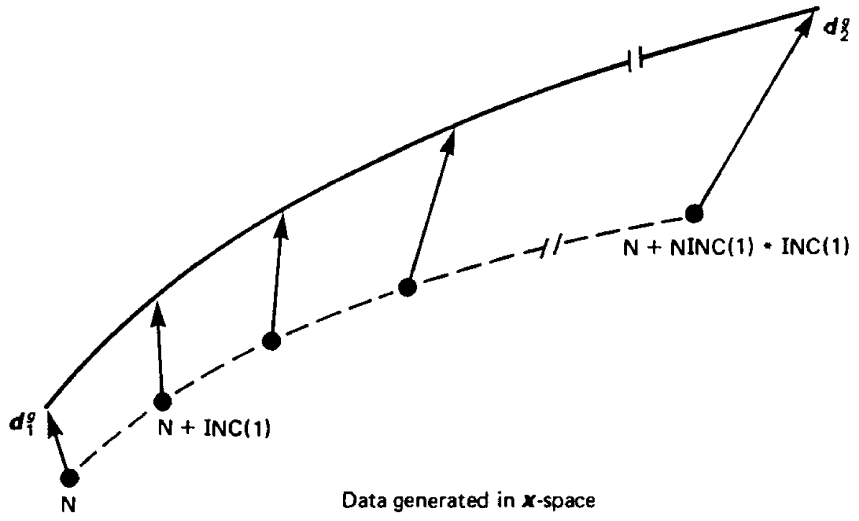
   In the case NUMGP = 2, linear interpolation takes place with respect to ξ-space. If the nodes are equally spaced in x-space, then the variation will also be linear in x-space. Otherwise, a nonlinear variation will be induced by the unequal nodal spacing.

   In the case NUMGP = 3, quadratic interpolation is performed with respect to ξ-space. Note that the third generation point does not generally coincide with any nodal point. The variation of the data may be determined from the following mapping:

$$\boldsymbol{d}_A = \boldsymbol{d}(\xi_A) = \tfrac{1}{2}\xi_A(\xi_A - 1)\boldsymbol{d}_1^g + \tfrac{1}{2}\xi_A(\xi_A + 1)\boldsymbol{d}_2^g + (1 - \xi_A^2)\boldsymbol{d}_3^g$$

   where ξ_A is the location of node number A in ξ-space (recall that nodes are assumed to be placed at equal intervals in ξ-space); $\boldsymbol{d}_1^g$, $\boldsymbol{d}_2^g$, and $\boldsymbol{d}_3^g$ are the data assigned to the three generation points in x-space; and $\boldsymbol{d}_A$ denotes the prescribed force/kinematic boundary condition data of the Ath node in x-space.

   The case in which NUMGP = 2 may be deduced from the case NUMGP = 3 by setting $\boldsymbol{d}_3^g = (\boldsymbol{d}_1^g + \boldsymbol{d}_2^g)/2$.



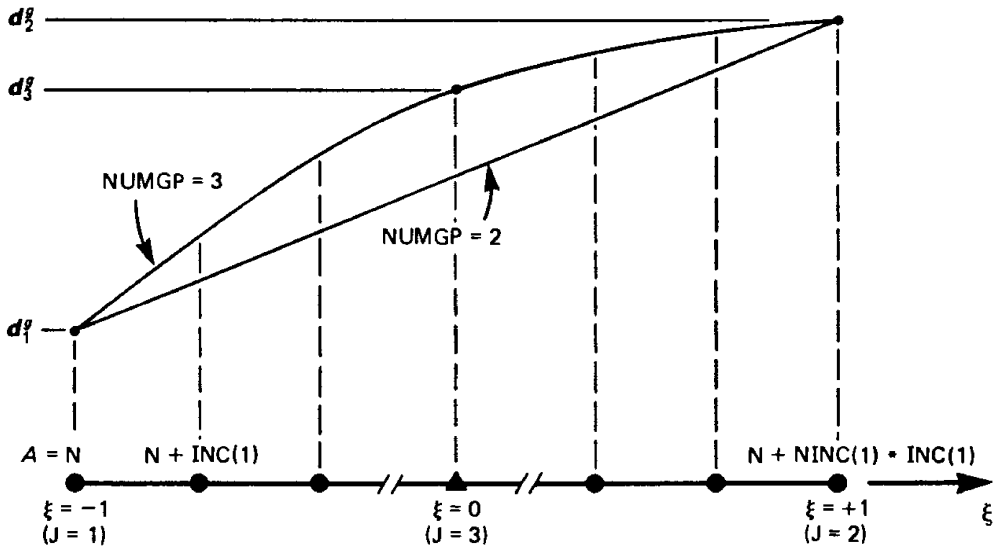


Figure 11.5.5 Load vector generation.

Generation over a surface

The surface may be defined by 4 or 8 generation points. The generation points and nodal patterns are the same as in coordinate generation (see Fig. 11.5.3).

In the case NUMGP = 4, bilinear interpolation is performed; for NUMGP = 8, biquadratic serendipity interpolation is performed. Note that generation points 5–8 do not generally coincide with any nodal points.

Generation within a volume

Generation of data within a brick-shaped volume may be performed using 8 or 20 generation points. The generation points and nodal patterns are the same as for coordinate generation (see Fig. 11.5.4).

If NUMGP = 8, trilinear interpolation is performed; if NUMGP = 20, triquadratic

serendipity interpolation is performed. Note that generation points 9–20 do not generally coincide with any nodal points.

3. The elements of the array F(NDOF,NUMNP,NLVECT) are initialized to zero. Prescribed forces/kinematic boundary conditions for node N are written within this array. If values for node N are defined more than once, the last value will take precedence.

In.7.2 Generation Point Prescribed Forces/Kinematic Boundary Conditions Cards (2I5,NDOF×F10.0)

Note	Columns	Variable	Description
1	1–5	M	Node number
	6–10	MGEN	Generation parameter; EQ.0, prescribed forces/kinematic boundary conditions of the Jth generation point are input on this card; M is ignored. EQ.1, prescribed forces/kinematic boundary conditions of the Jth generation point are set equal to the prescribed forces/kinematic boundary conditions of the Mth node which were previously defined; prescribed forces/ kinematic boundary conditions on this card are ignored.
	11–20	TEMP(1,J)	Degree of freedom 1 prescribed force/ kinematic boundary condition for generation point J
	21–30	TEMP(2,J)	Degree of freedom 2 prescribed force/ kinematic boundary condition for generation point J
	⋮	⋮	⋮
		TEMP(NDOF,J)	Degree of freedom NDOF prescribed force/kinematic boundary condition for generation point J

Notes

1. The prescribed forces/kinematic boundary conditions of each generation point are defined by a generation point prescribed forces/kinematic boundary conditions card. The cards must be read in order (J = 2, 3, . . . , NUMGP) following the nodal prescribed forces/kinematic boundary conditions card which initiated the generation sequence (J = 1). A nodal increments card (see Sec. In.5.3) follows the last generation point card (J = NUMGP) and completes the sequence.

In.7.3 Nodal Increments Card (6I5)

See instructions in Sec. In.5.3.

In.8.0 LOAD-TIME FUNCTIONS

If NLTFTN .EQ. 0 on the Execution Control Card (see Sec. In.2.0), no load-time function cards should be input; proceed directly to Sec. In.9.0.

If NLTFTN .GT. 0, the user must input NLTFTN load-time functions, each defined by NPTSLF pairs of time instants and function values, where NLTFTN and NPTSLF are defined on the Execution Control Card. A schematic representation of a typical load-time function is shown in Fig. 11.5.6. The time instants must be in

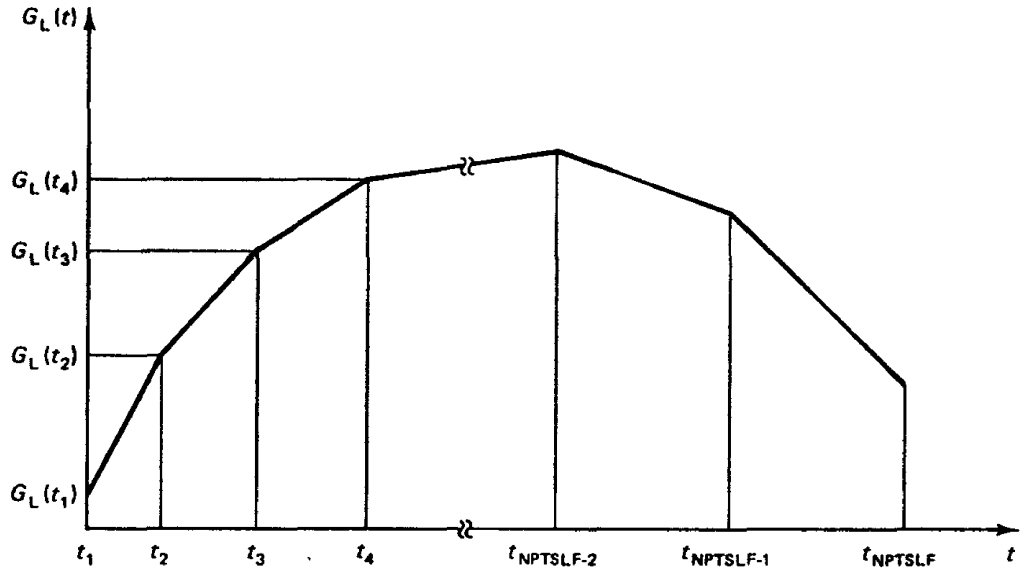


Figure 11.5.6 Schematic representation of load-time function L.

ascending order (i.e., $t_k < t_{k+1}$, $1 \leq k \leq \text{NPTSLF} - 1$). Load step intervals need not be equal and need not be the same from one load-time function to another. However, there must be the same number of load steps (i.e., NPTSLF) for each load-time function. As may be seen from Fig. 11.5.6, the load-time function is assumed to behave in a piecewise linear fashion between data points. For values of t outside the interval $[t_1, t_{\text{NPTSLF}}]$ we define G_i by constant extrapolation: $G_i(t) = G_i(t_1)$ for all $t \leq t_1$ and $G_i(t) = G_i(t_{\text{NPTSLF}})$ for all $t \geq t_{\text{NPTSLF}}$. For example, if the user wishes to define all load-time functions with a constant magnitude for all time, he may set NPTSLF = 1 and define a single point of $G_i(t)$ for any time t.

The load-time functions must be read in the order $G_1, G_2, \ldots, G_{\text{NLTFTN}}$, where the first NLVECT functions scale the corresponding load vectors specified in Sec. In.7.0. Element loads (e.g., pressure, gravity, etc.) are also multiplied by load-time functions. The applicable function number is defined in the element group data (see Sec. In.10).

Data cards for a typical load-time function are described next.

In.8.1 Load-time Function Cards (2F10.0)

Note	Columns	Variable	Description
(1)	1–10	G(K,1)	Time value K, i.e., t_k
	11–20	G(K,2)	Value of load-time function at time t_k

Notes

1. The local array G(K,I) is equivalent to global array G(K,I,L) for load-time function L.

In.9.0 KINEMATIC INITIAL CONDITION DATA

If the analysis is static (i.e., if IACODE .EQ. 1 on the Execution Control Card, see Sec. In.2.0) or if the program is instructed to read a restart file (i.e., if IREADR .EQ. 1 on the Execution Control Card), then no nodal initial condition data cards should be input; proceed directly to Sec. In.10.0.

Initial nodal displacements, velocities and accelerations are specified in separate data blocks having identical form. Each block is terminated with a blank card, which must be present even if zero initial conditions are desired at all nodes for one of the kinematic variables. For example, if the user wishes only to specify nonzero initial velocities, the data will have the form:

1. A blank card representing the absence of nonzero initial displacement data.
2. A block of initial velocity data input in the manner specified by Secs. In.9.1 to In.9.3 and terminating with a blank card.
3. A blank card representing the absence of nonzero initial acceleration data.

In.9.1 Nodal Initial Condition Cards (2I5,NSD×F10.0)

Note	Columns	Variable	Description
(1)	1–5	N	Node number; GE.1 and LE. NUMNP
(2)	6–10	NUMGP	Number of generation points EQ.0, no generation GT.0, generate initial conditions
(3)	11–20	D(1,N)*	Degree of freedom 1 initial condition
	21–30	D(2,N)	Degree of freedom 2 initial condition
	.	.	.
	.	.	.
	.	.	.
		D(NDOF,N)	Degree of freedom NDOF initial condition

* Here the array D is used generically to refer to either displacements (D), velocities (V), or accelerations (A).

Notes

1. Data must be included for each node subjected to nonzero initial displacements, velocities, or accelerations but need not be specified in any particular nodal order. *Terminate each data block with a blank card.*
2. If NUMGP is greater than zero, this card initiates an isoparametric data generation sequence. The scheme used is the same as that for prescribed nodal force/boundary condition data generation described in Sec. In.7.0. Cards 2 to NUMGP of the sequence define the initial condition data of the additional generation points and are described in Sec. In.9.2. The final card of the sequence defines the nodal increment information and is identical to the one used for coordinate generation (see Sec. In.5.3). After the generation sequence is completed, additional nodal initial condition cards, or generation sequences, may follow.

   The generation may be performed along a line, over a surface, or within a volume. For additional information concerning these options see Note (2) of Sec. In.7.1.
3. The elements of the arrays D(NDOF,NUMNP), V(NDOF,NUMNP) and A(NDOF, NUMNP) are initialized to zero. If the initial condition data of node N is input and/or generated more than one time, the last value takes precedence.

In.9.2 Generation Point Initial Condition Cards (2I5,NDOF×F10.0)

Note	Columns	Variable	Description
(1)	1–5	M	Node number
	6–10	MGEN	Generation parameter EQ.0, initial condition data of the Jth generation point are input on this card; M is ignored. EQ.1, initial condition data of the Jth generation point are set equal to the initial condition data of the Mth node which were previously defined; initial condition data on this card are ignored.
	11–20	TEMP(1,J)	Degree of freedom 1 initial condition for generation point J
	21–30	TEMP(2,J)	Degree of freedom 2 initial condition for generation point J
	.	.	.
	.	.	.
	.	.	.
		TEMP(NDOF,J)	Degree of freedom NDOF initial condition for generation point J

Notes

1. The initial condition data of each generation point are defined by a generation point initial condition card. The cards must be read in order (J = 2,3,...,NUMGP) following the nodal initial condition card that initiated the generation sequence (J = 1). A nodal increments card (see Sec. In.5.3), which completes the sequence, follows the last generation point card (J = NUMGP).

In.9.3 Nodal Increments Card (6I5)

See the discussion in Sec. In.5.3.

In.10.0 ELEMENT DATA

In.10.1 Two-Dimensional, Isotropic Elasticity Element

The present element may be used in triangular (three-node) or quadrilateral (four-node) form for plane stress, plane strain, or torsionless axisymmetric analysis. Two displacement degrees of freedom, in the x_1- and x_2-directions, are assigned to each node. Therefore, the user must specify NSD .GE. 2 and NDOF .GE. 2 on the Execution Control Card (see Sec. In.2.0).

Stress and strain output is computed at each of the integration points. Printed results include stresses and strains in the global coordinate system, principal stresses and strains, maximum shear stress or strain in the x_1, x_2-plane and angle of inclination, in degrees, of principal states (see Fig. 11.5.7). Any one of these quantities may also be output as a time history. All shear strains are reported according to the "engineering" convention (i.e., *twice* the value of the tensor components). See Sec. In.10.1.6 for precise definitions of the output quantities.

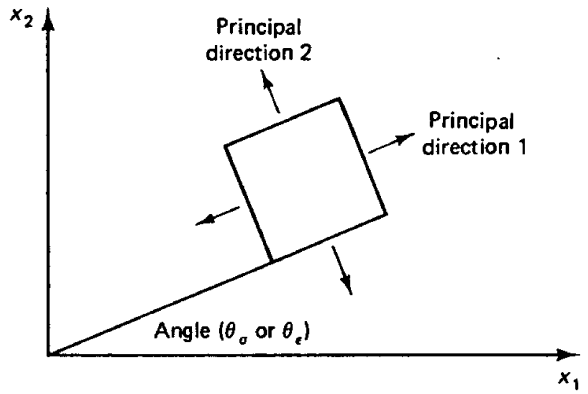


Figure 11.5.7 Orientation of principal axes.

The following sequence of cards is used to describe the elements in a group.

In.10.1.1 Element Group Control Card (13I5)

Note	Columns	Variable	Description
(1)	1–5	NTYPE	The number 1
	6–10	NUMEL	Number of elements in this group; GE. 1
	11–15	NUMAT	Number of material sets in this group; GE. 1
	16–20	NSURF	Number of surface force cards
(2)	21–25	NSOUT	Number of stress/strain time-history cards
	26–30	IOPT	Analysis option EQ.0, plane stress EQ.1, plane strain EQ.2, torsionless axisymmetric

Note	Columns	Variable	Description
(3)	31–35	ISTPRT	Stress output print code EQ.0, print stress output EQ.1, do not print stress output
(4)	36–40	LFSURF	Surface force load function number
(4)	41–45	LFBODY	Body force load function number
(5)	46–50	NICODE	Numerical integration code EQ.0, 2×2 Gaussian quadrature EQ.1, one-point Gaussian quadrature
(6)	51–55	IBBAR	Strain-displacement option EQ.0, standard formulation EQ.1, mean-dilatational $\bar{B}$ formulation
(7)	56–60	IMASS	Mass type code EQ.0, consistent mass matrix EQ.1, lumped mass matrix EQ.2, no mass matrix
	61–65	IMPEXP	Implicit/explicit code EQ.0, implicit element group EQ.1, explicit element group

Notes

1. All data on this card are stored in an array (NPAR(I),I=1,16):

$$\begin{aligned} &\text{NPAR(1)=NTYPE} \\ &\text{NPAR(2)=NUMEL} \\ &\vdots \\ &\text{NPAR(13)=IMPEXP} \end{aligned}$$

2. Output histories are ignored in static analysis as denoted by IACODE .EQ.1 on the Execution Control Card, Sec. In.2.0. See also Sec. In.10.1.6.
3. This option is useful, especially during dynamic analysis, for suppressing output from element groups in physically uninteresting regions of the mesh.
4. The user may specify any of the load-time functions defined in Sec. In.8.0.
5. The standard four-node quadrilateral employs 2 × 2 Gaussian quadrature. One-point Gaussian quadrature produces a dangerous element in that zero energy modes of deformation, so-called hourglass or keystone modes, may be present, resulting in a singular stiffness matrix. If enough displacement boundary conditions are present these modes may be eliminated. *However, one-point Gaussian quadrature should only be used if you know exactly what you are doing.* Internally, NINT denotes the number of integration points; NICODE .EQ. 0 ⇒ NINT .EQ. 4, and NICODE .EQ. 1 ⇒ NINT .EQ. 1.
6. The use of the mean-dilatational $\bar{B}$ strain-displacement matrix is effective in the analysis of nearly incompressible materials, where standard elements are susceptible to mesh "locking." Use *only* if IOPT .EQ. 1 or 2 (i.e., plane strain or axisymmetry).
7. IMASS .EQ. 2 will cause inertial and body forces to be neglected for this element group in both static and dynamic analysis.

In.10.1.2 Material Properties Cards (I5,5X,6F10.0)

Note	Columns	Variable	Description
(1)	1–5	M	Material identification number; GE. 1 and LE. NUMAT
	11–20	E	Young's modulus
(2)	21–30	POIS	Poisson's ratio
	31–40	RHO(M)	Mass density
(3)	41–50	RDAMPM(M)	Mass matrix Rayleigh damping coeffcient
(3)	51–60	RDAMPK(M)	Stiffness matrix Rayleigh damping coefficient
(4)	61–70	TH(M)	Thickness

Notes

1. Material property sets may be specified in any order. Note that E and POIS are not stored directly; rather they are used to compute the entries of the material stiffness matrix $\boldsymbol{C}$.
2. Poisson's ratio cannot be set equal to 0.5 since it results in division by zero. A value close to 0.5—for instance, 0.4999—can be employed for incompressible applications.
3. The element damping matrix is computed as:

$$\boldsymbol{c}^e = \text{RDAMPM(MAT(NEL))}*\boldsymbol{m}^e + \text{RDAMPK(MAT(NEL))}*\boldsymbol{k}^e.$$

4. For IOPT .EQ. 1 or 2, the thickness will default to 1.0.

In.10.1.3 Gravity Vector Card (2F10.0)

Note	Columns	Variable	Description
(1)	1–10	GRAV(1)	Component of gravity vector in x_1-direction
	11–20	GRAV(2)	Component of gravity vector in x_2-direction

Notes

1. These components will be scaled by the load-time function LFBODY when computing the element body force.

In.10.1.4 Element Definition

In.10.1.4.1 Nodal data cards (7I5)

Note	Columns	Variable	Description
(1)	1–5	N	Element number
	6–10	M	Element material number
(2)	11–15	IEN(1,N)	Number of first node

Note	Columns	Variable	Description
	16–20	IEN(2,N)	Number of second node
	21–25	IEN(3,N)	Number of third node
(3)	26–30	IEN(4,N)	Number of fourth node
(4)	31–35	NG	Generation parameter EQ.0, no generation EQ.1, generate data

Notes

1. All elements must be input on a nodal data card or generated. Each group begins with an element number 1. *Terminate with a blank card.*
2. Element nodes must be listed in counterclockwise order. See Fig. 11.5.8.
3. For triangular elements, set node number IEN (4,N) equal to IEN(3,N).
4. If the generation parameter is set to 1, an Element Generation Data Card must be input next.

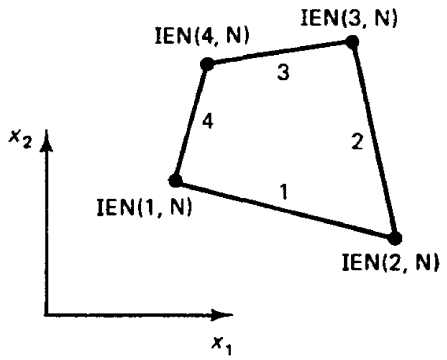


Figure 11.5.8 Nodal ordering for element number N.

In.10.1.4.2 Element generation data cards (6I5)

Note	Columns	Variable	Description
(1)	1–5	NEL(1)	Number of elements in direction 1; GE. 0 If EQ.0, set internally to 1.
	6–10	INCEL(1)	Element number increment for direction 1 If EQ.0, set internally to 1.
	11–15	INC(1)	Node number increment for direction 1 If EQ.0, set internally to 1.
	16–20	NEL(2)	Number of elements in direction 2; GE. 0 If EQ.0, set internally to 1.
	21–25	INCEL(2)	Element number increment for direction 2 If EQ.0, set internally to NEL(1).
	26–30	INC(2)	Node number increment for direction 2 If EQ.0, set internally to (NEL(1) + 1)*INC(1).

Notes

1. See Fig. 11.5.9 for a schematic representation of the generation algorithm.

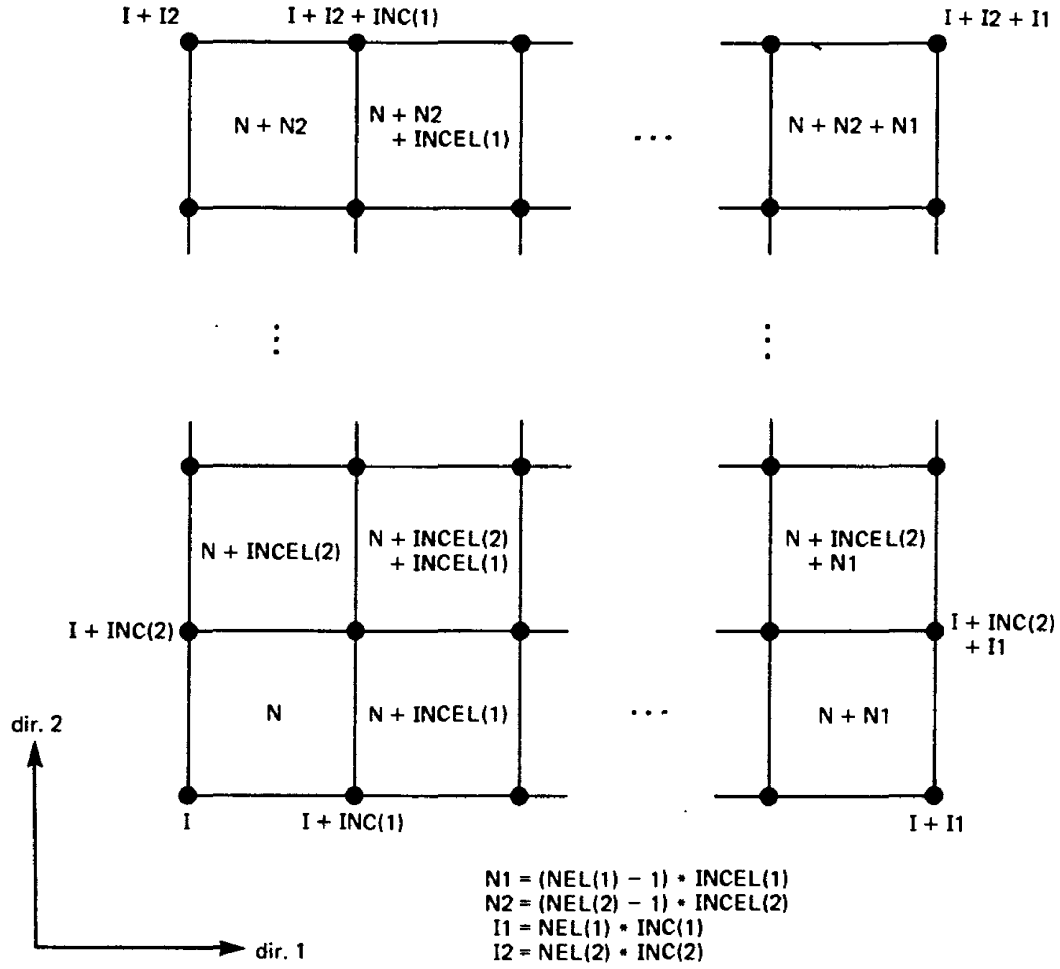


Figure 11.5.9 Element generation over a logically rectangular region.

In.10.1.5 Element Surface Load Cards (2I5,4F10.0)

Note	Columns	Variable	Description
(1)	1–5	IELNO(K)	Element number; GE. 1 and LE. NUMEL
(2)	6–10	ISIDE(K)	Element side number; GE. 1 and LE. 4
(3)	11–20	PRESS(1,K)	Pressure at node I
	21–30	PRESS(2,K)	Pressure at node J
(4)	31–40	SHEAR(1,K)	Shear stress at node I
	41–50	SHEAR(2,K)	Shear stress at node J

Notes

1. Each element side subjected to a surface load must be specified. If more than one side of the element is loaded, one card must be input for each loaded side. The index K in the arrays IELNO and ISIDE corresponds to the order in which the cards are read: 1 .LE. K .LE. NSURF. The cards need not be specified in any particular order. *The total number*

of Element Surface Load Cards must equal NSURF, *as specified on the Element Group Control Card* (see Sec. In.10.1.1).

2. The element side number is deduced as follows: Consider the element of Fig. 11.5.8. Let N = IELNO(K). Side 1 connects nodes IEN(1,N) and IEN(2,N), side 2 connects nodes IEN(2,N) and IEN(3,N), etc. For the purpose of element surface force specification, locally we refer to the node beginning the loaded side as I and the node ending the side as J.
3. The pressure is assumed to be positive pointing inward, and is linearly interpolated between nodal values; see Fig. 11.5.10. In computing the element surface forces, pressure is scaled by load-time function LFSURF.

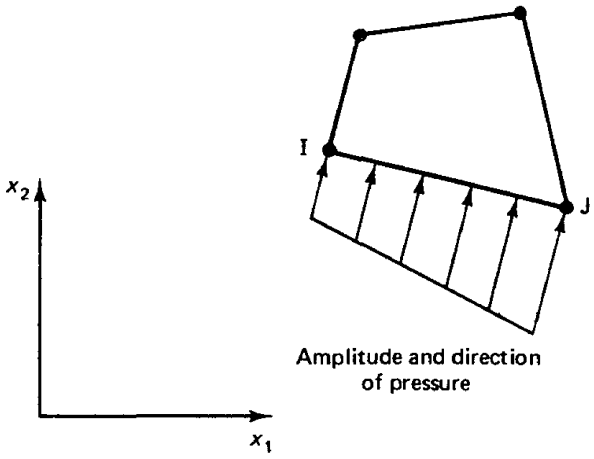


Figure 11.5.10

4. The shear stress is assumed to be positive pointing in a counterclockwise fashion and is linearly interpolated between nodal values; see Fig. 11.5.11. In computing the element surface forces, the shear stress is scaled by load-time function LFSURF.

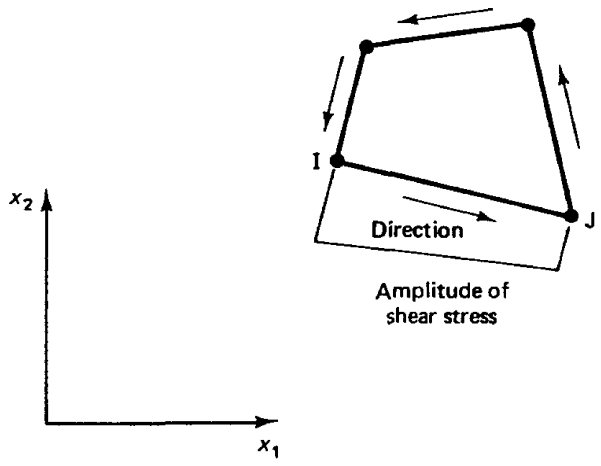


Figure 11.5.11

In.10.1.6 Element Time History Cards (3I5)

If the analysis is static (i.e., IACODE .EQ. 1 on the Execution Control Card; see Sec. In.2.0), then output history cards are read but subsequently ignored.

Note	Columns	Variable	Description
(1)	1–5	ISHIST(1,N)	Element number; GE. 1 and LE. NUMEL
(2)	6–10	ISHIST(2,N)	Quadrature point number; GE. 1 and LE. NINT
(3)	11–15	ISHIST(3,N)	Stress/strain component number, GE. 1 and LE. 16

Notes

1. Each desired element time history must be specified; thus an element for which the user requests multiple time histories will engender multiple data cards. The index N in the array ISHIST corresponds to the order in which the cards are read: 1 .LE. N .LE. NSOUT. The cards need not be specified in any particular order. *The total number of Element Time History Cards must equal* NSOUT *as specified on the Element Group Control Card* (see Sec. In.10.1.1).
2. NINT denotes the number of integration points; NICODE .EQ. 0 $\Rightarrow$ NINT .EQ. 4, and NICODE .EQ. 1 $\Rightarrow$ NINT .EQ. 1. In local coordinates (i.e., ξ, η), the quadrature point number defines the following locations:

NINT	l	$\tilde{\xi}_l$	$\tilde{\eta}_l$
1	1	0	0
4	1	$\frac{-1}{\sqrt{3}}$	$\frac{-1}{\sqrt{3}}$
	2	$\frac{1}{\sqrt{3}}$	$\frac{-1}{\sqrt{3}}$
	3	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$
	4	$\frac{-1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$

3. The following element components may be specified:

NCOMP	Description	Output Label
1	σ_{11}, stress 11	S 11
2	σ_{22}, stress 22	S 22
3	σ_{12}, stress 12	S 12
4	σ_{33}, stress 33	S 33
5	σ_1, principal stress 1	PS 1
6	σ_2, principal stress 2	PS 2
7	$\tau = \frac{1}{2}\lvert\sigma_1 - \sigma_2\rvert$, shear stress	TAU
8	θ_σ, stress angle (i.e., angle between x_1 and σ_1 axes)	SANG
9	ϵ_{11}, strain 11	E 11
10	ϵ_{22}, strain 22	E 22
11	$\gamma_{12} = 2\epsilon_{12}$ (i.e., engineering shear strain)	G 12
12	ϵ_{33}, strain 33	E 33

13	ϵ_1, principal strain 1	PE 1
14	ϵ_2, principal strain 2	PE 2
15	$\gamma = \lvert\epsilon_1 - \epsilon_2\rvert$, (i.e., engineering shear strain)	GAM
16	θ_ϵ, strain angle (i.e., angle between x_1 and ϵ_1 axes)	EANG

In.10.2 Three-Dimensional, Elastic Truss Element

The present element connects two or three nodes in space and transmits axial force only. There are three degrees of freedom at each node, i.e., the x_1, x_2, and x_3 translations. When employing truss elements, the user must specify NSD .EQ. 3 and NDOF .GE. 3 on the Execution Control Card (see Sec. In.2.0). The following sequence of cards is used to describe truss elements:

In.10.2.1 Element Group Control Card (10I5)

Note	Columns	Variable	Description
(1)	1–5	NTYPE	The number 2
	6–10	NUMEL	Number of elements in this group; GE. 1
	11–15	NUMAT	Number of material sets in this group; GE. 1
	16–20	NEN	Maximum number of nodes for all elements in this group EQ. 2 or 3
(2)	21–25	NSOUT	Number of stress/strain time-history cards
(3)	26–30	ISTPRT	Stress output print code EQ.0, print stress output EQ.1, do not print stress output
(4)	31–35	LFBODY	Body force load function number
(5)	36–40	NINT	Number of integration points EQ.1, one-point Gaussian quadrature EQ.2, two-point Gaussian quadrature EQ.3, three-point Gaussian quadrature
(6)	41–45	IMASS	Mass type code EQ.0, consistent mass matrix EQ.1, lumped mass matrix EQ.2, no mass matrix
	46–50	IMPEXP	Implicit/explicit code EQ.0, implicit element group EQ.1, explicit element group

Notes

1. All data on this card are stored in an array (NPAR(I),I = 1,16):

$$\begin{aligned} \text{NPAR(1)} &= \text{NTYPE} \\ \text{NPAR(2)} &= \text{NUMEL} \\ &\vdots \\ \text{NPAR(10)} &= \text{IMPEXP} \end{aligned}$$

2. Output histories are ignored in static analysis as denoted by IACODE .EQ. 1 on the Execution Control Card, Sec. In.2.0. See also Sec. In.10.2.5.
3. This option is useful, especially during dynamic analysis, for suppressing output from element groups in physically uninteresting regions of the mesh.
4. The user may specify any of the load-time functions defined in Sec. In.8.0.
5. The following facts may be used to guide the selection of an appropriate integration rule:

   For two-node truss elements, the stiffness is exactly integrated by the one-point Gauss rule and the mass is exactly integrated by the two-point Gauss rule.

   For three-node truss elements, if the location of the "middle node," x_3, satisfies $x_3 = (x_1 + x_2)/2$, then the stiffness is exactly integrated by the two-point Gauss rule and the mass is exactly integrated by the three-point Gauss rule.
6. IMASS .EQ. 2 will cause inertial and body forces to be neglected for this element group in both static and dynamic analysis.

In.10.2.2 Material Properties Cards (I5,5X,5F10.0)

Note	Columns	Variable	Description
(1)	1–5	M	Material identification number GE. 1 and LE. NUMAT
	11–20	E	Young's modulus
	21–30	RHO(M)	Mass density
(2)	31–40	RDAMPM(M)	Mass matrix Rayleigh damping coefficient
(2)	41–50	RDAMPK(M)	Stiffness matrix Rayleigh damping coefficient
	51–60	AREA(M)	Area

Notes

1. Material property sets may be specified in any order. Note that E is stored as the single entry in the material stiffness matrix $\boldsymbol{C}$.
2. The element damping matrix is computed as

$$\boldsymbol{c}^e = \text{RDAMPM(MAT(NEL))}*\boldsymbol{m}^e + \text{RDAMPK(MAT(NEL))}*\boldsymbol{k}^e.$$

In.10.2.3 Gravity Vector Card (3F10.0)

Note	Columns	Variable	Description
(1)	1–10	GRAV(1)	Component of gravity vector in x_1-direction
	11–20	GRAV(2)	Component of gravity vector in x_2-direction
	21–30	GRAV(3)	Component of gravity vector in x_3-direction

Notes

1. These components will be scaled by the load-time function LFBODY when computing the element body force.

In.10.2.4 Element Definition

In.10.2.4.1. Nodal Data Cards (6I5)

If NEN .EQ. 2, all nodal data cards in this group have the following form.

Note	Columns	Variable	Description
(1)	1–5	N	Element number
	6–10	M	Element material number
(2)	11–15	IEN(1,N)	Number of first node
	16–20	IEN(2,N)	Number of second node
(3)	21–25	NG	Generation parameter EQ.0, no generation EQ.1, generate data

If NEN .EQ. 3, all nodal data cards in this group have the following form.

Note	Columns	Variable	Description
(1)	1–5	N	Element number
	6–10	M	Element material number
(2)	11–15	IEN(1,N)	Number of first node
	16–20	IEN(2,N)	Number of second node
(4)	21–25	IEN(3,N)	Number of third node
(3)	26–30	NG	Generation parameter EQ.0, no generation EQ.1, generate data

Notes

1. All elements must be input on a nodal data card or generated. Each group begins with an element number 1. *Terminate with a blank card.*
2. See Fig. 11.5.12 for definition of nodal ordering.
3. If the generation parameter is set to 1, a generation data card must be input next.
4. Two-node elements may be defined within a group of three-node elements (i.e., NEN = 3) by setting IEN(3,N) equal to IEN(2,N).

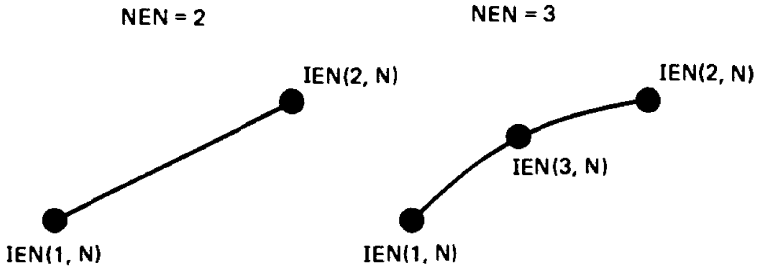


Figure 11.5.12 Nodal ordering for the truss element.

In.10.2.4.2 Element Generation Data Cards (3I5)

Note	Columns	Variable	Description
(1)	1–5	NEL(1)	Number of elements to be generated
	6–10	INCEL(1)	Element number increment If EQ.0, set internally to 1.
	11–15	INC(1)	Node number increment If EQ.0, set internally to 1.

Notes

1. See Fig. 11.5.13 for a schematic representation of the generation algorithm.

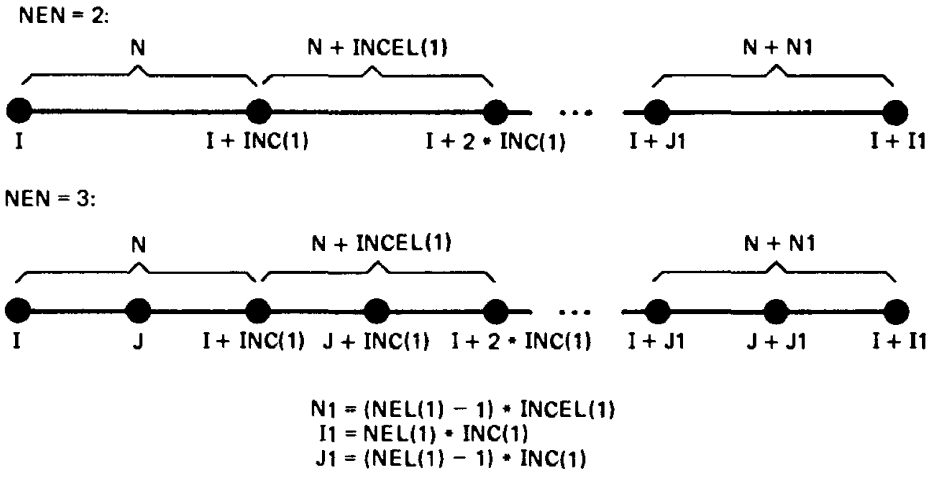


Figure 11.5.13 Generation of truss elements along a line.

In.10.2.5 Element Time History Cards (3I5)

If the analysis is static (i.e., IACODE .EQ. 1 on the Execution Control Card; see Sec. In.2.0), then output history cards are read, but subsequently ignored.

Note	Columns	Variable	Description
(1)	1–5	ISHIST(1,N)	Element number; GE. 1 and LE. NUMEL
(2)	6–10	ISHIST(2,N)	Quadrature point number; GE. 1 and LE. NINT
(3)	11–15	ISHIST(3,N)	Stress/strain component number, GE. 1 and LE. 3

Notes

1. Each desired element time history must be specified; thus an element for which the user requests multiple time histories will engender multiple data cards. The index N in the array ISHIST corresponds to the order in which the cards are read: 1 .LE. N .LE. NSOUT. The cards need not be specified in any particular order. *The total number of Element Time History Cards must equal* NSOUT, *as specified on the Element Group Control Card* (see Sec. In.10.2.1).
2. In the local coordinate (i.e., ξ), the quadrature point number corresponds to the following locations:

NINT	l	$\tilde{\xi}_l$
1	1	0
2	1	$\frac{-1}{\sqrt{3}}$
	2	$\frac{1}{\sqrt{3}}$
3	1	$\frac{-1}{\sqrt{3/5}}$
	2	$\frac{1}{\sqrt{3/5}}$
	3	0

3. The following element components may be specified:

NCOMP	Description	Output Label
1	Axial stress	STRS
2	Axial force	FORC
3	Axial strain	STRN

11.5.3 Examples

Example 1 (Planar Truss)

Here we consider a planar truss, shown schematically with node numbers in Fig. 11.5.14, subjected to multiple static loadings. Looking at the input file given in

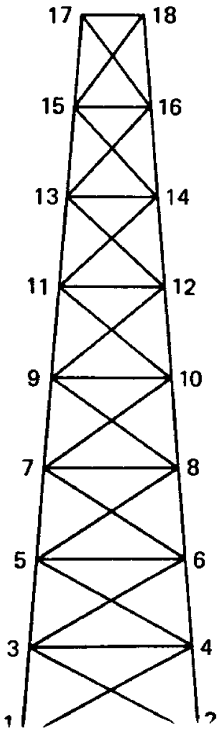


Table 11.5.1, note the use of generation capabilities, including the definition of the truss elements. In the time-sequence data we request three time steps and have set $\Delta t = 1.0$, although the code would default to this value. Two load vectors and load-time functions have been specified. The first load vector corresponds to a linearly varying lateral load, as might be used for preliminary seismic design. The second defines a uniform vertical load acting at every node. The load-time functions, shown schematically in Fig. 11.5.15, produce three different load cases combining varying proportions of the two load vectors. The output on pages 693 through 704 can be used to verify DLEARN upon installation on a new computer system.

TABLE 11.5.1 Input File for Planar Truss Problem

```
In 0.0      0
In 1.0    TRUSS TOWER WITH STATIC VERTICAL AND LATERAL LOADS
In 2.0        1     1     0     0     0     0     1     0     3    18     3     2     2     4     1
In 3.0        1     3     1     1     0     1          0.00          1.00          0.00          1.00
In 5.1        1     2                 0.                  0.
In 5.2       17     0                48.                768.
In 5.3        8     2
In 5.1        2     2               144.                  0.
In 5.2       18     0                96.                768.                                        Coordinates
In 5.3        8     2

              1    18     1     0     0     1
In 6.0        1     0     0     1     1     1
              2     0     0     1     1     1                                                        Boundary conditions

In 7.1        3     2         500.0           0.00          0.00
In 7.2       17     0        4000.0           0.00          0.00                                     Load vector 1
In 7.3        7     2

In 7.1        3     2           0.00       -3000.0          0.00
In 7.2       18     0           0.00       -3000.0          0.00                                     Load vector 2
In 7.3       15     1

                0.00            0.0
                1.00            1.0
                2.00            0.0                                                                 Load-time function 1
In 8.1          3.00            1.7
                0.00            0.0
                1.00            0.0                                                                 Load-time function 2
                2.00            1.0
                3.00            1.3
In 10.2.1     2    40     2     2     0     0     0     1     1     0
In 10.2.2     1           30000000.   7.30E-4          0.00          0.00         10.0
              2           30000000.   7.30E-4          0.00          0.00          6.0
In 10.2.3           0.00      -386.4          0.00
              1     1     1     3     1
              4     1     2                                                                                                  Element group
              5     1     2     4     1
              4     1     2
              9     1     3     4     1
              4     1     2
             13     2     1     4     1
              4     1     2
             17     2     2     3     1
              4     1     2
In 10.2.4.1  21     2     9    11     1                                                                 Element nodes
In 10.2.4.2   4     1     2
             25     2    10    12     1
              4     1     2
             29     2    11    12     1
              4     1     2
             33     2     9    12     1
              4     1     2
             37     2    10    11     1
              4     1     2
            *END
```

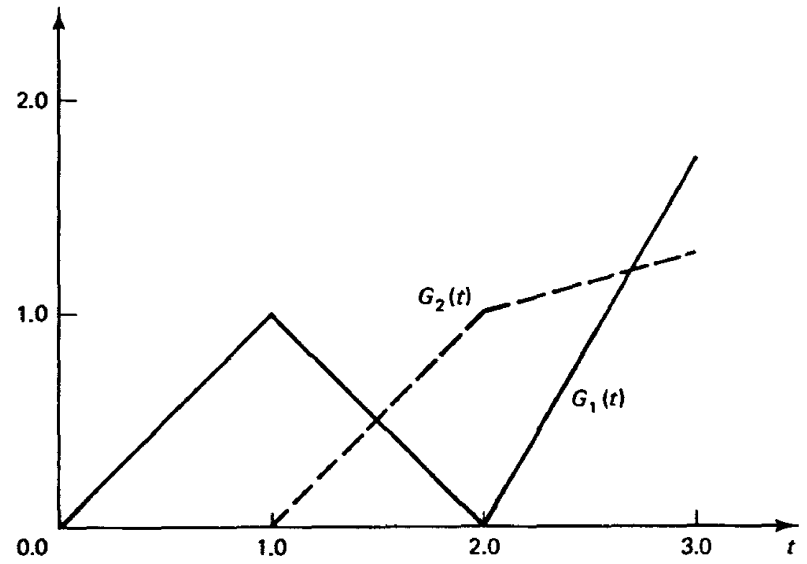


Figure 11.5.15 Load-time functions for planar truss problem.

```
TRUSS TOWER WITH STATIC VERTICAL AND LATERAL LOADS

E X E C U T I O N   C O N T R O L   I N F O R M A T I O N
     EXECUTION CODE . . . . . . . . . . . . . . . (IEXEC ) =

        EQ. 0, DATA CHECK
        EQ. 1, EXECUTION

     ANALYSIS CODE . . . . . . . . . . . . . . . . (IACODE) =

        EQ. 0, DYNAMIC ANALYSIS
        EQ. 1, STATIC ANALYSIS

     READ RESTART FILE CODE . . . . . . . . . . . (IREADR) =      0

        EQ. 0, DO NOT READ RESTART FILE
        EQ. 1, READ RESTART FILE

     WRITE RESTART FILE CODE . . . . . . . . . . . (IWRITR) =      0

        EQ. 0, DO NOT WRITE RESTART FILE
        EQ. 1, WRITE RESTART FILE

     INPUT DATA PRINT CODE . . . . . . . . . . . . (IPRTIN) =      0

        EQ. 0, PRINT NODAL AND ELEMENT INPUT DATA
        EQ. 1, DO NOT PRINT NODAL AND ELEMENT INPUT DATA

     RANK CHECK CODE . . . . . . . . . . . . . . . (IRANK ) =      0

        EQ. 0, DO NOT PERFORM RANK CHECK
        EQ. 1, PRINT NUMBERS OF ZERO AND NEGATIVE PIVOTS
        EQ. 2, PRINT ALL PIVOTS

     NUMBER OF TIME SEQUENCES . . . . . . . . . . (NUMSEQ) =      1

     NUMBER OF NODAL OUTPUT TIME-HISTORIES . . . (NDOUT ) =      0

     NUMBER OF SPACE DIMENSIONS . . . . . . . . . (NSD   ) =      3

     NUMBER OF NODAL POINTS . . . . . . . . . . . (NUMNP ) =     18

     NUMBER OF NODAL DEGREES-OF-FREEDOM . . . . (NDOF  ) =      3

     NUMBER OF LOAD VECTORS . . . . . . . . . . . (NLVECT) =      2

     NUMBER OF LOAD-TIME FUNCTIONS . . . . . . . (NLTFTN) =      2

     NUMBER OF POINTS ON LOAD-TIME FUNCTIONS . . (NPTSLF) =      4

     NUMBER OF ELEMENT GROUPS . . . . . . . . . . (NUMEG ) =      1
```

T I M E S E Q U E N C E D A T A

NUMBER OF TIME SEQUENCES . . . . . (NUMSEQ) = 1

TIME SEQUENCE NUMBER . . . . . (N) = 1
NUMBER OF TIME STEPS . . . . . (NSTEP(N)) = 3
KINEMATIC PRINT INCREMENT . . . . . (NDPRT(N)) = 1
STRESS/STRAIN PRINT INCREMENT . . . (NSPRT(N)) = 1
TIME HISTORY PLOT INCREMENT . . . . (NHPLT(N)) = 0
NUMBER OF ITERATIONS . . . . . (NITER(N)) = 1
FIRST INTEGRATION PARAMETER . . . . (ALPHA(N)) = 0.00000E+00
SECOND INTEGRATION PARAMETER . . . . (BETA(N)) = 1.00000E+00
THIRD INTEGRATION PARAMETER . . . . (GAMMA(N)) = 0.00000E+00
TIME STEP . . . . . (DT(N)) = 1.00000E+00

N O D A L C O O R D I N A T E D A T A

NODE NO.	X1	X2	X3
1	0.00000000E+00	0.00000000E+00	0.00000000E+00
2	1.44000000E+02	0.00000000E+00	0.00000000E+00
3	6.00000000E+00	9.60000000E+01	0.00000000E+00
4	1.38000000E+02	9.60000000E+01	0.00000000E+00
5	1.20000000E+01	1.92000000E+02	0.00000000E+00
6	1.32000000E+02	1.92000000E+02	0.00000000E+00
7	1.80000000E+01	2.88000000E+02	0.00000000E+00
8	1.26000000E+02	2.88000000E+02	0.00000000E+00
9	2.40000000E+01	3.84000000E+02	0.00000000E+00
10	1.20000000E+02	3.84000000E+02	0.00000000E+00
11	3.00000000E+01	4.80000000E+02	0.00000000E+00
12	1.14000000E+02	4.80000000E+02	0.00000000E+00
13	3.60000000E+01	5.76000000E+02	0.00000000E+00
14	1.08000000E+02	5.76000000E+02	0.00000000E+00
15	4.20000000E+01	6.72000000E+02	0.00000000E+00
16	1.02000000E+02	6.72000000E+02	0.00000000E+00
17	4.80000000E+01	7.68000000E+02	0.00000000E+00
18	9.60000000E+01	7.68000000E+02	0.00000000E+00

N O D A L B O U N D A R Y C O N D I T I O N C O D E S

NODE NO.	DOF1	DOF2	DOF3
1	1	1	1
2	1	1	1
3	0	0	1
4	0	0	1
5	0	0	1
6	0	0	1
7	0	0	1
8	0	0	1
9	0	0	1
10	0	0	1
11	0	0	1
12	0	0	1
13	0	0	1
14	0	0	1
15	0	0	1
16	0	0	1
17	0	0	1
18	0	0	1

P R E S C R I B E D F O R C E S A N D K I N E M A T I C
B O U N D A R Y C O N D I T I O N S

LOAD VECTOR NUMBER = 1

NODE NO.	DOF1	DOF2	DOF3
3	5.00000000E+02	0.00000000E+00	0.00000000E+00
5	1.00000000E+03	0.00000000E+00	0.00000000E+00
7	1.50000000E+03	0.00000000E+00	0.00000000E+00
9	2.00000000E+03	0.00000000E+00	0.00000000E+00
11	2.50000000E+03	0.00000000E+00	0.00000000E+00
13	3.00000000E+03	0.00000000E+00	0.00000000E+00
15	3.50000000E+03	0.00000000E+00	0.00000000E+00
17	4.00000000E+03	0.00000000E+00	0.00000000E+00

PRESCRIBED FORCES AND KINEMATIC BOUNDARY CONDITIONS

LOAD VECTOR NUMBER = 2

NODE NO.	DOF1	DOF2	DOF3
3	0.00000000E+00	-3.00000000E+03	0.00000000E+00
4	0.00000000E+00	-3.00000000E+03	0.00000000E+00
5	0.00000000E+00	-3.00000000E+03	0.00000000E+00
6	0.00000000E+00	-3.00000000E+03	0.00000000E+00
7	0.00000000E+00	-3.00000000E+03	0.00000000E+00
8	0.00000000E+00	-3.00000000E+03	0.00000000E+00
9	0.00000000E+00	-3.00000000E+03	0.00000000E+00
10	0.00000000E+00	-3.00000000E+03	0.00000000E+00
11	0.00000000E+00	-3.00000000E+03	0.00000000E+00
12	0.00000000E+00	-3.00000000E+03	0.00000000E+00
13	0.00000000E+00	-3.00000000E+03	0.00000000E+00
14	0.00000000E+00	-3.00000000E+03	0.00000000E+00
15	0.00000000E+00	-3.00000000E+03	0.00000000E+00
16	0.00000000E+00	-3.00000000E+03	0.00000000E+00
17	0.00000000E+00	-3.00000000E+03	0.00000000E+00
18	0.00000000E+00	-3.00000000E+03	0.00000000E+00

LOAD-TIME FUNCTION DATA

NUMBER OF LOAD-TIME FUNTIONS . . . . . (NLTFTN) = 2

FUNCTION NUMBER 1

TIME	LOAD FACTOR
0.00000000E+00	0.00000000E+00
1.00000000E+00	1.00000000E+00
2.00000000E+00	0.00000000E+00
3.00000000E+00	1.70000000E+00

FUNCTION NUMBER 2

TIME	LOAD FACTOR
0.00000000E+00	0.00000000E+00
1.00000000E+00	0.00000000E+00
2.00000000E+00	1.00000000E+00
3.00000000E+00	1.30000000E+00

ELEMENT GROUP DATA

ELEMENT GROUP NUMBER . . . . . (NEG) = 1

TWO/THREE-NODE TRUSS ELEMENTS

ELEMENT TYPE NUMBER . . . . . (NTYPE) = 2

NUMBER OF ELEMENTS . . . . . (NUMEL) = 40

NUMBER OF ELEMENT MATERIAL SETS . . . . . (NUMAT) = 2

NUMBER OF ELEMENT NODES . . . . . (NEN) = 2

NUMBER OF STRESS/STRAIN TIME HISTORIES . . (NSOUT) = 0

STRESS OUTPUT PRINT CODE . . . . . (ISTPRT) = 0

EQ.0, STRESS OUTPUT PRINTED
EQ.1, STRESS OUTPUT NOT PRINTED

BODY FORCE LOAD-TIME FUNCTION NUMBER . . . (LFBODY) = 0

INTEGRATION CODE . . . . . (NINT) = 1

EQ.1, 1-POINT GAUSSIAN QUADRATURE
EQ.2, 2-POINT GAUSSIAN QUADRATURE
EQ.3, 3-POINT GAUSSIAN QUADRATURE

M A T E R I A L S E T D A T A

NUMBER OF MATERIAL SETS . . . . . (NUMAT) = 2

SET NUMBER	YOUNG'S MODULUS	MASS DENSITY	MASS DAMPING	STIFFNESS DAMPING	AREA
1	3.0000E+07	7.3000E-04	0.0000E+00	0.0000E+00	1.0000E+01
2	3.0000E+07	7.3000E-04	0.0000E+00	0.0000E+00	6.0000E+00

G R A V I T Y V E C T O R C O M P O N E N T S

X-1 DIRECTION . . . . . = 0.00000000E+00

X-2 DIRECTION . . . . . = -3.86400000E+02

X-3 DIRECTION . . . . . = 0.00000000E+00

E L E M E N T D A T A

ELEMENT	MATERIAL	NODE 1	NODE 2
1	1	1	3
2	1	3	5
3	1	5	7
4	1	7	9
5	1	2	4
6	1	4	6
7	1	6	8
8	1	8	10
9	1	3	4
10	1	5	6
11	1	7	8
12	1	9	10
13	2	1	4
14	2	3	6
15	2	5	8
16	2	7	10
17	2	2	3
18	2	4	5
19	2	6	7
20	2	8	9
21	2	9	11
22	2	11	13
23	2	13	15
24	2	15	17
25	2	10	12
26	2	12	14
27	2	14	16
28	2	16	18
29	2	11	12
30	2	13	14
31	2	15	16
32	2	17	18
33	2	9	12
34	2	11	14
35	2	13	16
36	2	15	18
37	2	10	11
38	2	12	13
39	2	14	15
40	2	16	17

TRUSS TOWER WITH STATIC VERTICAL AND LATERAL LOADS

E Q U A T I O N S Y S T E M D A T A

NUMBER OF EQUATIONS	(NEQ) =	32
NUMBER OF TERMS IN LEFT-HAND-SIDE MATRIX .	(NALHS) =	192
MEAN HALF BANDWIDTH	(MEANBW) =	6
TOTAL LENGTH OF BLANK COMMON REQUIRED . . .	(NWORDS) =	2012

D I S P L A C E M E N T S

STEP NUMBER = 1

TIME = 1.000E+00

NODE NO.	DOF1	DOF2	DOF3
3	-3.55458742E-02	-3.80776188E-02	0.00000000E+00
4	-3.39701332E-02	2.82149641E-02	0.00000000E+00
5	-1.18072704E-01	-6.85865605E-02	0.00000000E+00
6	-1.16727240E-01	4.99146888E-02	0.00000000E+00
7	-2.46087075E-01	-9.10080677E-02	0.00000000E+00
8	-2.44821303E-01	6.49165276E-02	0.00000000E+00
9	-4.15844342E-01	-1.05185369E-01	0.00000000E+00
10	-4.14547637E-01	7.29927551E-02	0.00000000E+00
11	-6.33329338E-01	-1.22704195E-01	0.00000000E+00
12	-6.31285639E-01	8.31843360E-02	0.00000000E+00
13	-9.00366267E-01	-1.27236953E-01	0.00000000E+00
14	-8.98780674E-01	8.21569828E-02	0.00000000E+00
15	-1.20224430E+00	-1.20066166E-01	0.00000000E+00
16	-1.20101352E+00	7.13913000E-02	0.00000000E+00
17	-1.52011008E+00	-1.04034181E-01	0.00000000E+00
18	-1.51917308E+00	5.35084016E-02	0.00000000E+00

E N T S T R E S S E S A N D S T R A I N S

ENT GROUP NUMBER . . . . . (NEG) = 1

INT. PT. NUMBER	X1 COORD.	X2 COORD.	X3 COORD.	STRESS	FORCE	STRAIN
1	3.00E+00	4.80E+01	0.00E+00	-1.25E+04	-1.25E+05	-4.18E-04
1	9.00E+00	1.44E+02	0.00E+00	-1.11E+04	-1.11E+05	-3.70E-04
1	1.50E+01	2.40E+02	0.00E+00	-9.47E+03	-9.47E+04	-3.16E-04
1	2.10E+01	3.36E+02	0.00E+00	-7.72E+03	-7.72E+04	-2.57E-04
1	1.41E+02	4.80E+01	0.00E+00	9.44E+03	9.44E+04	3.15E-04
1	1.35E+02	1.44E+02	0.00E+00	8.36E+03	8.36E+04	2.79E-04
1	1.29E+02	2.40E+02	0.00E+00	7.16E+03	7.16E+04	2.39E-04
1	1.23E+02	3.36E+02	0.00E+00	5.82E+03	5.82E+04	1.94E-04
1	7.20E+01	9.60E+01	0.00E+00	3.58E+02	3.58E+03	1.19E-05
1	7.20E+01	1.92E+02	0.00E+00	3.36E+02	3.36E+03	1.12E-05
1	7.20E+01	2.88E+02	0.00E+00	3.52E+02	3.52E+03	1.17E-05
1	7.20E+01	3.84E+02	0.00E+00	4.05E+02	4.05E+03	1.35E-05
1	6.90E+01	4.80E+01	0.00E+00	-2.10E+03	-1.26E+04	-7.00E-05
1	6.90E+01	1.44E+02	0.00E+00	-2.13E+03	-1.28E+04	-7.10E-05
1	6.90E+01	2.40E+02	0.00E+00	-2.21E+03	-1.32E+04	-7.35E-05
1	6.90E+01	3.36E+02	0.00E+00	-2.20E+03	-1.32E+04	-7.33E-05
1	7.50E+01	4.80E+01	0.00E+00	1.33E+03	7.96E+03	4.42E-05
1	7.50E+01	1.44E+02	0.00E+00	1.56E+03	9.35E+03	5.20E-05
1	7.50E+01	2.40E+02	0.00E+00	1.65E+03	9.87E+03	5.49E-05
1	7.50E+01	3.36E+02	0.00E+00	1.70E+03	1.02E+04	5.68E-05
1	2.70E+01	4.32E+02	0.00E+00	-9.68E+03	-5.81E+04	-3.23E-04
1	3.30E+01	5.28E+02	0.00E+00	-6.61E+03	-3.96E+04	-2.20E-04
1	3.90E+01	6.24E+02	0.00E+00	-3.64E+03	-2.18E+04	-1.21E-04
1	4.50E+01	7.20E+02	0.00E+00	-1.19E+03	-7.16E+03	-3.98E-05

E L E M E N T S T R E S S E S A N D S T R A I N S

ELEMENT GROUP NUMBER . . . . . (NEG) = 1

ELEMENT NUMBER	INT. PT. NUMBER	X1 COORD.	X2 COORD.	X3 COORD.	STRESS	FORCE	STRA
25	1	1.17E+02	4.32E+02	0.00E+00	7.39E+03	4.43E+04	2.46E
26	1	1.11E+02	5.28E+02	0.00E+00	4.88E+03	2.93E+04	1.63E
27	1	1.05E+02	6.24E+02	0.00E+00	2.53E+03	1.52E+04	8.43E
28	1	9.90E+01	7.20E+02	0.00E+00	6.23E+02	3.74E+03	2.08E
29	1	7.20E+01	4.80E+02	0.00E+00	7.30E+02	4.38E+03	2.43E
30	1	7.20E+01	5.76E+02	0.00E+00	6.61E+02	3.96E+03	2.20E
31	1	7.20E+01	6.72E+02	0.00E+00	6.15E+02	3.69E+03	2.05E
32	1	7.20E+01	7.68E+02	0.00E+00	5.86E+02	3.51E+03	1.95E
33	1	6.90E+01	4.32E+02	0.00E+00	-2.26E+03	-1.36E+04	-7.54E
34	1	6.90E+01	5.28E+02	0.00E+00	-2.04E+03	-1.22E+04	-6.79E
35	1	6.90E+01	6.24E+02	0.00E+00	-1.71E+03	-1.03E+04	-5.71E
36	1	6.90E+01	7.20E+02	0.00E+00	-1.12E+03	-6.69E+03	-3.72E
37	1	7.50E+01	4.32E+02	0.00E+00	1.57E+03	9.39E+03	5.22E
38	1	7.50E+01	5.28E+02	0.00E+00	1.54E+03	9.27E+03	5.15E
39	1	7.50E+01	6.24E+02	0.00E+00	1.36E+03	8.16E+03	4.53E
40	1	7.50E+01	7.20E+02	0.00E+00	9.65E+02	5.79E+03	3.22E

D I S P L A C E M E N T S

STEP NUMBER = 2

TIME = 2.000E+00

NODE NO.	DOF1	DOF2	DOF3
3	-1.03904457E-03	-7.03705120E-03	0.00000000E+00
4	1.03904457E-03	-7.03705120E-03	0.00000000E+00
5	-7.97810718E-04	-1.32947618E-02	0.00000000E+00
6	7.97810718E-04	-1.32947618E-02	0.00000000E+00
7	-6.72223927E-04	-1.85330487E-02	0.00000000E+00
8	6.72223927E-04	-1.85330487E-02	0.00000000E+00
9	-6.43972712E-04	-2.28047225E-02	0.00000000E+00
10	6.43972712E-04	-2.28047225E-02	0.00000000E+00
11	-9.38067044E-04	-2.78375422E-02	0.00000000E+00
12	9.38067044E-04	-2.78375422E-02	0.00000000E+00
13	-5.64200278E-04	-3.15277555E-02	0.00000000E+00
14	5.64200278E-04	-3.15277555E-02	0.00000000E+00
15	-2.80376309E-04	-3.38077792E-02	0.00000000E+00
16	2.80376309E-04	-3.38077792E-02	0.00000000E+00
17	-6.37576302E-05	-3.48593216E-02	0.00000000E+00
18	6.37576302E-05	-3.48593216E-02	0.00000000E+00

: N T S T R E S S E S A N D S T R A I N S

:NT GROUP NUMBER . . . . . (NEG) = 1

INT. PT. NUMBER	X1 COORD.	X2 COORD.	X3 COORD.	STRESS	FORCE	STRAIN
1	3.00E+00	4.80E+01	0.00E+00	-2.21E+03	-2.21E+04	-7.37E-05
1	9.00E+00	1.44E+02	0.00E+00	-1.94E+03	-1.94E+04	-6.48E-05
1	1.50E+01	2.40E+02	0.00E+00	-1.63E+03	-1.63E+04	-5.43E-05
1	2.10E+01	3.36E+02	0.00E+00	-1.33E+03	-1.33E+04	-4.43E-05
1	1.41E+02	4.80E+01	0.00E+00	-2.21E+03	-2.21E+04	-7.37E-05
1	1.35E+02	1.44E+02	0.00E+00	-1.94E+03	-1.94E+04	-6.48E-05
1	1.29E+02	2.40E+02	0.00E+00	-1.63E+03	-1.63E+04	-5.43E-05
1	1.23E+02	3.36E+02	0.00E+00	-1.33E+03	-1.33E+04	-4.43E-05
1	7.20E+01	9.60E+01	0.00E+00	4.72E+02	4.72E+03	1.57E-05
1	7.20E+01	1.92E+02	0.00E+00	3.99E+02	3.99E+03	1.33E-05
1	7.20E+01	2.88E+02	0.00E+00	3.73E+02	3.73E+03	1.24E-05
1	7.20E+01	3.84E+02	0.00E+00	4.02E+02	4.02E+03	1.34E-05
1	6.90E+01	4.80E+01	0.00E+00	-5.65E+02	-3.39E+03	-1.88E-05
1	6.90E+01	1.44E+02	0.00E+00	-4.42E+02	-2.65E+03	-1.47E-05
1	6.90E+01	2.40E+02	0.00E+00	-4.53E+02	-2.72E+03	-1.51E-05
1	6.90E+01	3.36E+02	0.00E+00	-4.22E+02	-2.53E+03	-1.41E-05
1	7.50E+01	4.80E+01	0.00E+00	-5.65E+02	-3.39E+03	-1.88E-05
1	7.50E+01	1.44E+02	0.00E+00	-4.42E+02	-2.65E+03	-1.47E-05
1	7.50E+01	2.40E+02	0.00E+00	-4.53E+02	-2.72E+03	-1.51E-05
1	7.50E+01	3.36E+02	0.00E+00	-4.22E+02	-2.53E+03	-1.41E-05
1	2.70E+01	4.32E+02	0.00E+00	-1.57E+03	-9.43E+03	-5.24E-05
1	3.30E+01	5.28E+02	0.00E+00	-1.14E+03	-6.85E+03	-3.80E-05
1	3.90E+01	6.24E+02	0.00E+00	-7.04E+02	-4.23E+03	-2.35E-05
1	4.50E+01	7.20E+02	0.00E+00	-3.23E+02	-1.94E+03	-1.08E-05

E L E M E N T S T R E S S E S A N D S T R A I N S

ELEMENT GROUP NUMBER . . . . . (NEG) = 1

ELEMENT NUMBER	INT. PT. NUMBER	X1 COORD.	X2 COORD.	X3 COORD.	STRESS	FORCE	STRAI
25	1	1.17E+02	4.32E+02	0.00E+00	-1.57E+03	-9.43E+03	-5.24E-
26	1	1.11E+02	5.28E+02	0.00E+00	-1.14E+03	-6.85E+03	-3.80E-
27	1	1.05E+02	6.24E+02	0.00E+00	-7.04E+02	-4.23E+03	-2.35E-
28	1	9.90E+01	7.20E+02	0.00E+00	-3.23E+02	-1.94E+03	-1.08E-
29	1	7.20E+01	4.80E+02	0.00E+00	6.70E+02	4.02E+03	2.23E-
30	1	7.20E+01	5.76E+02	0.00E+00	4.70E+02	2.82E+03	1.57E-
31	1	7.20E+01	6.72E+02	0.00E+00	2.80E+02	1.68E+03	9.35E-
32	1	7.20E+01	7.68E+02	0.00E+00	7.97E+01	4.78E+02	2.66E-
33	1	6.90E+01	4.32E+02	0.00E+00	-5.90E+02	-3.54E+03	-1.97E-
34	1	6.90E+01	5.28E+02	0.00E+00	-4.65E+02	-2.79E+03	-1.55E-
35	1	6.90E+01	6.24E+02	0.00E+00	-3.61E+02	-2.16E+03	-1.20E-
36	1	6.90E+01	7.20E+02	0.00E+00	-2.04E+02	-1.22E+03	-6.79E-
37	1	7.50E+01	4.32E+02	0.00E+00	-5.90E+02	-3.54E+03	-1.97E-
38	1	7.50E+01	5.28E+02	0.00E+00	-4.65E+02	-2.79E+03	-1.55E-
39	1	7.50E+01	6.24E+02	0.00E+00	-3.61E+02	-2.16E+03	-1.20E-
40	1	7.50E+01	7.20E+02	0.00E+00	-2.04E+02	-1.22E+03	-6.79E-

D I S P L A C E M E N T S

STEP NUMBER = 3

TIME = 3.000E+00

NODE NO.	DOF1	DOF2	DOF3
3	3.34677851E-02	2.40035164E-02	0.00000000E+00
4	3.60482223E-02	-4.22890665E-02	0.00000000E+00
5	1.16477083E-01	4.19970369E-02	0.00000000E+00
6	1.18322861E-01	-7.65042123E-02	0.00000000E+00
7	2.44742627E-01	5.39419704E-02	0.00000000E+00
8	2.46165751E-01	-1.01982625E-01	0.00000000E+00
9	4.14556396E-01	5.95759242E-02	0.00000000E+00
10	4.15835582E-01	-1.18602200E-01	0.00000000E+00
11	6.31453204E-01	6.70291111E-02	0.00000000E+00
12	6.33161773E-01	-1.38859420E-01	0.00000000E+00
13	8.99237866E-01	6.41814424E-02	0.00000000E+00
14	8.99909074E-01	-1.45212494E-01	0.00000000E+00
15	1.20168354E+00	5.24506079E-02	0.00000000E+00
16	1.20157427E+00	-1.39006859E-01	0.00000000E+00
17	1.51998256E+00	3.43155377E-02	0.00000000E+00
18	1.51930059E+00	-1.23227045E-01	0.00000000E+00

E M E N T S T R E S S E S A N D S T R A I N S

ELEMENT GROUP NUMBER . . . . . (NEG) = 1

MENT MBER	INT. PT. NUMBER	X1 COORD.	X2 COORD.	X3 COORD.	STRESS	FORCE	STRAIN
1	1	3.00E+00	4.80E+01	0.00E+00	8.12E+03	8.12E+04	2.71E-04
2	1	9.00E+00	1.44E+02	0.00E+00	7.22E+03	7.22E+04	2.41E-04
3	1	1.50E+01	2.40E+02	0.00E+00	6.21E+03	6.21E+04	2.07E-04
4	1	2.10E+01	3.36E+02	0.00E+00	5.06E+03	5.06E+04	1.69E-04
5	1	1.41E+02	4.80E+01	0.00E+00	-1.39E+04	-1.39E+05	-4.62E-04
6	1	1.35E+02	1.44E+02	0.00E+00	-1.23E+04	-1.23E+05	-4.08E-04
7	1	1.29E+02	2.40E+02	0.00E+00	-1.04E+04	-1.04E+05	-3.47E-04
8	1	1.23E+02	3.36E+02	0.00E+00	-8.47E+03	-8.47E+04	-2.82E-04
9	1	7.20E+01	9.60E+01	0.00E+00	5.86E+02	5.86E+03	1.95E-05
10	1	7.20E+01	1.92E+02	0.00E+00	4.61E+02	4.61E+03	1.54E-05
11	1	7.20E+01	2.88E+02	0.00E+00	3.95E+02	3.95E+03	1.32E-05
12	1	7.20E+01	3.84E+02	0.00E+00	4.00E+02	4.00E+03	1.33E-05
13	1	6.90E+01	4.80E+01	0.00E+00	9.71E+02	5.83E+03	3.24E-05
14	1	6.90E+01	1.44E+02	0.00E+00	1.25E+03	7.48E+03	4.16E-05
15	1	6.90E+01	2.40E+02	0.00E+00	1.30E+03	7.80E+03	4.33E-05
16	1	6.90E+01	3.36E+02	0.00E+00	1.36E+03	8.14E+03	4.52E-05
17	1	7.50E+01	4.80E+01	0.00E+00	-2.46E+03	-1.47E+04	-8.19E-05
18	1	7.50E+01	1.44E+02	0.00E+00	-2.44E+03	-1.47E+04	-8.14E-05
19	1	7.50E+01	2.40E+02	0.00E+00	-2.55E+03	-1.53E+04	-8.50E-05
20	1	7.50E+01	3.36E+02	0.00E+00	-2.55E+03	-1.53E+04	-8.49E-05
21	1	2.70E+01	4.32E+02	0.00E+00	6.54E+03	3.92E+04	2.18E-04
22	1	3.30E+01	5.28E+02	0.00E+00	4.32E+03	2.59E+04	1.44E-04
23	1	3.90E+01	6.24E+02	0.00E+00	2.23E+03	1.34E+04	7.44E-05
24	1	4.50E+01	7.20E+02	0.00E+00	5.47E+02	3.28E+03	1.82E-05

E L E M E N T S T R E S S E S A N D S T R A I N S

ELEMENT GROUP NUMBER . . . . . (NEG) = 1

ELEMENT NUMBER	INT. PT. NUMBER	X1 COORD.	X2 COORD.	X3 COORD.	STRESS	FORCE	STRAI
25	1	1.17E+02	4.32E+02	0.00E+00	-1.05E+04	-6.32E+04	-3.51E-
26	1	1.11E+02	5.28E+02	0.00E+00	-7.17E+03	-4.30E+04	-2.39E-
27	1	1.05E+02	6.24E+02	0.00E+00	-3.94E+03	-2.36E+04	-1.31E-
28	1	9.90E+01	7.20E+02	0.00E+00	-1.27E+03	-7.62E+03	-4.23E-
29	1	7.20E+01	4.80E+02	0.00E+00	6.10E+02	3.66E+03	2.03E-
30	1	7.20E+01	5.76E+02	0.00E+00	2.80E+02	1.68E+03	9.32E-
31	1	7.20E+01	6.72E+02	0.00E+00	-5.46E+01	-3.28E+02	-1.82E-
32	1	7.20E+01	7.68E+02	0.00E+00	-4.26E+02	-2.56E+03	-1.42E-
33	1	6.90E+01	4.32E+02	0.00E+00	1.08E+03	6.49E+03	3.61E-
34	1	6.90E+01	5.28E+02	0.00E+00	1.11E+03	6.64E+03	3.69E-
35	1	6.90E+01	6.24E+02	0.00E+00	9.91E+02	5.94E+03	3.30E-
36	1	6.90E+01	7.20E+02	0.00E+00	7.08E+02	4.25E+03	2.36E-
37	1	7.50E+01	4.32E+02	0.00E+00	-2.75E+03	-1.65E+04	-9.15E-
38	1	7.50E+01	5.28E+02	0.00E+00	-2.47E+03	-1.48E+04	-8.25E-
39	1	7.50E+01	6.24E+02	0.00E+00	-2.08E+03	-1.25E+04	-6.94E-
40	1	7.50E+01	7.20E+02	0.00E+00	-1.37E+03	-8.24E+03	-4.58E-

A M I C S T O R A G E A L L O C A T I O N I N F O R M A T I O N

ARRAY NO.	ARRAY	ADDRESS	DIM1	DIM2	DIM3	PREC.
1	NSTEP	1	1	0	0	1
2	NDPRT	3	1	0	0	1
3	NSPRT	5	1	0	0	1
4	NMPLT	7	1	0	0	1
5	NITER	9	1	0	0	1
6	ALPHA	11	1	0	0	2
7	BETA	13	1	0	0	2
8	GAMMA	15	1	0	0	2
9	DT	17	1	0	0	2
10	D	19	3	18	0	2
11	X	127	3	18	0	2
12	ID	235	3	18	0	1
13	F	289	3	18	2	2
14	G	505	4	2	2	2
15	G1	537	2	0	0	2
16	IDIAG	541	32	0	0	1
17	NGRP	573	1	0	0	1
*****	BEGIN ELEMENT GROUP NUMBER 1					
18	NPAR	575	16	0	0	1
19	MP	591	29	0	0	1
20	W	621	1	0	0	2
21	DET	623	1	0	0	2
22	SHL	625	2	2	1	2
23	SHG	633	2	2	1	2
24	XS	641	3	1	0	2
25	RHO	647	2	0	0	2
26	RDAMPM	651	2	0	0	2
27	RDAMPK	655	2	0	0	2
28	AREA	659	2	0	0	2
29	C	663	1	1	2	2
30	GRAV	667	3	0	0	2
31	IEN	673	2	40	0	1
32	MAT	753	40	0	0	1
33	LM	793	3	2	40	1
34	ELEFFM	1033	6	6	0	2
35	XL	1105	3	2	0	2
36	WORK	1117	16	0	0	2
37	B	1149	1	6	0	2
38	DMAT	1161	1	1	0	2
39	DB	1163	1	6	0	2
40	VL	1175	3	2	0	2
41	AL	1187	3	2	0	2
42	ELRESF	1199	6	0	0	2
43	DL	1211	3	2	0	2
44	STRAIN	1223	1	0	0	2
45	STRESS	1225	1	0	0	2
46	FORCE	1227	1	0	0	2
*****	END ELEMENT GROUP DATA					
47	ALHS	1229	192	0	0	2
48	BRHS	1613	32	0	0	2

TRUSS TOWER WITH STATIC VERTICAL AND LATERAL LOADS

E X E C U T I O N T I M I N G I N F O R M A T I O N

I N I T I A L I Z A T I O N P H A S E	=	1.578E+00
S O L U T I O N P H A S E	=	2.461E+00
FORMATION OF LEFT-HAND-SIDE MATRICES	=	1.797E-01
FACTORIZATIONS	=	1.953E-02
FORMATION OF RIGHT-HAND-SIDE VECTORS	=	5.859E-02
FORWARD REDUCTIONS/BACK SUBSTITUTIONS	=	1.953E-02
CALCULATION OF ELEMENT OUTPUT	=	1.781E+00
SUBTOTAL	=	2.059E+00

Example 2 (Static Analysis of a Plane Strain Cantilever Beam)

This problem illustrates the use of the four-node quadrilateral, elastic continuum element. The mathematical model and corresponding mesh are shown in Figs. 4.4.1 and 4.4.2. In Table 11.5.2 we present the input file needed for one of the static analyses discussed in Chapter 4. Notice that the nodes have been numbered in a manner that minimizes the bandwidth of the effective mass matrix M^* (i.e., the stiffness matrix K). Observe that the boundary condition codes are set to allow warping of the root section of the cantilever, which is consistent with the analytic solution. The single load vector specified represents the consistent nodal loads $f_a^e = \int N_a \ell \, d\Omega$ computed from the surface tractions defined in Chapter 4. In this example, the nodal loads were computed by hand. An alternative procedure is to use the element surface loads described in Sec. In.10.1.5. However, note that the element surface loads are restricted to piecewise linear variation. The parabolic variation of end shear cannot be exactly represented by *continuous* piecewise linear variation, but it can be exactly represented by *discontinuous* piecewise linear interpolation (see Sec. 1.15, Exercise 1).

TABLE 11.5.2 Input File for Static Analysis of Cantilever Beam

```
ln 0.0      0
ln 1.0      PLANE STRAIN CANTILEVER BEAM: STATIC ANALYSIS WITH CONSISTENT NODAL FORCES
ln 2.0          1     1     0     0     0     0     1     0     2    45     2     1     1     1     1
ln 3.0          1     1     1     1     0     1          0.0         0.0         0.0         1.0
ln 5.1          1     4           0.0           0.0
                                  0.0           0.0
ln 5.2                            0.0           2.0                  } Coordinates
                                  0.0           2.0
ln 5.3          8     5     4     1
                1                 1     1
ln 6.0          5                 1     0                            } Boundary conditions
                6    41     5     1     0
                2       -7.500E-01    0.17383
                3       -1.500E+00    0.13867
                4       -2.250E+00    0.08008
                5        0.000E+00    0.01465
               41        0.000E+00   -0.09280
ln 7.1         42        0.000E+00   -0.17383                         } Load vector
               43        0.000E+00   -0.13867
               44        0.000E+00   -0.08008
               45        0.000E+00   -0.01465
ln 8.1              0.0         1.0
ln 10.1.1       1    32     1     0     0     1     0     0     0     0     0     0     0
ln 10.1.2       1                1.00          0.300         0.0           0.0           0.0
ln 10.1.3                                                                                    } Element group
ln 10.1.4.1     1     1     1     6     7     2     1
ln 10.1.4.2     8     4     5     4     1     1                          } Element nodes
                *END
```

Example 3 (Dynamic Analysis of a Plane Strain Cantilever Beam)

This problem, like the previous one, employs the configuration illustrated in Figs. 4.4.1 and 4.4.2. However, here the beam is forced dynamically. The beam is assumed initially at rest and all loads, namely, those of the static linear elasticity solution, are applied instantaneously at $t = 0^+$. The following data were employed in this calculation:

$$P(t) = 1000H(t) \qquad \text{(load factor, see Sec. 4.4)}$$

$$\left.\begin{aligned} \lambda &= 1.2 \times 10^6 \\ \mu &= 0.8 \times 10^6 \end{aligned}\right\} \qquad \text{(Lamé parameters)}$$

$$\rho = 2.0 \times 10^{-3} \qquad \text{(density)}$$

where $H(t)$ denotes the Heaviside step function (i.e., $H(t) = 1$ if $t > 0$; otherwise, $H(t) = 0$).

The dynamic response is calculated with the Hilber-Hughes-Taylor α-method. The parameters employed are

$$\alpha = -0.1, \qquad \beta = 0.3025, \qquad \gamma = 0.60$$

which results in an implicit, second-order accurate, unconditionally stable algorithm with some high-frequency numerical dissipation. One hundred equal time steps of length $\Delta t = 2.5 \times 10^{-4}$ were used in the analysis. This amounts to about 50 time steps for the fundamental period* and ensures an accurate integration of the response in the fundamental mode. By comparing the input files of Tables 11.5.2 and 11.5.3, it may be seen that little additional information is required to specify the dynamic problem. The same load vector is utilized as in the static case, but it is now scaled by a factor of 1000 by way of a constant load-time function. Three blank lines are inserted to indicate zero kinematic initial conditions. Notice that setting NDPRT and NSPRT equal to zero minimizes the printed output. Time-history plots of the vertical displacement of the tip (i.e., node 45) and the σ_{11}-component of stress in element 20 are requested. The output file is presented on pages 707 through 715.

TABLE 11.5.3 Input File for Dynamic Analysis of a Plane Strain Cantilever Beam

```
ln 0.0      0
ln 1.0      PLANE STRAIN CANTILEVER BEAM: DYNAMIC ANALYSIS WITH ALPHA METHOD
ln 2.0          1     0     0     0     0     0     1     1     2    45     2     1     1     1     1
ln 3.0          1   100     0     0     2     1          -0.10       0.3025          0.60     2.50E-4
ln 4.0         45     2     1
ln 5.1          1     4                0.0                0.0                          } Coordinates
ln 5.2                                16.0                0.0                          }
                                      16.0                2.0                          }
                                       0.0                2.0                          }
ln 5.3          8     5     4     1                                                      }
ln 6.0          1                       1     1                                        } Boundary conditions
                5                       1     0                                        }
                6    41     5     1     0                                              }
ln 7.1          2          -7.500E-01          0.17383                                 } Load vector
                3          -1.500E+00          0.13867                                 }
                4          -2.250E+00          0.08008                                 }
                5           0.000E+00          0.01465                                 }
               41           0.000E+00         -0.09280                                 }
               42           0.000E+00         -0.17383                                 }
               43           0.000E+00         -0.13867                                 }
               44           0.000E+00         -0.08008                                 }
               45           0.000E+00         -0.01465                                 }
ln 8.1                0.0          1000.0                                             } Load-time function
ln 9.1                                                                                } Initial conditions
ln 10.1.1       1    32     1     0     1     1     0     0     0     0     0     0     0
ln 10.1.2       1           2.08E+6           0.30      0.0020          0.00          0.00
ln 10.1.3                                                                             } Element group
ln 10.1.4.1     1     1     1     6     7     2     1                                 } Element nodes
ln 10.1.4.2     8     4     5     4     1     1                                       }
ln 10.1.6      20     4     1                                                         } Element time-history
            *END
```

*The fundamental period can be estimated from classical beam theory:

$$T_1 = \frac{2\pi}{\omega_1} \cong \frac{2\pi}{1.875^2}\sqrt{\frac{ml^4}{EI}} = \frac{2\pi}{1.875^2}\sqrt{\frac{(0.008)(16)^4}{(2.08 \times 10^6)(5.3333)}} = 1.23 \times 10^{-2}$$

where E is Young's modulus, I is the cross-sectional moment of inertia, l is the length and m denotes the mass per unit length. (See W. C. Hurty and M. F. Rubinstein, *Dynamics of Structures*, p. 203. Englewood Cliffs, N.J.: Prentice-Hall, 1964.)

```
PLANE STRAIN CANTILEVER BEAM: DYNAMIC ANALYSIS WITH ALPHA METHOD

E X E C U T I O N   C O N T R O L   I N F O R M A T I O N
     EXECUTION CODE . . . . . . . . . . . . . . . (IEXEC ) =      1

        EQ. 0, DATA CHECK
        EQ. 1, EXECUTION

     ANALYSIS CODE . . . . . . . . . . . . . . . . (IACODE) =      0

        EQ. 0, DYNAMIC ANALYSIS
        EQ. 1, STATIC ANALYSIS

     READ RESTART FILE CODE . . . . . . . . . . . (IREADR) =      0

        EQ. 0, DO NOT READ RESTART FILE
        EQ. 1, READ RESTART FILE

     WRITE RESTART FILE CODE . . . . . . . . . . . (IWRITR) =      0

        EQ. 0, DO NOT WRITE RESTART FILE
        EQ. 1, WRITE RESTART FILE

     INPUT DATA PRINT CODE . . . . . . . . . . . . (IPRTIN) =      0

        EQ. 0, PRINT NODAL AND ELEMENT INPUT DATA
        EQ. 1, DO NOT PRINT NODAL AND ELEMENT INPUT DATA

     RANK CHECK CODE . . . . . . . . . . . . . . . (IRANK ) =      0

        EQ. 0, DO NOT PERFORM RANK CHECK
        EQ. 1, PRINT NUMBERS OF ZERO AND NEGATIVE PIVOTS
        EQ. 2, PRINT ALL PIVOTS

     NUMBER OF TIME SEQUENCES . . . . . . . . . . (NUMSEQ) =      1
     NUMBER OF NODAL OUTPUT TIME-HISTORIES . . . (NDOUT ) =      1
     NUMBER OF SPACE DIMENSIONS . . . . . . . . . (NSD   ) =      2
     NUMBER OF NODAL POINTS . . . . . . . . . . . (NUMNP ) =     45
     NUMBER OF NODAL DEGREES-OF-FREEDOM . . . . . (NDOF  ) =      2
     NUMBER OF LOAD VECTORS . . . . . . . . . . . (NLVECT) =      1
     NUMBER OF LOAD-TIME FUNCTIONS . . . . . . . . (NLTFTN) =      1
     NUMBER OF POINTS ON LOAD-TIME FUNCTIONS . . . (NPTSLF) =      1
     NUMBER OF ELEMENT GROUPS . . . . . . . . . . (NUMEG ) =      1


T I M E   S E Q U E N C E   D A T A
     NUMBER OF TIME SEQUENCES . . . . . . (NUMSEQ  ) =      1


     TIME SEQUENCE NUMBER . . . . . . . . (N       ) =      1
     NUMBER OF TIME STEPS . . . . . . . . (NSTEP(N)) =    100
     KINEMATIC PRINT INCREMENT . . . . . (NDPRT(N)) =      0
     STRESS/STRAIN PRINT INCREMENT . . . (NSPRT(N)) =      0
     TIME HISTORY PLOT INCREMENT . . . . (NHPLT(N)) =      2
     NUMBER OF ITERATIONS . . . . . . . . (NITER(N)) =      1
     FIRST INTEGRATION PARAMETER . . . . (ALPHA(N)) = -1.00000E-01
     SECOND INTEGRATION PARAMETER . . . . (BETA(N) ) =  3.02500E-01
     THIRD INTEGRATION PARAMETER . . . . (GAMMA(N)) =  6.00000E-01
     TIME STEP . . . . . . . . . . . . . (DT(N)    ) =  2.50000E-04


N O D A L   T I M E - H I S T O R Y   I N F O R M A T I O N
     NUMBER OF NODAL TIME HISTORIES . . . . . . (NDOUT ) =      1

       NODE       DOF      KINEMATIC
      NUMBER     NUMBER      TYPE
        45         2         DISP
```

N O D A L C O O R D I N A T E D A T A

NODE NO.	X1	X2
1	0.00000000E+00	0.00000000E+00
2	0.00000000E+00	5.00000000E-01
3	0.00000000E+00	1.00000000E+00
4	0.00000000E+00	1.50000000E+00
5	0.00000000E+00	2.00000000E+00
6	2.00000000E+00	0.00000000E+00
7	2.00000000E+00	5.00000000E-01
8	2.00000000E+00	1.00000000E+00
9	2.00000000E+00	1.50000000E+00
10	2.00000000E+00	2.00000000E+00
11	4.00000000E+00	0.00000000E+00
12	4.00000000E+00	5.00000000E-01
13	4.00000000E+00	1.00000000E+00
14	4.00000000E+00	1.50000000E+00
15	4.00000000E+00	2.00000000E+00
16	6.00000000E+00	0.00000000E+00
17	6.00000000E+00	5.00000000E-01
18	6.00000000E+00	1.00000000E+00
19	6.00000000E+00	1.50000000E+00
20	6.00000000E+00	2.00000000E+00
21	8.00000000E+00	0.00000000E+00
22	8.00000000E+00	5.00000000E-01
23	8.00000000E+00	1.00000000E+00
24	8.00000000E+00	1.50000000E+00
25	8.00000000E+00	2.00000000E+00
26	1.00000000E+01	0.00000000E+00
27	1.00000000E+01	5.00000000E-01
28	1.00000000E+01	1.00000000E+00
29	1.00000000E+01	1.50000000E+00
30	1.00000000E+01	2.00000000E+00
31	1.20000000E+01	0.00000000E+00
32	1.20000000E+01	5.00000000E-01
33	1.20000000E+01	1.00000000E+00
34	1.20000000E+01	1.50000000E+00
35	1.20000000E+01	2.00000000E+00
36	1.40000000E+01	0.00000000E+00
37	1.40000000E+01	5.00000000E-01
38	1.40000000E+01	1.00000000E+00
39	1.40000000E+01	1.50000000E+00
40	1.40000000E+01	2.00000000E+00
41	1.60000000E+01	0.00000000E+00
42	1.60000000E+01	5.00000000E-01
43	1.60000000E+01	1.00000000E+00
44	1.60000000E+01	1.50000000E+00
45	1.60000000E+01	2.00000000E+00

```
N O D A L   B O U N D A R Y   C O N D I T I O N   C O D E S

    NODE NO.          DOF1       DOF2
        1               1          1
        5               1          0
        6               1          0
       11               1          0
       16               1          0
       21               1          0
       26               1          0
       31               1          0
       36               1          0
       41               1          0


P R E S C R I B E D   F O R C E S   A N D   K I N E M A T I C
  B O U N D A R Y   C O N D I T I O N S

    LOAD VECTOR NUMBER =      1

    NODE NO.              DOF1                DOF2
        2          -7.50000000E-01     1.73830000E-01
        3          -1.50000000E+00     1.38670000E-01
        4          -2.25000000E+00     8.00800000E-02
        5           0.00000000E+00     1.46500000E-02
       41           0.00000000E+00    -9.28000000E-02
       42           0.00000000E+00    -1.73830000E-01
       43           0.00000000E+00    -1.38670000E-01
       44           0.00000000E+00    -8.00800000E-02
       45           0.00000000E+00    -1.46500000E-02


L O A D - T I M E   F U N C T I O N   D A T A
    NUMBER OF LOAD-TIME FUNTIONS . . . . . (NLTFTN  ) =      1


    FUNCTION NUMBER      1
              TIME                LOAD FACTOR
         0.00000000E+00       1.00000000E+03

THERE ARE NO NONZERO INITIAL DISPLACEMENTS

THERE ARE NO NONZERO INITIAL VELOCITIES

THERE ARE NO NONZERO INITIAL ACCELERATIONS
```

E L E M E N T G R O U P D A T A

ELEMENT GROUP NUMBER . . . . . (NEG) = 1

F O U R - N O D E Q U A D R I L A T E R A L E L E M E N T S

ELEMENT TYPE NUMBER . . . . . (NTYPE) = 1

NUMBER OF ELEMENTS . . . . . (NUMEL) = 32

NUMBER OF ELEMENT MATERIAL SETS . . . . . (NUMAT) = 1

NUMBER OF SURFACE FORCE CARDS . . . . . (NSURF) = 0

NUMBER OF STRESS/STRAIN TIME HISTORIES . . (NSOUT) = 1

ANALYSIS OPTION . . . . . (IOPT) = 1

EQ.0, PLANE STRESS
EQ.1, PLANE STRAIN
EQ.2, AXISYMMETRIC

STRESS OUTPUT PRINT CODE . . . . . (ISTPRT) = 0

EQ.0, STRESS OUTPUT PRINTED
EQ.1, STRESS OUTPUT NOT PRINTED

SURFACE FORCE LOAD-TIME FUNCTION NUMBER . . (LFSURF) = 0

BODY FORCE LOAD-TIME FUNCTION NUMBER . . . (LFBODY) = 0

NUMERICAL INTEGRATION CODE . . . . . (NICODE) = 0

EQ.0, 2 X 2 GAUSSIAN QUADRATURE
EQ.1, 1-POINT GAUSSIAN QUADRATURE

STRAIN-DISPLACEMENT OPTION . . . . . (IBBAR) = 0

EQ.0, STANDARD FORMULATION
EQ.1, B-BAR FORMULATION

MASS TYPE CODE . . . . . (IMASS) = 0

EQ.0, CONSISTENT MASS MATRIX
EQ.1, LUMPED MASS MATRIX
EQ.2, NO MASS MATRIX

IMPLICIT/EXPLICIT CODE . . . . . (IMPEXP) = 0

EQ.0, IMPLICIT ELEMENT GROUP
EQ.1, EXPLICIT ELEMENT GROUP

M A T E R I A L S E T D A T A

NUMBER OF MATERIAL SETS . . . . . (NUMAT) = 1

SET NUMBER	YOUNG'S MODULUS	POISSON'S RATIO	MASS DENSITY	MASS DAMPING	STIFFNESS DAMPING	THICKNESS
1	2.0800E+06	3.0000E-01	2.0000E-03	0.0000E+00	0.0000E+00	1.0000E+00

G R A V I T Y V E C T O R C O M P O N E N T S

X-1 DIRECTION . . . . . = 0.00000000E+00

X-2 DIRECTION . . . . . = 0.00000000E+00

E L E M E N T D A T A

ELEMENT	MATERIAL	NODE 1	NODE 2	NODE 3	NODE 4
1	1	1	6	7	2
2	1	2	7	8	3
3	1	3	8	9	4
4	1	4	9	10	5
5	1	6	11	12	7
6	1	7	12	13	8
7	1	8	13	14	9
8	1	9	14	15	10
9	1	11	16	17	12
10	1	12	17	18	13
11	1	13	18	19	14
12	1	14	19	20	15
13	1	16	21	22	17
14	1	17	22	23	18
15	1	18	23	24	19
16	1	19	24	25	20
17	1	21	26	27	22
18	1	22	27	28	23
19	1	23	28	29	24
20	1	24	29	30	25
21	1	26	31	32	27
22	1	27	32	33	28
23	1	28	33	34	29
24	1	29	34	35	30
25	1	31	36	37	32
26	1	32	37	38	33
27	1	33	38	39	34
28	1	34	39	40	35
29	1	36	41	42	37
30	1	37	42	43	38
31	1	38	43	44	39
32	1	39	44	45	40

E L E M E N T T I M E H I S T O R Y I N F O R M A T I O N

NUMBER OF STRESS/STRAIN TIME HISTORIES . . (NSOUT) = 1

ELEMENT NUMBER	INT PT NUMBER	COMPONENT
20	4	S 11

PLANE STRAIN CANTILEVER BEAM: DYNAMIC ANALYSIS WITH ALPHA METHOD

E Q U A T I O N S Y S T E M D A T A

NUMBER OF EQUATIONS . . . . . (NEQ) = 79

NUMBER OF TERMS IN LEFT-HAND-SIDE MATRIX . (NALHS) = 872

MEAN HALF BANDWIDTH . . . . . (MEANBW) = 11

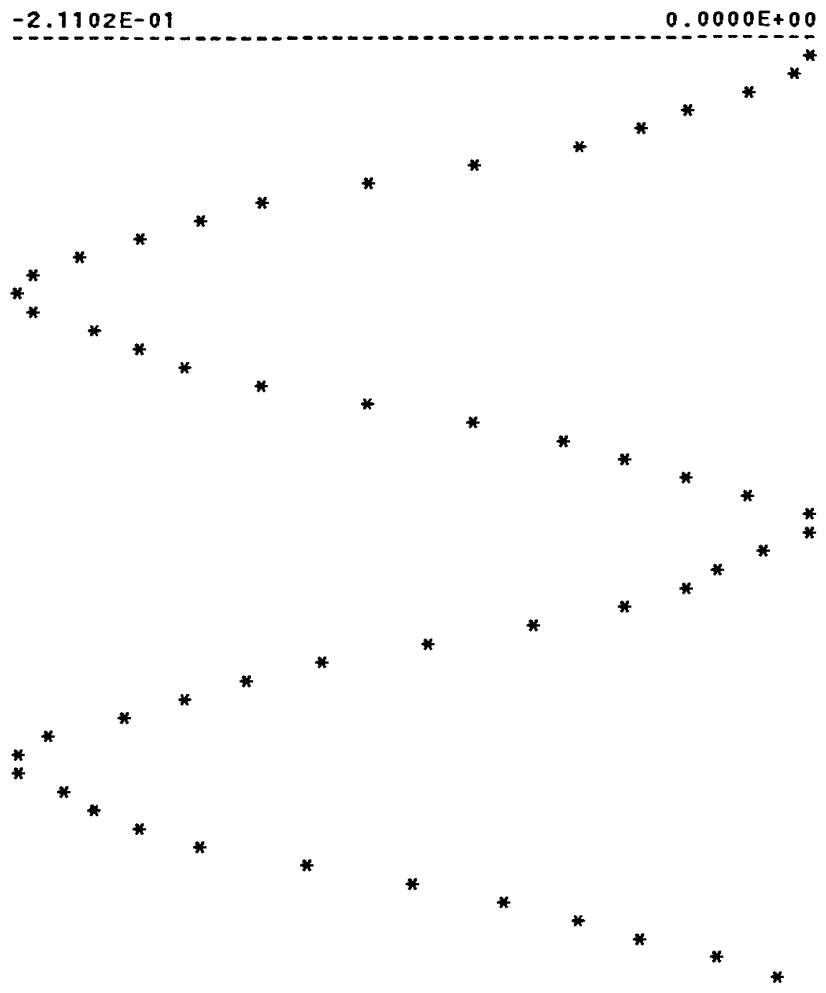
TOTAL LENGTH OF BLANK COMMON REQUIRED . . . (NWORDS) = 5196

PLANE STRAIN CANTILEVER BEAM: DYNAMIC ANALYSIS WITH ALPHA METHOD

NODE NUMBER = 45

DOF NUMBER = 2 OUTPUT: DISP

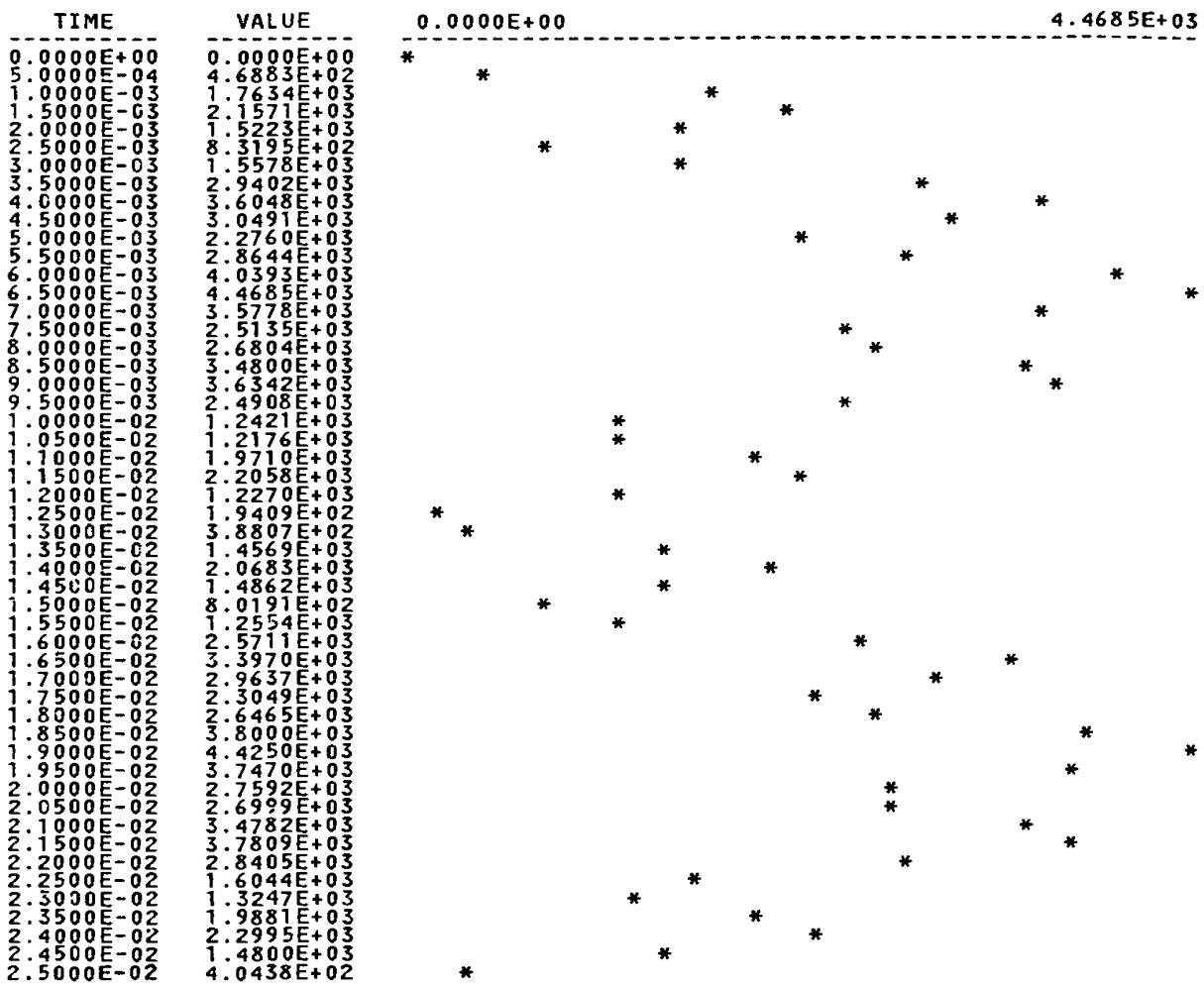
TIME	VALUE
0.0000E+00	0.0000E+00
5.0000E-04	-5.4078E-03
1.0000E-03	-1.8582E-02
1.5000E-03	-3.3184E-02
2.0000E-03	-4.6195E-02
2.5000E-03	-6.2725E-02
3.0000E-03	-8.8417E-02
3.5000E-03	-1.1853E-01
4.0000E-03	-1.4416E-01
4.5000E-03	-1.6146E-01
5.0000E-03	-1.7541E-01
5.5000E-03	-1.9146E-01
6.0000E-03	-2.0595E-01
6.5000E-03	-2.1102E-01
7.0000E-03	-2.0382E-01
7.5000E-03	-1.9046E-01
8.0000E-03	-1.7812E-01
8.5000E-03	-1.6520E-01
9.0000E-03	-1.4554E-01
9.5000E-03	-1.1748E-01
1.0000E-02	-8.8231E-02
1.0500E-02	-6.5911E-02
1.1000E-02	-4.9822E-02
1.1500E-02	-3.4154E-02
1.2000E-02	-1.6705E-02
1.2500E-02	-3.9318E-03
1.3000E-02	-2.8680E-03
1.3500E-02	-1.1958E-02
1.4000E-02	-2.3960E-02
1.4500E-02	-3.4801E-02
1.5000E-02	-4.9081E-02
1.5500E-02	-7.2129E-02
1.6000E-02	-1.0136E-01
1.6500E-02	-1.2838E-01
1.7000E-02	-1.4806E-01
1.7500E-02	-1.6424E-01
1.8000E-02	-1.8210E-01
1.8500E-02	-1.9972E-01
1.9000E-02	-2.0945E-01
1.9500E-02	-2.0710E-01
2.0000E-02	-1.9753E-01
2.0500E-02	-1.8743E-01
2.1000E-02	-1.7690E-01
2.1500E-02	-1.6005E-01
2.2000E-02	-1.3415E-01
2.2500E-02	-1.0517E-01
2.3000E-02	-8.0990E-02
2.3500E-02	-6.2825E-02
2.4000E-02	-4.5339E-02
2.4500E-02	-2.5752E-02
2.5000E-02	-9.3527E-03



PLANE STRAIN CANTILEVER BEAM: DYNAMIC ANALYSIS WITH ALPHA METHOD

ELEMENT NUMBER = 20

INTEGRATION POINT NUMBER = 4 OUTPUT: S 11



TIME	VALUE
0.0000E+00	0.0000E+00
5.0000E-04	4.6883E+02
1.0000E-03	1.7634E+03
1.5000E-03	2.1571E+03
2.0000E-03	1.5223E+03
2.5000E-03	8.3195E+02
3.0000E-03	1.5578E+03
3.5000E-03	2.9402E+03
4.0000E-03	3.6048E+03
4.5000E-03	3.0491E+03
5.0000E-03	2.2760E+03
5.5000E-03	2.8644E+03
6.0000E-03	4.0393E+03
6.5000E-03	4.4685E+03
7.0000E-03	3.5778E+03
7.5000E-03	2.5135E+03
8.0000E-03	2.6804E+03
8.5000E-03	3.4800E+03
9.0000E-03	3.6342E+03
9.5000E-03	2.4908E+03
1.0000E-02	1.2421E+03
1.0500E-02	1.2176E+03
1.1000E-02	1.9710E+03
1.1500E-02	2.2058E+03
1.2000E-02	1.2270E+03
1.2500E-02	1.9409E+02
1.3000E-02	3.8807E+02
1.3500E-02	1.4569E+03
1.4000E-02	2.0683E+03
1.4500E-02	1.4862E+03
1.5000E-02	8.0191E+02
1.5500E-02	1.2554E+03
1.6000E-02	2.5711E+03
1.6500E-02	3.3970E+03
1.7000E-02	2.9637E+03
1.7500E-02	2.3049E+03
1.8000E-02	2.6465E+03
1.8500E-02	3.8000E+03
1.9000E-02	4.4250E+03
1.9500E-02	3.7470E+03
2.0000E-02	2.7592E+03
2.0500E-02	2.6999E+03
2.1000E-02	3.4782E+03
2.1500E-02	3.7809E+03
2.2000E-02	2.8405E+03
2.2500E-02	1.6044E+03
2.3000E-02	1.3247E+03
2.3500E-02	1.9881E+03
2.4000E-02	2.2995E+03
2.4500E-02	1.4800E+03
2.5000E-02	4.0438E+02

IC STORAGE ALLOCATION INFORMATION

ARRAY NO.	ARRAY	ADDRESS	DIM1	DIM2	DIM3	PREC.
1	NSTEP	1	1	0	0	1
2	NDPRT	3	1	0	0	1
3	NSPRT	5	1	0	0	1
4	NHPLT	7	1	0	0	1
5	NITER	9	1	0	0	1
6	ALPHA	11	1	0	0	2
7	BETA	13	1	0	0	2
8	GAMMA	15	1	0	0	2
9	DT	17	1	0	0	2
10	IDHIST	19	3	1	0	1
11	DOUT	23	2	51	0	1
12	VPRED	125	2	45	0	2
13	DPRED	305	2	45	0	2
14	A	485	2	45	0	2
15	V	665	2	45	0	2
16	D	845	2	45	0	2
17	X	1025	2	45	0	2
18	ID	1205	2	45	0	1
19	F	1295	2	45	1	2
20	G	1475	1	2	1	2
21	G1	1479	1	0	0	2
22	IDIAG	1481	79	0	0	1
23	NGRP	1561	1	0	0	1
*****	BEGIN ELEMENT GROUP NUMBER 1					
24	NPAR	1563	16	0	0	1
25	MP	1579	35	0	0	1
26	N	1615	4	0	0	2
27	DET	1623	4	0	0	2
28	R	1631	4	0	0	2
29	SHL	1639	3	4	4	2
30	SHG	1735	3	4	4	2
31	SHGBAR	1831	3	4	1	2
32	RHO	1855	1	0	0	2
33	RDAMPM	1857	1	0	0	2
34	RDAMPK	1859	1	0	0	2
35	TH	1861	1	0	0	2
36	C	1863	4	4	1	2
37	GRAV	1895	2	0	0	2
38	IEN	1899	4	32	0	1
39	MAT	2027	32	0	0	1
40	LM	2059	2	4	32	1
41	IELNO	2315	0	0	0	1
42	ISIDE	2315	0	0	0	1
43	PRESS	2315	2	0	0	2
44	SHEAR	2319	2	0	0	2
45	ISHIST	2323	3	1	0	1
46	SOUT	2327	2	51	0	1
47	ELEFFM	2429	8	8	0	2
48	XL	2557	2	4	0	2
49	WORK	2573	16	0	0	2
50	B	2605	4	8	0	2

```
D Y N A M I C   S T O R A G E   A L L O C A T I O N   I N F O R M A T I O N

      ARRAY NO.      ARRAY        ADDRESS        DIM1        DIM2        DIM3        PREC.

           51        DMAT           2669           4           4           0           2
           52        DB             2701           4           8           0           2
           53        VL             2765           2           4           0           2
           54        AL             2781           2           4           0           2
           55        ELRESF         2797           8           0           0           2
           56        DL             2813           2           4           0           2
           57        STRAIN         2829           4           0           0           2
           58        STRESS         2837           4           0           0           2
           59        PSTRN          2845           4           0           0           2
           60        PSTRS          2853           4           0           0           2

        *****        END ELEMENT GROUP DATA

           61        ALHS           2861         872           0           0           2
           62        BRHS           4605          79           0           0           2

PLANE STRAIN CANTILEVER BEAM: DYNAMIC ANALYSIS WITH ALPHA METHOD

    E X E C U T I O N   T I M I N G   I N F O R M A T I O N

    I N I T I A L I Z A T I O N   P H A S E          =  1.340E+00

    S O L U T I O N   P H A S E                      =  9.855E+01

        FORMATION OF LEFT-HAND-SIDE MATRICES         =  1.891E+00
        FACTORIZATIONS                               =  1.289E-01
        FORMATION OF RIGHT-HAND-SIDE VECTORS         =  8.767E+01
        FORWARD REDUCTIONS/BACK SUBSTITUTIONS        =  3.773E+00
        CALCULATION OF ELEMENT OUTPUT                =  1.006E+00
                                                        ---------
        SUBTOTAL                                     =  9.447E+01
```

Example 4 (Implicit-Explicit Dynamic Analysis of a Rod)

The initial-value problem consists of a rod, modeled with two-node truss elements, fixed at the left end, and subjected to an initial displacement. The initial velocity and external forces are assumed equal to 0. The length of the rod is 10.5 and there are 21 equal-length elements. Young's modulus equals 100 in all elements except the last, in which it equals 10^7. The initial displacement varies linearly between 0 on the left end of the rod (i.e., $x = 0$) and 0.1 at $x = 10.0$ and is constant over the last element. The cross-sectional area of the rod is 1.0 and its density is 0.01. The implicit-explicit algorithm described in Sec. 9.4 is employed. The algorithmic parameters used are

$$\alpha = 0.0, \qquad \beta = 0.3025, \qquad \gamma = 0.60$$

The critical time step of explicit integration may be computed from (9.4.29), namely,

$$\omega^h \Delta t \leq \Omega_{\text{bif}} = \frac{2(1 - \xi)}{\gamma + \frac{1}{2}}$$

There is no viscous damping, so $\xi = 0$. Assuming lumped mass, the maximum element frequency is (see (9.2.1)):

$$\omega^h_{\max} = \frac{2c}{h} = \frac{2\sqrt{E/\rho}}{h}$$

Evaluating these formulas reveals that the stable critical time step for the last element is $\frac{1}{316}$ that of the first 20 elements. This suggests a mesh partition in which the last element is treated implicitly and the first 20 elements are treated explicitly, thus enabling the

larger critical time step to be employed. Specifically, $\Delta t = 4.5 \times 10^{-3}$ was used, and the analysis was run for 100 steps. Output histories are requested for displacement at the end of the rod (i.e., node 22) and stress in element 10. The initial displacements require that the initial accelerations be computed. This is facilitated by a one-step, "zero Δt" time sequence as described in Sec. In.3.0. The data for the analysis is presented in Table 11.5.4 and the output follows on pages 717 through 729.

Exercise 1. The addition of Rayleigh damping can significantly reduce critical time steps in explicit integration. This can be seen from (9.4.28):

$$\Omega_{\text{crit}} = \frac{(\xi^2 + 2\gamma)^{1/2} - \xi}{\gamma}$$

In the physically undamped case,

$$\Omega_{\text{crit}} = \sqrt{\frac{2}{\gamma}}$$

Estimate the fundamental mode of the rod from

$$\omega_1 = \frac{c\pi}{2L}$$

Select the stiffness-proportional damping factor RDAMPK such that the damping ratio in the fundamental mode, namely, ξ_1, is 10%. Determine the maximum damping factor from

$$\xi_{\max} = \tfrac{1}{2}(\text{RDAMPM}/\omega^h_{\max} + \text{RDAMPK} * \omega^h_{\max})$$

Consider the setup of Example 4. Due to *implicit* treatment of the single stiff element, it may be ignored in calculating $\omega^h_{\max}$, i.e., employ $E = 100.0$ and $h = 0.5$. Estimate ω_1 using $E = 100.0$ and $L = 10.0$. Assuming $\gamma = \frac{1}{2}$, calculate the reduction in critical time step compared with the physically undamped case, namely,

$$\frac{[(\xi^2_{\max} + 2\gamma)^{1/2} - \xi_{\max}]/\gamma}{\sqrt{2/\gamma}}$$

[*Ans.* 0.189]

Repeat the calculation assuming the rod is discretized by 210 elements of length 0.05.

TABLE 11.5.4 Input File for Implicit-Explicit Dynamic Analysis of a Rod

```
In 0.0      0
In 1.0  IMPLICIT-EXPLICIT DYNAMIC ANALYSIS OF A ROD
In 2.0      1     0     0     0     0     0     2     1     3    22     3     0     0     0     2
In 3.0      1     1     1     1     1     1           0.0          0.0          0.0          0.0
            2   100     0     0     2     1           0.0       0.3025          0.6       0.0045
In 4.0     22     1     1
In 5.1      1     2           0.0           0.0           0.0
In 5.2                      10.5           0.0           0.0
In 5.3     21     1
In 6.0      1     0     0     1     1     1
            2    22     1     0     1     1
In 9.1      1     2           0.0
In 9.2            0           0.1
In 9.3     20     1
           22     0           0.1
In 9.1
In 10.2.1   2     1     1     2     0     0     0     1     1     0
In 10.2.2   1           1.0E+7           0.01           0.0          0.0          1.0
In 10.2.3
In 10.2.4.1 1     1    21    22
In 10.2.1   2    20     1     2     1     0     0     1     1     1
In 10.2.2   1                100.0           0.01           0.0          0.0          1.0
In 10.2.3
In 10.2.4.1 1     1     1     2     1
In 10.2.4.2 20    1     1
In 10.2.5  10     1     1
           *END
```

Coordinates

Boundary conditions

Initial displacements

Initial velocities/accelerations

El. group 1: Element nodes

El. group 2: Element nodes; El. time-history

```
IMPLICIT-EXPLICIT DYNAMIC ANALYSIS OF A ROD

E X E C U T I O N   C O N T R O L   I N F O R M A T I O N

     EXECUTION CODE . . . . . (IEXEC ) =      1

        EQ. 0, DATA CHECK
        EQ. 1, EXECUTION

     ANALYSIS CODE . . . . . (IACODE) =      0

        EQ. 0, DYNAMIC ANALYSIS
        EQ. 1, STATIC ANALYSIS

     READ RESTART FILE CODE . . . . . (IREADR) =      0

        EQ. 0, DO NOT READ RESTART FILE
        EQ. 1, READ RESTART FILE

     WRITE RESTART FILE CODE . . . . . (IWRITR) =      0

        EQ. 0, DO NOT WRITE RESTART FILE
        EQ. 1, WRITE RESTART FILE

     INPUT DATA PRINT CODE . . . . . (IPRTIN) =      0

        EQ. 0, PRINT NODAL AND ELEMENT INPUT DATA
        EQ. 1, DO NOT PRINT NODAL AND ELEMENT INPUT DATA

     RANK CHECK CODE . . . . . (IRANK ) =      0

        EQ. 0, DO NOT PERFORM RANK CHECK
        EQ. 1, PRINT NUMBERS OF ZERO AND NEGATIVE PIVOTS
        EQ. 2, PRINT ALL PIVOTS

     NUMBER OF TIME SEQUENCES . . . . . (NUMSEQ) =      2
     NUMBER OF NODAL OUTPUT TIME-HISTORIES . . . (NDOUT ) =      1
     NUMBER OF SPACE DIMENSIONS . . . . . (NSD   ) =      3
     NUMBER OF NODAL POINTS . . . . . (NUMNP ) =     22
     NUMBER OF NODAL DEGREES-OF-FREEDOM . . . . (NDOF  ) =      3
     NUMBER OF LOAD VECTORS . . . . . (NLVECT) =      0
     NUMBER OF LOAD-TIME FUNCTIONS . . . . . (NLTFTN) =      0
     NUMBER OF POINTS ON LOAD-TIME FUNCTIONS . . (NPTSLF) =      0
     NUMBER OF ELEMENT GROUPS . . . . . (NUMEG ) =      2
```

```
E  S E Q U E N C E  D A T A
UMBER OF TIME SEQUENCES . . . . . (NUMSEQ  ) =     2

IME SEQUENCE NUMBER . . . . . (N       ) =     1
UMBER OF TIME STEPS . . . . . (NSTEP(N)) =     1
INEMATIC PRINT INCREMENT . . . . . (NDPRT(N)) =     1
TRESS/STRAIN PRINT INCREMENT . . . (NSPRT(N)) =     1
IME HISTORY PLOT INCREMENT . . . . (NHPLT(N)) =     1
UMBER OF ITERATIONS . . . . . (NITER(N)) =     1
IRST INTEGRATION PARAMETER . . . . (ALPHA(N)) =  0.00000E+00
ECOND INTEGRATION PARAMETER . . . . (BETA(N) ) =  0.00000E+00
HIRD INTEGRATION PARAMETER . . . . (GAMMA(N)) =  0.00000E+00
IME STEP . . . . . (DT(N)   ) =  0.00000E+00

IME SEQUENCE NUMBER . . . . . (N       ) =     2
UMBER OF TIME STEPS . . . . . (NSTEP(N)) =   100
INEMATIC PRINT INCREMENT . . . . . (NDPRT(N)) =     0
TRESS/STRAIN PRINT INCREMENT . . . (NSPRT(N)) =     0
IME HISTORY PLOT INCREMENT . . . . (NHPLT(N)) =     2
UMBER OF ITERATIONS . . . . . (NITER(N)) =     1
IRST INTEGRATION PARAMETER . . . . (ALPHA(N)) =  0.00000E+00
ECOND INTEGRATION PARAMETER . . . . (BETA(N) ) =  3.02500E-01
HIRD INTEGRATION PARAMETER . . . . (GAMMA(N)) =  6.00000E-01
IME STEP . . . . . (DT(N)   ) =  4.50000E-03

 A L  T I M E - H I S T O R Y  I N F O R M A T I O N
UMBER OF NODAL TIME HISTORIES . . . . . (NDOUT ) =     1

  NODE      DOF      KINEMATIC
 NUMBER    NUMBER      TYPE
   22         1        DISP
```

N O D A L C O O R D I N A T E D A T A

NODE NO.	X1	X2	X3
1	0.00000000E+00	0.00000000E+00	0.00000000E+00
2	5.00000000E-01	0.00000000E+00	0.00000000E+00
3	1.00000000E+00	0.00000000E+00	0.00000000E+00
4	1.50000000E+00	0.00000000E+00	0.00000000E+00
5	2.00000000E+00	0.00000000E+00	0.00000000E+00
6	2.50000000E+00	0.00000000E+00	0.00000000E+00
7	3.00000000E+00	0.00000000E+00	0.00000000E+00
8	3.50000000E+00	0.00000000E+00	0.00000000E+00
9	4.00000000E+00	0.00000000E+00	0.00000000E+00
10	4.50000000E+00	0.00000000E+00	0.00000000E+00
11	5.00000000E+00	0.00000000E+00	0.00000000E+00
12	5.50000000E+00	0.00000000E+00	0.00000000E+00
13	6.00000000E+00	0.00000000E+00	0.00000000E+00
14	6.50000000E+00	0.00000000E+00	0.00000000E+00
15	7.00000000E+00	0.00000000E+00	0.00000000E+00
16	7.50000000E+00	0.00000000E+00	0.00000000E+00
17	8.00000000E+00	0.00000000E+00	0.00000000E+00
18	8.50000000E+00	0.00000000E+00	0.00000000E+00
19	9.00000000E+00	0.00000000E+00	0.00000000E+00
20	9.50000000E+00	0.00000000E+00	0.00000000E+00
21	1.00000000E+01	0.00000000E+00	0.00000000E+00
22	1.05000000E+01	0.00000000E+00	0.00000000E+00

N O D A L B O U N D A R Y C O N D I T I O N C O D E S

NODE NO.	DOF1	DOF2	DOF3
1	1	1	1
2	0	1	1
3	0	1	1
4	0	1	1
5	0	1	1
6	0	1	1
7	0	1	1
8	0	1	1
9	0	1	1
10	0	1	1
11	0	1	1
12	0	1	1
13	0	1	1
14	0	1	1
15	0	1	1
16	0	1	1
17	0	1	1
18	0	1	1
19	0	1	1
20	0	1	1
21	0	1	1
22	0	1	1

```
I N I T I A L   D I S P L A C E M E N T S
     STEP NUMBER =          0
     TIME        =  0.000E+00

     NODE NO.          DOF1            DOF2            DOF3
          2      5.000000000E-03  0.000000000E+00  0.000000000E+00
          3      1.000000000E-02  0.000000000E+00  0.000000000E+00
          4      1.500000000E-02  0.000000000E+00  0.000000000E+00
          5      2.000000000E-02  0.000000000E+00  0.000000000E+00
          6      2.500000000E-02  0.000000000E+00  0.000000000E+00
          7      3.000000000E-02  0.000000000E+00  0.000000000E+00
          8      3.500000000E-02  0.000000000E+00  0.000000000E+00
          9      4.000000000E-02  0.000000000E+00  0.000000000E+00
         10      4.500000000E-02  0.000000000E+00  0.000000000E+00
         11      5.000000000E-02  0.000000000E+00  0.000000000E+00
         12      5.500000000E-02  0.000000000E+00  0.000000000E+00
         13      6.000000000E-02  0.000000000E+00  0.000000000E+00
         14      6.500000000E-02  0.000000000E+00  0.000000000E+00
         15      7.000000000E-02  0.000000000E+00  0.000000000E+00
         16      7.500000000E-02  0.000000000E+00  0.000000000E+00
         17      8.000000000E-02  0.000000000E+00  0.000000000E+00
         18      8.500000000E-02  0.000000000E+00  0.000000000E+00
         19      9.000000000E-02  0.000000000E+00  0.000000000E+00
         20      9.500000000E-02  0.000000000E+00  0.000000000E+00
         21      1.000000000E-01  0.000000000E+00  0.000000000E+00
         22      1.000000000E-01  0.000000000E+00  0.000000000E+00

THERE ARE NO NONZERO INITIAL VELOCITIES

THERE ARE NO NONZERO INITIAL ACCELERATIONS

E L E M E N T   G R O U P   D A T A
     ELEMENT GROUP NUMBER . . . . . (NEG   ) =      1

T W O / T H R E E - N O D E   T R U S S   E L E M E N T S
     ELEMENT TYPE NUMBER . . . . . (NTYPE ) =      2
     NUMBER OF ELEMENTS  . . . . . (NUMEL ) =      1
     NUMBER OF ELEMENT MATERIAL SETS . . . . . (NUMAT ) =      1
     NUMBER OF ELEMENT NODES . . . . . (NEN   ) =      2
     NUMBER OF STRESS/STRAIN TIME HISTORIES  . . (NSOUT ) =      0
```

```
      STRESS OUTPUT PRINT CODE . . . . . (ISTPRT) =      0

         EQ.0, STRESS OUTPUT PRINTED
         EQ.1, STRESS OUTPUT NOT PRINTED

      BODY FORCE LOAD-TIME FUNCTION NUMBER  . . . (LFBODY) =      0

      INTEGRATION CODE  . . . . . (NINT  ) =      1

         EQ.1, 1-POINT GAUSSIAN QUADRATURE
         EQ.2, 2-POINT GAUSSIAN QUADRATURE
         EQ.3, 3-POINT GAUSSIAN QUADRATURE


      MASS TYPE CODE  . . . . . (IMASS ) =      1

         EQ.0, CONSISTENT MASS MATRIX
         EQ.1, LUMPED MASS MATRIX
         EQ.2, NO MASS MATRIX

      IMPLICIT/EXPLICIT CODE  . . . . . (IMPEXP) =      0

         EQ.0, IMPLICIT ELEMENT GROUP
         EQ.1, EXPLICIT ELEMENT GROUP


M A T E R I A L   S E T   D A T A

      NUMBER OF MATERIAL SETS . . . . . (NUMAT ) =      1

       SET      YOUNG'S       MASS          MASS        STIFFNESS       AREA
      NUMBER    MODULUS      DENSITY       DAMPING      DAMPING

        1     1.0000E+07  1.0000E-02  0.0000E+00  0.0000E+00  1.0000E+00

G R A V I T Y   V E C T O R   C O M P O N E N T S

      X-1 DIRECTION . . . . . =  0.00000000E+00

      X-2 DIRECTION . . . . . =  0.00000000E+00

      X-3 DIRECTION . . . . . =  0.00000000E+00

E L E M E N T   D A T A

      ELEMENT    MATERIAL   NODE 1     NODE 2
          1          1         21         22
```

```
E L E M E N T   G R O U P   D A T A
    ELEMENT GROUP NUMBER . . . . . (NEG   ) =     2

T W O / T H R E E - N O D E   T R U S S   E L E M E N T S
    ELEMENT TYPE NUMBER . . . . . (NTYPE ) =     2
    NUMBER OF ELEMENTS . . . . . (NUMEL ) =    20
    NUMBER OF ELEMENT MATERIAL SETS . . . . . (NUMAT ) =     1
    NUMBER OF ELEMENT NODES . . . . . (NEN   ) =     2
    NUMBER OF STRESS/STRAIN TIME HISTORIES . . (NSOUT ) =     1
    STRESS OUTPUT PRINT CODE . . . . . (ISTPRT) =     0
       EQ.0, STRESS OUTPUT PRINTED
       EQ.1, STRESS OUTPUT NOT PRINTED
    BODY FORCE LOAD-TIME FUNCTION NUMBER . . . (LFBODY) =     0
    INTEGRATION CODE . . . . . (NINT  ) =     1
       EQ.1, 1-POINT GAUSSIAN QUADRATURE
       EQ.2, 2-POINT GAUSSIAN QUADRATURE
       EQ.3, 3-POINT GAUSSIAN QUADRATURE
    MASS TYPE CODE . . . . . (IMASS ) =     1
       EQ.0, CONSISTENT MASS MATRIX
       EQ.1, LUMPED MASS MATRIX
       EQ.2, NO MASS MATRIX
    IMPLICIT/EXPLICIT CODE . . . . . (IMPEXP) =     1
       EQ.0, IMPLICIT ELEMENT GROUP
       EQ.1, EXPLICIT ELEMENT GROUP

M A T E R I A L   S E T   D A T A
    NUMBER OF MATERIAL SETS . . . . . (NUMAT ) =     1

    SET      YOUNG'S       MASS          MASS        STIFFNESS       AREA
    NUMBER   MODULUS       DENSITY       DAMPING      DAMPING
      1    1.0000E+02  1.0000E-02  0.0000E+00  0.0000E+00  1.0000E+00
```

```
G R A V I T Y   V E C T O R   C O M P O N E N T S
    X-1 DIRECTION . . . . . . . . . . . . . . . =  0.00000000E+00
    X-2 DIRECTION . . . . . . . . . . . . . . . =  0.00000000E+00
    X-3 DIRECTION . . . . . . . . . . . . . . . =  0.00000000E+00

E L E M E N T   D A T A
    ELEMENT   MATERIAL  NODE 1    NODE 2
        1          1         1         2
        2          1         2         3
        3          1         3         4
        4          1         4         5
        5          1         5         6
        6          1         6         7
        7          1         7         8
        8          1         8         9
        9          1         9        10
       10          1        10        11
       11          1        11        12
       12          1        12        13
       13          1        13        14
       14          1        14        15
       15          1        15        16
       16          1        16        17
       17          1        17        18
       18          1        18        19
       19          1        19        20
       20          1        20        21

E L E M E N T   T I M E   H I S T O R Y   I N F O R M A T I O N
    NUMBER OF STRESS/STRAIN TIME HISTORIES  . . (NSOUT ) =     1

      ELEMENT   INT PT   COMPONENT
      NUMBER    NUMBER
        10         1       STRS

IMPLICIT-EXPLICIT DYNAMIC ANALYSIS OF A ROD

E Q U A T I O N   S Y S T E M   D A T A
    NUMBER OF EQUATIONS . . . . . . . . . . . . (NEQ   ) =       21
    NUMBER OF TERMS IN LEFT-HAND-SIDE MATRIX  . (NALHS ) =       22
    MEAN HALF BANDWIDTH . . . . . . . . . . . . (MEANBW) =        1
    TOTAL LENGTH OF BLANK COMMON REQUIRED . . . (NWORDS) =     2542
```

D I S P L A C E M E N T S

STEP NUMBER = 1

TIME = 0.000E+00

NODE NO.	DOF1	DOF2	DOF3
2	5.00000000E-03	0.00000000E+00	0.00000000E+00
3	1.00000000E-02	0.00000000E+00	0.00000000E+00
4	1.50000000E-02	0.00000000E+00	0.00000000E+00
5	2.00000000E-02	0.00000000E+00	0.00000000E+00
6	2.50000000E-02	0.00000000E+00	0.00000000E+00
7	3.00000000E-02	0.00000000E+00	0.00000000E+00
8	3.50000000E-02	0.00000000E+00	0.00000000E+00
9	4.00000000E-02	0.00000000E+00	0.00000000E+00
10	4.50000000E-02	0.00000000E+00	0.00000000E+00
11	5.00000000E-02	0.00000000E+00	0.00000000E+00
12	5.50000000E-02	0.00000000E+00	0.00000000E+00
13	6.00000000E-02	0.00000000E+00	0.00000000E+00
14	6.50000000E-02	0.00000000E+00	0.00000000E+00
15	7.00000000E-02	0.00000000E+00	0.00000000E+00
16	7.50000000E-02	0.00000000E+00	0.00000000E+00
17	8.00000000E-02	0.00000000E+00	0.00000000E+00
18	8.50000000E-02	0.00000000E+00	0.00000000E+00
19	9.00000000E-02	0.00000000E+00	0.00000000E+00
20	9.50000000E-02	0.00000000E+00	0.00000000E+00
21	1.00000000E-01	0.00000000E+00	0.00000000E+00
22	1.00000000E-01	0.00000000E+00	0.00000000E+00

V E L O C I T I E S

STEP NUMBER = 1

TIME = 0.000E+00

NODE NO.	DOF1	DOF2	DOF3

THERE ARE NO NONZERO COMPONENTS

A C C E L E R A T I O N S

STEP NUMBER = 1

TIME = 0.000E+00

NODE NO.	DOF1	DOF2	DOF3
3	-5.55111512E-15	0.00000000E+00	0.00000000E+00
4	5.55111512E-15	0.00000000E+00	0.00000000E+00
5	-2.22044605E-14	0.00000000E+00	0.00000000E+00
6	-1.94289029E-14	0.00000000E+00	0.00000000E+00
7	3.60822483E-14	0.00000000E+00	0.00000000E+00
8	-1.66533454E-14	0.00000000E+00	0.00000000E+00
9	-3.60822483E-14	0.00000000E+00	0.00000000E+00
10	6.93889390E-14	0.00000000E+00	0.00000000E+00
11	-1.02695630E-13	0.00000000E+00	0.00000000E+00
12	1.02695630E-13	0.00000000E+00	0.00000000E+00
13	-7.21644966E-14	0.00000000E+00	0.00000000E+00
14	7.21644966E-14	0.00000000E+00	0.00000000E+00
16	-6.93889390E-14	0.00000000E+00	0.00000000E+00
17	1.41553436E-13	0.00000000E+00	0.00000000E+00
18	-1.41553436E-13	0.00000000E+00	0.00000000E+00
19	6.93889390E-14	0.00000000E+00	0.00000000E+00
21	-2.00000000E+02	0.00000000E+00	0.00000000E+00
22	2.77555756E-08	0.00000000E+00	0.00000000E+00

E L E M E N T S T R E S S E S A N D S T R A I N S

ELEMENT GROUP NUMBER . . . . . (NEG) = 1

ELEMENT NUMBER	INT. PT. NUMBER	X1 COORD.	X2 COORD.	X3 COORD.	STRESS	FORCE	STRAIN
1	1	1.02E+01	0.00E+00	0.00E+00	-6.94E-11	-6.94E-11	-6.94E-18

E L E M E N T S T R E S S E S A N D S T R A I N S

ELEMENT GROUP NUMBER . . . . . (NEG) = 2

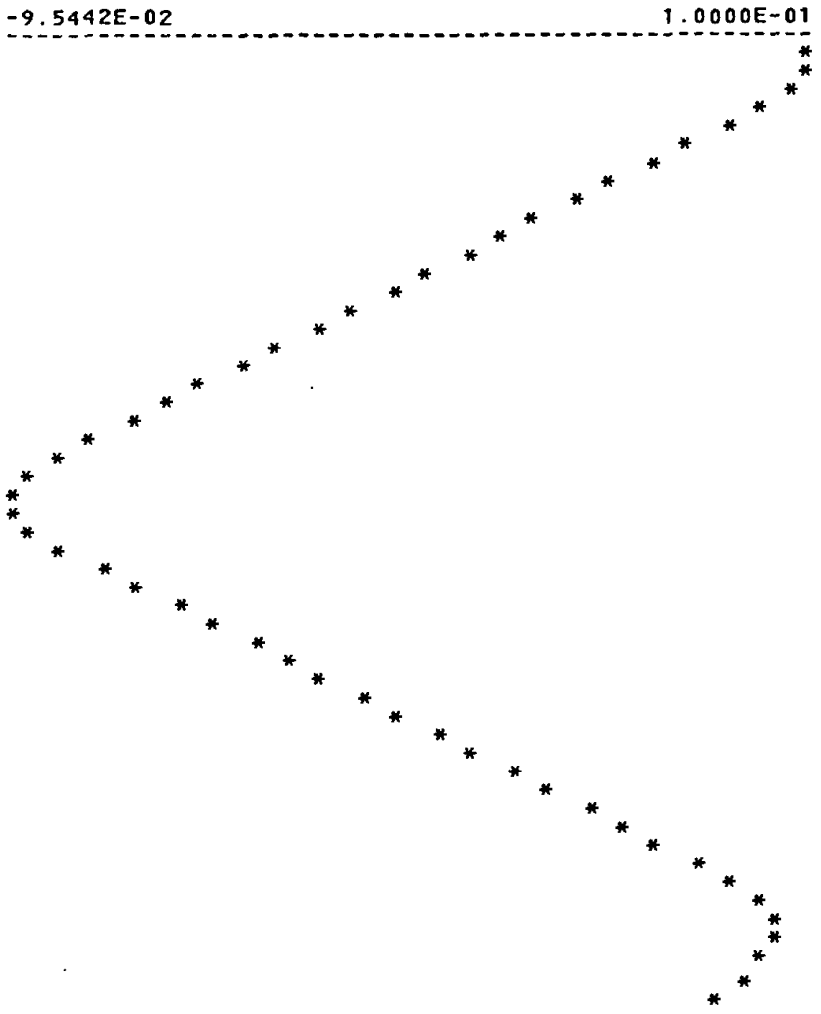
ELEMENT NUMBER	INT. PT. NUMBER	X1 COORD.	X2 COORD.	X3 COORD.	STRESS	FORCE	STRAIN
1	1	2.50E-01	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
2	1	7.50E-01	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
3	1	1.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
4	1	1.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
5	1	2.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
6	1	2.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
7	1	3.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
8	1	3.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
9	1	4.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
10	1	4.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
11	1	5.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
12	1	5.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
13	1	6.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
14	1	6.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
15	1	7.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
16	1	7.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
17	1	8.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
18	1	8.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
19	1	9.25E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02
20	1	9.75E+00	0.00E+00	0.00E+00	1.00E+00	1.00E+00	1.00E-02

IMPLICIT-EXPLICIT DYNAMIC ANALYSIS OF A ROD

NODE NUMBER = 22

DOF NUMBER = 1 OUTPUT: DISP

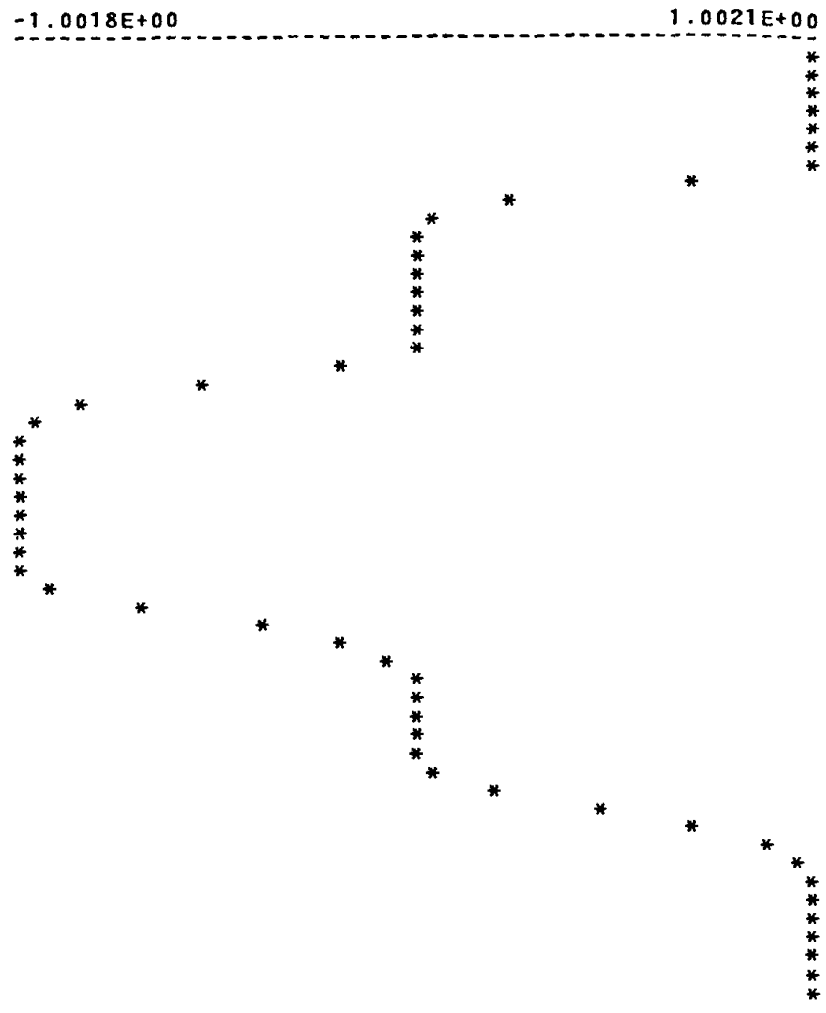
TIME	VALUE
0.0000E+00	1.0000E-01
0.0000E+00	1.0000E-01
9.0000E-03	9.5603E-02
1.8000E-02	8.7134E-02
2.7000E-02	7.8211E-02
3.6000E-02	6.9223E-02
4.5000E-02	6.0225E-02
5.4000E-02	5.1225E-02
6.3000E-02	4.2225E-02
7.2000E-02	3.3225E-02
8.1000E-02	2.4225E-02
9.0000E-02	1.5225E-02
9.9000E-02	6.2250E-03
1.0800E-01	-2.7750E-03
1.1700E-01	-1.1775E-02
1.2600E-01	-2.0775E-02
1.3500E-01	-2.9775E-02
1.4400E-01	-3.8775E-02
1.5300E-01	-4.7775E-02
1.6200E-01	-5.6775E-02
1.7100E-01	-6.5775E-02
1.8000E-01	-7.4774E-02
1.8900E-01	-8.3659E-02
1.9800E-01	-9.1395E-02
2.0700E-01	-9.5442E-02
2.1600E-01	-9.4162E-02
2.2500E-01	-8.8593E-02
2.3400E-01	-8.0782E-02
2.4300E-01	-7.2120E-02
2.5200E-01	-6.3202E-02
2.6100E-01	-5.4220E-02
2.7000E-01	-4.5224E-02
2.7900E-01	-3.6225E-02
2.8800E-01	-2.7225E-02
2.9700E-01	-1.8225E-02
3.0600E-01	-9.2250E-03
3.1500E-01	-2.2500E-04
3.2400E-01	8.7750E-03
3.3300E-01	1.7775E-02
3.4200E-01	2.6775E-02
3.5100E-01	3.5775E-02
3.6000E-01	4.4775E-02
3.6900E-01	5.3776E-02
3.7800E-01	6.2782E-02
3.8700E-01	7.1777E-02
3.9600E-01	8.0459E-02
4.0500E-01	8.7779E-02
4.1400E-01	9.2123E-02
4.2300E-01	9.2408E-02
4.3200E-01	8.8829E-02
4.4100E-01	8.2499E-02
4.5000E-01	7.4625E-02



IMPLICIT-EXPLICIT DYNAMIC ANALYSIS OF A ROD

ELEMENT NUMBER = 10
INTEGRATION POINT NUMBER = 1 OUTPUT: STRS

TIME	VALUE
0.0000E+00	1.0000E+00
0.0000E+00	1.0000E+00
9.0000E-03	1.0000E+00
1.8000E-02	1.0000E+00
2.7000E-02	1.0000E+00
3.6000E-02	1.0000E+00
4.5000E-02	9.8613E-01
5.4000E-02	6.9567E-01
6.3000E-02	2.1295E-01
7.2000E-02	3.7574E-02
8.1000E-02	5.5897E-03
9.0000E-02	8.4809E-04
9.9000E-02	1.4168E-04
1.0800E-01	2.8074E-05
1.1700E-01	6.3156E-06
1.2600E-01	8.1778E-07
1.3500E-01	-1.4638E-02
1.4400E-01	-1.7536E-01
1.5300E-01	-5.3637E-01
1.6200E-01	-8.3211E-01
1.7100E-01	-9.5730E-01
1.8000E-01	-9.9151E-01
1.8900E-01	-9.9855E-01
1.9800E-01	-9.9977E-01
2.0700E-01	-9.9996E-01
2.1600E-01	-9.9999E-01
2.2500E-01	-1.0000E+00
2.3400E-01	-1.0018E+00
2.4300E-01	-9.9973E-01
2.5200E-01	-9.2160E-01
2.6100E-01	-6.9174E-01
2.7000E-01	-3.9669E-01
2.7900E-01	-1.7502E-01
2.8800E-01	-6.1613E-02
2.9700E-01	-1.8053E-02
3.0600E-01	-4.5810E-03
3.1500E-01	-1.1041E-03
3.2400E-01	-1.2314E-03
3.3300E-01	1.1990E-03
3.4200E-01	4.6505E-02
3.5100E-01	1.9879E-01
3.6000E-01	4.4648E-01
3.6900E-01	6.9268E-01
3.7800E-01	8.6165E-01
3.8700E-01	9.4836E-01
3.9600E-01	9.8359E-01
4.0500E-01	9.9545E-01
4.1400E-01	9.9886E-01
4.2300E-01	9.9975E-01
4.3200E-01	1.0009E+00
4.4100E-01	1.0021E+00
4.5000E-01	9.8302E-01



IC STORAGE ALLOCATION INFORMATION

ARRAY NO.	ARRAY	ADDRESS	DIM1	DIM2	DIM3	PREC.
1	NSTEP	1	2	0	0	1
2	NDPRT	3	2	0	0	1
3	NSPRT	5	2	0	0	1
4	NHPLT	7	2	0	0	1
5	NITER	9	2	0	0	1
6	ALPHA	11	2	0	0	2
7	BETA	15	2	0	0	2
8	GAMMA	19	2	0	0	2
9	DT	23	2	0	0	2
10	IDHIST	27	3	1	0	1
11	DOUT	31	2	52	0	1
12	VPRED	135	3	22	0	2
13	DPRED	267	3	22	0	2
14	A	399	3	22	0	2
15	V	531	3	22	0	2
16	D	663	3	22	0	2
17	X	795	3	22	0	2
18	ID	927	3	22	0	1
19	IDIAG	993	21	0	0	1
20	NGRP	1015	2	0	0	1
*****	BEGIN ELEMENT GROUP NUMBER		1			
21	NPAR	1017	16	0	0	1
22	MP	1033	29	0	0	1
23	W	1063	1	0	0	2
24	DET	1065	1	0	0	2
25	SHL	1067	2	2	1	2
26	SHG	1075	2	2	1	2
27	XS	1083	3	1	0	2
28	RHO	1089	1	0	0	2
29	RDAMPM	1091	1	0	0	2
30	RDAMPK	1093	1	0	0	2
31	AREA	1095	1	0	0	2
32	C	1097	1	1	1	2
33	GRAV	1099	3	0	0	2
34	IEN	1105	2	1	0	1
35	MAT	1107	1	0	0	1
36	LM	1109	3	2	1	1
37	ELEFFM	1115	6	6	0	2
38	XL	1187	3	2	0	2
39	WORK	1199	16	0	0	2
40	B	1231	1	6	0	2
41	DMAT	1243	1	1	0	2
42	DB	1245	1	6	0	2
43	VL	1257	3	2	0	2
44	AL	1269	3	2	0	2
45	ELRESF	1281	6	0	0	2
46	DL	1293	3	2	0	2
47	STRAIN	1305	1	0	0	2
48	STRESS	1307	1	0	0	2
49	FORCE	1309	1	0	0	2
*****	BEGIN ELEMENT GROUP NUMBER		2			
50	NPAR	1311	16	0	0	1

```
D Y N A M I C   S T O R A G E   A L L O C A T I O N   I N F O R M A T I O N

   ARRAY NO.    ARRAY        ADDRESS      DIM1        DIM2        DIM3        PREC.
      51        MP           1327           29          0           0           1
      52        W            1357            1          0           0           2
      53        DET          1359            1          0           0           2
      54        SHL          1361            2          2           1           2
      55        SHG          1369            2          2           1           2
      56        XS           1377            3          1           0           2
      57        RHO          1383            1          0           0           2
      58        RDAMPM       1385            1          0           0           2
      59        RDAMPK       1387            1          0           0           2
      60        AREA         1389            1          0           0           2
      61        C            1391            1          1           1           2
      62        GRAV         1393            3          0           0           2
      63        IEN          1399            2         20           0           1
      64        MAT          1439           20          0           0           1
      65        LM           1459            3          2          20           1
      66        ISHIST       1579            3          1           0           1
      67        SOUT         1583            2         52           0           1
      68        ELEFFM       1687            6          6           0           2
      69        XL           1759            3          2           0           2
      70        WORK         1771           16          0           0           2
      71        B            1803            1          6           0           2
      72        DMAT         1815            1          1           0           2
      73        DB           1817            1          6           0           2
      74        VL           1829            3          2           0           2
      75        AL           1841            3          2           0           2
      76        ELRESF       1853            6          0           0           2
      77        DL           1865            3          2           0           2
      78        STRAIN       1877            1          0           0           2
      79        STRESS       1879            1          0           0           2
      80        FORCE        1881            1          0           0           2

   *****        END ELEMENT GROUP DATA

      81        ALHS         1883           22          0           0           2
      82        BRHS         1927           21          0           0           2


   IMPLICIT-EXPLICIT DYNAMIC ANALYSIS OF A ROD

        E X E C U T I O N   T I M I N G   I N F O R M A T I O N

        I N I T I A L I Z A T I O N   P H A S E        =  1.711E+00

        S O L U T I O N   P H A S E                    =  1.584E+01

            FORMATION OF LEFT-HAND-SIDE MATRICES       =  1.211E-01
            FACTORIZATIONS                             =  0.000E+00
            FORMATION OF RIGHT-HAND-SIDE VECTORS       =  1.122E+01
            FORWARD REDUCTIONS/BACK SUBSTITUTIONS      =  1.973E-01
            CALCULATION OF ELEMENT OUTPUT              =  4.473E-01
                                                          ---------
            SUBTOTAL                                   =  1.199E+01
```

11.5.4 Subroutine Index for Program Listing

Routine	Description	Page
MAIN	Program to set storage capacity, precision and input/output units	734
SUBROUTINE DLEARN	Global driver program	735
SUBROUTINE DRIVER	Solution driver program	737
SUBROUTINE ADDLHS	Program to add element left-hand-side matrix to global left-hand-side matrix	739
SUBROUTINE ADDRHS	Program to add element residual-force vector to global right-hand-side vector	740
SUBROUTINE BACK	Program to perform forward reduction and back substitution	740

11.5.4 Subroutine Index for Program Listing (cont.)

Routine	Description	Page
JBROUTINE BC	Program to read, generate and write boundary condition data and establish equation numbers	741
_OCK DATA	Program to define output labels and numerical constants	741
JBROUTINE BTDB	Program to multiply $B^T DB$, taking account of symmetry, and accumulate into element stiffness matrix	742
JBROUTINE CLEAR	Program to clear a floating-point array	742
JNCTION COLDOT	Program to compute the dot product of vectors stored column-wise	742
JBROUTINE COLHT	Program to compute column heights in global left-hand-side matrix	742
JBROUTINE COMPBC	Program to compute displacement, velocity and acceleration boundary conditions	743
JBROUTINE CONTM	Program to form mass matrix for a continuum element with NEN nodes	743
JBROUTINE CONTMA	Program to calculate inertial and gravity/body force $(-m^e * (a^e - g^e))$ for a continuum element with NEN nodes	745
JBROUTINE COORD	Program to read, generate, and write coordinate data	745
JBROUTINE CORRCT	Program to perform corrector update of displacements and velocities	746
JBROUTINE DCTNRY	Program to store pointer information in dictionary	746
JBROUTINE DHIST	Program to read, write, and store nodal time history input	746
JBROUTINE DIAG	Program to compute diagonal addresses of left-hand-side matrix	747
JBROUTINE DYNPTS	Program to set memory pointers for dynamic analysis data arrays	747
JBROUTINE ECHO	Program to echo input data	747
JBROUTINE ELEMNT	Program to calculate element task number	747
JBROUTINE ELCARD	Program to read element group control card	748
JBROUTINE ELMLIB	Program to call element routines	748
JBROUTINE EQSET	Program to allocate storage for global equation system	748
JBROUTINE FACTOR	Program to perform Crout factorization $A = U^T DU$	749
JBROUTINE FORMLM	Program to form LM array	750
JBROUTINE GENEL	Program to read and generate element node and material numbers	750
JBROUTINE GENEL1	Program to generate element node and material numbers	750
UBROUTINE GENELD	Program to set defaults for element node and material number generation	751
UBROUTINE GENELI	Program to increment element node numbers	751
UBROUTINE GENFL	Program to read and generate floating-point nodal data	751

11.5.4 Subroutine Index for Program Listing (cont.)

Routine	Description	Page
SUBROUTINE GENFL1	Program to generate floating-point nodal data via isoparametric interpolation	752
SUBROUTINE GENSH	Program to call shape function routines for isoparametric generation	753
SUBROUTINE GENSH1	Program to compute one-dimensional shape functions for isoparametric generation	753
SUBROUTINE GENSH2	Program to compute two-dimensional shape functions for isoparametric generation	753
SUBROUTINE GENSH3	Program to compute three-dimensional shape functions for isoparametric generation	754
SUBROUTINE HPLOT	Program to plot output histories	754
SUBROUTINE ICLEAR	Program to clear an integer array	755
SUBROUTINE IGEN	Program to read and generate integer nodal data	755
SUBROUTINE IMOVE	Program to move an integer array	756
SUBROUTINE INPUT	Program to read, generate and write nodal input data	756
SUBROUTINE INTERP	Program to perform linear interpolation	757
SUBROUTINE ITERUP	Program to perform intermediate update of displacements, velocities, and accelerations during iterative loop in predictor/corrector algorithm	757
SUBROUTINE LFAC	Program to compute load factors at time t	758
SUBROUTINE LOAD	Program to accumulate nodal forces and transfer into right-hand-side vector	758
SUBROUTINE LOCAL	Program to localize a global array	758
FUNCTION LOUT	Program to determine logical switch	758
SUBROUTINE LTIMEF	Program to read, write, and store load-time functions	758
SUBROUTINE MATADD	Program to add rectangular matrices	759
SUBROUTINE MEANSH	Program to calculate mean values of shape function global derivatives for mean-dilatational $\bar{B}$ formulation	760
SUBROUTINE MINMAX	Program to compute the min and max in the row of a matrix	760
SUBROUTINE MOVE	Program to move a floating-point array	760
FUNCTION MPOINT	Program to calculate storage pointer	760
SUBROUTINE MULTAB	Program to multiply two matrices	761
SUBROUTINE PIVOTS	Program to determine the numbers of zero and negative terms in diagonal of array D of Crout factorization $A = U^T DU$	761
SUBROUTINE PREDCT	Program to calculate predictor for displacements, velocities, and accelerations	762
SUBROUTINE PRINC	Program to compute principal values of symmetric second-rank tensor	762
SUBROUTINE PRINTD	Program to print kinematic data	763
SUBROUTINE PRINTF	Program to print prescribed force and boundary condition data	763

11.5.4 Subroutine Index for Program Listing (cont.)

Routine	Description	Page
BROUTINE PRINTP	Program to print diagonal of array D after Crout factorization $A = U^T DU$	764
BROUTINE PRNTEL	Program to print data for element with NEN nodes	764
BROUTINE PROP2D	Program to read, write and store properties for two-dimensional continuum elements	764
BROUTINE PRTDC	Program to print memory-pointer dictionary	765
BROUTINE PRTDC1	Program to print memory-pointer information for an array	765
BROUTINE PRTS2D	Program to print stress, strain, and principal values for two-dimensional continuum elements	766
NCTION RCDOT	Program to compute the dot product of a vector stored row-wise with a vector stored column-wise	766
NCTION ROWDOT	Program to compute the dot product of vectors stored row-wise	766
BROUTINE RSIN	Program to read restart file	767
BROUTINE RSOUT	Program to write restart file	767
BROUTINE SERROR	Program to print error message if available storage is exceeded	767
BROUTINE SETUPD	Program to calculate the D matrix	767
BROUTINE SHIST	Program to read, write, and store element time-history input data	768
BROUTINE SMULT	Program to perform scalar multiplication of a matrix	768
BROUTINE STATIN	Program to set memory pointers for static analysis data arrays and call associated input routines	768
BROUTINE STORED	Program to store nodal time histories as single-precision data	769
BROUTINE TIMCON	Program to compute current time sequence parameters and time-integration coefficients	770
BROUTINE TIMING	Program to determine elapsed cpu time	770
BROUTINE TIMLOG	Program to print log of execution times	770
BROUTINE TSEQ	Program to set memory pointers for time sequence and nodal time history data arrays	770
BROUTINE TSEQIN	Program to read, write and store time sequence data	771
BROUTINE ZTEST	Program to determine if an array contains only zero entries	772
BROUTINE QUADC	Program to set storage and call tasks for the four-node quadrilateral, elastic continuum element	772
BROUTINE QDCT1	Program to read, generate, and write data for the four-node quadrilateral, elastic continuum element	775
BROUTINE QDCT2	Program to calculate effective mass matrix for the four-node quadrilateral, elastic continuum element and assemble into the global left-hand-side matrix	776

11.5.4 Subroutine Index for Program Listing (cont.)

Routine	Description	Page
SUBROUTINE QDCT3	Program to calculate residual-force vector for the four-node quadrilateral, elastic continuum element and assemble into the global right-hand-side vector	777
SUBROUTINE QDCT4	Program to calculate and print stress, strain, and principal values for the four-node quadrilateral, elastic continuum element	779
SUBROUTINE QDCT5	Program to calculate and store element time histories for the four-node quadrilateral, elastic continuum element	780
SUBROUTINE QDCB	Program to set up the strain-displacement matrix B for two-dimensional continuum elements	780
SUBROUTINE QDCK	Program to form stiffness matrix for a continuum element with NEN nodes	781
SUBROUTINE QDCKD	Program to form internal force $(-k^e * d^e)$ for a continuum element with NEN nodes	782
SUBROUTINE QDCRSF	Program to read, write and store surface force data for the four-node quadrilateral, elastic continuum element	782
SUBROUTINE QDCSHG	Program to calculate global derivatives of shape functions and Jacobian determinants for a four-node quadrilateral element	783
SUBROUTINE QDCSHL	Program to calculate integration-rule weights, shape functions, and local derivatives for a four-node quadrilateral element	784
SUBROUTINE QDCSTR	Program to calculate stress, strain, and principal values at an integration point for a two-dimensional continuum element	784
SUBROUTINE QDCSUF	Program to compute consistent surface loads for the four-node quadrilateral, elastic continuum element	785
SUBROUTINE TRUSS	Program to set storage and call tasks for the three-dimensional, elastic truss element	785
SUBROUTINE TRUST1	Program to read, generate, and write element data for the three-dimensional, elastic truss element	788
SUBROUTINE TRUST2	Program to calculate effective mass matrix for the three-dimensional, elastic truss element and assemble into the global left-hand-side matrix	789
SUBROUTINE TRUST3	Program to calculate residual-force vector for the three-dimensional, elastic truss element and assemble into the global right-hand-side vector	790
SUBROUTINE TRUST4	Program to calculate and print stress, strain, and force for the three-dimensional, elastic truss element	791
SUBROUTINE TRUST5	Program to calculate and store element time histories for the three-dimensional, elastic truss element	792
SUBROUTINE TRUSB	Program to set up the strain-displacement matrix B for the three-dimensional, elastic truss element	792

11.5.4 Subroutine Index for Program Listing (cont.)

Routine	Description	Page
BROUTINE TRUSHG	Program to calculate global derivatives of shape functions and Jacobian determinants for the three-dimensional, elastic truss element	793
BROUTINE TRUSHL	Program to calculate integration-rule weights, shape functions and local derivatives for a two or three node one-dimensional element	793
BROUTINE TRUSK	Program to form stiffness matrix for the three-dimensional, elastic truss element	794
BROUTINE TRUSKD	Program to form internal force ($-k^e * d^e$) for the three-dimensional, elastic truss element	795
BROUTINE TRUSPR	Program to read, write, and store properties for the three-dimensional, elastic truss element	795
BROUTINE TRUSPT	Program to print stress, strain, and force for the three-dimensional, elastic truss element	795
BROUTINE TRUSTR	Program to calculate stress, strain and force at an integration point for the three-dimensional, elastic truss element	796

11.5.5 Program Listing

```
PROGRAM MAIN                                                          MAIN      1
                                                                      MAIN      2
*************************************************************         MAIN      3
*                                                           *         MAIN      4
*                 * * *   D L E A R N   * * *               *         MAIN      5
*                                                           *         MAIN      6
*                                                           *         MAIN      7
*                 A LINEAR STATIC AND DYNAMIC               *         MAIN      8
*                                                           *         MAIN      9
*               FINITE ELEMENT ANALYSIS PROGRAM             *         MAIN     10
*                                                           *         MAIN     11
*                                                           *         MAIN     12
*                 T. J. R.   H U G H E S                    *         MAIN     13
*                                                           *         MAIN     14
*                    R. M.   F E R E N C Z                  *         MAIN     15
*                                                           *         MAIN     16
*                    A. M.   R A E F S K Y                  *         MAIN     17
*                                                           *         MAIN     18
*************************************************************         MAIN     19
                                                                      MAIN     20
PROGRAM TO SET STORAGE CAPACITY, PRECISION AND INPUT/OUTPUT UNITS      MAIN     21
                                                                      MAIN     22
COMMON /BPOINT/ MFIRST,MLAST,MTOT,IPREC                               MAIN     23
COMMON /IOUNIT/ IIN,IOUT,IRSIN,IRSOUT                                 MAIN     24
COMMON A(50000)                                                       MAIN     25
                                                                      MAIN     26
   MFIRST = ADDRESS OF FIRST AVAILABLE WORD IN BLANK COMMON            MAIN     27
   MLAST  = ADDRESS OF LAST AVAILABLE WORD IN BLANK COMMON             MAIN     28
   MTOT   = TOTAL STORAGE ALLOCATED TO BLANK COMMON                    MAIN     29
   IPREC  = PRECISION FLAG; EQ.1, SINGLE PRECISION                     MAIN     30
                            EQ.2, DOUBLE PRECISION                    MAIN     31
                                                                      MAIN     32
MFIRST = 1                                                            MAIN     33
MTOT   = 50000                                                        MAIN     34
MLAST  = MTOT                                                         MAIN     35
IPREC  = 2                                                            MAIN     36
                                                                      MAIN     37
   IIN    = INPUT UNIT NUMBER                                         MAIN     38
   IOUT   = OUTPUT UNIT NUMBER                                        MAIN     39
   IRSIN  = RESTART INPUT UNIT NUMBER                                 MAIN     40
   IRSOUT = RESTART OUTPUT UNIT NUMBER                                MAIN     41
```